

Supplementary Material for

Mean and most probable first-passage times of lazy random walks in reflecting hypercubic lattices: exact results, modal scaling and inert defects

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Equations, theorems, figures and tables without the prefix S refer to the main text. The companion notes R0–R5 cited as [R2, Thm 9.1] and the like are in the folder `proofs/` of the repository <https://github.com/zhouyi-xiaoxiao/master-dissertation>, which also contains all code and data.

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S1 Model: notation, proofs and checks

S1.1 The walk

The cancelled-move rule (1) must be distinguished from the bounce-back rule, which puts a walker that tries to leave on the interior neighbour: for $N \geq 3$ that rule gives $P(\mathbf{x}, \mathbf{x} + \mathbf{e}_i) = q/d \neq P(\mathbf{x} + \mathbf{e}_i, \mathbf{x})$ at a wall, a non-symmetric matrix with a non-uniform stationary distribution. Under the cancelled-move rule $\eta = 0$ in the interior and $\eta = d$ at a corner.

By irreducibility every $\mathbf{x} \neq \mathbf{a}$ has a path to $\mathbf{a}$ of positive probability and length at most $n - 1$, so every row sum of Q^{n-1} is below one: the spectral radius of Q is below one, $I - Q$ is invertible, and (3) follows from $T = \sum_t t f(t) = \sum_{t \geq 0} S(t)$. The maximum in Definition 2.1 exists because $f \geq 0$ and $\sum_t f(t) = 1$.

Table S1: Main notation. Sites are zero-based; s without a subscript is the Laplace variable.

Symbol	Meaning
Ω, n, q	lattice $\{0, \dots, N-1\}^d$; number of sites N^d ; activity
$\mathbf{o}, \mathbf{a}$	start site; target site
P, P_1	transition matrix (1); the same at $q = 1$
Q, r	P without the row and column of $\mathbf{a}$; one-step absorption vector $(I - Q)\mathbf{1}$
$X_t, \mathcal{T}$	position of the walker after t steps; first-passage time $\min\{t \geq 0 : X_t = \mathbf{a}\}$
$S(t), f(t)$	survival probability $\mathbb{P}(\mathcal{T} > t)$; first-passage PMF (2)
h	vector of the mean first-passage times $\mathbb{E}_{\mathbf{x}}[\mathcal{T}]$ over the start sites $\mathbf{x}$
T, t^*, ϱ	mean first-passage time; mode (Definition 2.1); t^*/T
$g(t), t_c^*$	unit-rate continuous-time density; its mode (at activity q the mode is t_c^*/q)
t_{cert}	time beyond which f is certified to stay below its maximum (Proposition 3.1)
$\hat{F}(z), \hat{F}(s)$	generating function of f ; Laplace transform of g
$\hat{P}(\mathbf{x}, s \mathbf{y}), \mu_{\mathbf{k}}$	reflecting propagator (7); its rates
$\psi_k(m), \phi_{\mathbf{k}}(\mathbf{x})$	one-dimensional cosine eigenvector; product eigenvector $\prod_i \psi_{k_i}(x_i)$
μ_1, X	slowest reflecting rate (8); $\mu_1 qT$
ν_j, b_j	decay rates and residues of the first-passage law (6)
L, L^+, G, R	$I - P$; its pseudo-inverse; pseudo-inverse of the grid Laplacian; effective resistance
s_k, c_k, α_k	$\sin \frac{\pi k}{2N}$; $\cos \frac{\pi k}{2N}$; 1 if $k = 0$, else 2
$\vartheta_k, \vartheta_k(m)$	$\frac{\pi k}{N}$; $\frac{\pi k(2m+1)}{2N} = (m + \frac{1}{2})\vartheta_k$
$\theta_3(x), \theta_4(x)$	Jacobi theta functions of the nome x : $\sum_{n \in \mathbb{Z}} x^{n^2}$, $\sum_{n \in \mathbb{Z}} (-1)^n x^{n^2}$
$\mathcal{B}, \mathcal{C}, p$	blocked sites; open cluster of the start; blocked fraction (Section 8)
$\Theta(p), D_{\text{eff}}, \Sigma$	time-dilation factor D_0/D_{eff} ; effective diffusivity; conductivity of the diluted lattice

S1.2 Proof of Proposition 2.2

(a) By (1) the off-diagonal entries of P are q times those of P_1 and both matrices have unit row sums, so $P = (1 - q)I + qP_1$. Deleting the row and column of $\mathbf{a}$ gives $Q = (1 - q)I + qQ_1$, and $r = (I - Q)\mathbf{1} = q r_1$. Then $T = \mathbf{e}_{\mathbf{o}}^T(I - Q)^{-1}\mathbf{1} = q^{-1}\mathbf{e}_{\mathbf{o}}^T(I - Q_1)^{-1}\mathbf{1}$ by (3).

(b) Summing (2) gives $\hat{F}_q(z) = z\mathbf{e}_{\mathbf{o}}^T(I - zQ)^{-1}r$ for small $|z|$, and integrating g gives $\hat{F}(s) = \mathbf{e}_{\mathbf{o}}^T(sI + I - Q_1)^{-1}r_1$ for $\text{Re } s \geq 0$; both are rational functions and are continued as such. By (a),

$$I - zQ = (1 - z)I + zq(I - Q_1) = zq\left[\frac{1 - z}{zq}I + (I - Q_1)\right],$$

so $z(I - zQ)^{-1}r = z(zq)^{-1}(sI + I - Q_1)^{-1}q r_1$ with $s = (1 - z)/(zq)$, which is (5).

(c) Insert $\mathbf{e}^{-(I - Q_1)t} = \sum_{\nu} \mathbf{e}^{-\nu t} \Pi_{\nu}$, $Q^{t-1} = (I - q(I - Q_1))^{t-1} = \sum_{\nu} (1 - q\nu)^{t-1} \Pi_{\nu}$ and $(sI + I - Q_1)^{-1} = \sum_{\nu} (s + \nu)^{-1} \Pi_{\nu}$ into the definitions and use $r = q r_1$. Because $(I - Q_1)^{-1}r_1 = \mathbf{1}$, we get $\sum_{\nu} b_{\nu}/\nu = \mathbf{e}_{\mathbf{o}}^T \mathbf{1} = 1$ and $\sum_{\nu} b_{\nu}/\nu^2 = \mathbf{e}_{\mathbf{o}}^T(I - Q_1)^{-1}\mathbf{1} = qT$. The formula for S follows from $S(t) = \sum_{\tau > t} f(\tau)$.

(d) $g_q(t) = \mathbf{e}_{\mathbf{o}}^T \mathbf{e}^{-q(I - Q_1)t} q r_1 = q g(qt)$ by (a). Its mean is $\mathbf{e}_{\mathbf{o}}^T(I - Q)^{-2}r = \mathbf{e}_{\mathbf{o}}^T(I - Q)^{-1}\mathbf{1} = T$. The eigenvalues of Q are $1 - q\nu$ with $\nu \in (0, 2)$, hence larger than $1 - 2q$. $\square$

In double precision the two sides of (5) agree to 3×10^{-16} at 15 test points in $d = 1, 2, 3$, and to 8×10^{-16} in a separately written implementation. Part (d) also bounds parity effects: at $q = 0.8$ any alternating component of f decays at least as fast as 0.6^t .

S1.3 Proof of Proposition 2.3

The eigenvalues of P are $1 - q\mu_{\mathbf{k}}$, and relaxation across the box takes about $2dN^2/(\pi^2 q)$ steps. Solve $(sI + I - P_1)c = \mathbf{e}_{\mathbf{a}}$ for the column $c = \hat{P}(\cdot, s | \mathbf{a})$, with the target written last and c' the restriction of c to the other sites:

$$(sI + I - Q_1)c' = r_1 \hat{P}(\mathbf{a}, s | \mathbf{a}), \quad [s + 1 - P_1(\mathbf{a}, \mathbf{a})] \hat{P}(\mathbf{a}, s | \mathbf{a}) - r_1^T c' = 1. \quad (\text{S1.1})$$

The first equation, the formula $\hat{F}(s) = \mathbf{e}_0^\top (sI + I - Q_1)^{-1} r_1$ and the symmetry of the propagator give (9). The algebra of Proposition 2.2(b) applied to P gives $\tilde{P}(\mathbf{x}, z | \mathbf{y}) = [(I - zP)^{-1}]_{\mathbf{xy}} = \tilde{P}(\mathbf{x}, s | \mathbf{y})/(qz)$ at $s = (1 - z)/(qz)$, so the same ratio of discrete-time propagators equals $\tilde{F}_q(z)$. Eliminating c' from (S1.1) gives

$$\frac{1}{\tilde{P}(\mathbf{a}, s | \mathbf{a})} = s + 1 - P_1(\mathbf{a}, \mathbf{a}) - \sum_{\nu} \frac{\|\Pi_{\nu} r_1\|^2}{s + \nu}, \quad (\text{S1.2})$$

the sum running over the distinct eigenvalues of $I - Q_1$.

(i) By (7), $\tilde{P}(\mathbf{a}, s | \mathbf{a}) = \sum_{i=0}^m w_i / (s + \mu_{(i)})$ with all $w_i > 0$. For non-real s its imaginary part is $-\text{Im } s \sum_i w_i / |s + \mu_{(i)}|^2 \neq 0$. On the real axis its derivative $-\sum_i w_i / (s + \mu_{(i)})^2$ is negative, and on each interval $(-\mu_{(i+1)}, -\mu_{(i)})$ the function falls from $+\infty$ to $-\infty$, so it has exactly one zero there, which is simple. It is positive for $s > 0$ and negative for $s < -\mu_{(m)}$.

(ii) The zeros of $\tilde{P}(\mathbf{a}, s | \mathbf{a})$ are the poles of its reciprocal, which by (S1.2) are the points $s = -\nu$ with $\Pi_{\nu} r_1 \neq 0$. If $b_{\nu} = \mathbf{e}_0^\top \Pi_{\nu} r_1 \neq 0$ then $\Pi_{\nu} r_1 \neq 0$.

(iii) Q_1 is then a symmetric, irreducible, non-negative matrix, so by the Perron–Frobenius theorem its largest eigenvalue $1 - \nu_{\min}$ is simple with an eigenvector $v > 0$. Since $r_1 \geq 0$ and $r_1 \neq 0$, $\Pi_{\nu_{\min}} r_1 = v (v^\top r_1) / \|v\|^2 \neq 0$ and $b_{\nu_{\min}} = v(\mathbf{o}) (v^\top r_1) / \|v\|^2 > 0$; by (ii) $\nu_{\min}$ is the smallest of the $\nu_{(i)}$. With $h = (I - Q_1)^{-1} \mathbf{1}$, $h(\mathbf{x}) = qT_{\mathbf{x} \rightarrow \mathbf{a}}$, we have $v^\top \mathbf{1} = v^\top (I - Q_1)h = \nu_{\min} v^\top h$. For $d = 1$ the hypothesis holds when $\mathbf{a}$ is an end site. $\square$

The eigenvectors of Q_1 that are orthogonal to r_1 are exactly the eigenvectors of the reflecting lattice that vanish at the target (extended by zero at $\mathbf{a}$; they are combinations of cosine modes of equal rate). They survive the removal of the target unchanged and carry no first-passage probability. All other decay rates are shifted, by (10), to values strictly between the weighted reflecting rates. For a corner target every $\phi_{\mathbf{k}}(\mathbf{a})$ is non-zero, so $\mu_{(1)} = \mu_1$; for a centre target (N odd) the modes with an odd index vanish there and $\mu_{(1)} = \frac{2}{d} \sin^2 \frac{\pi}{N} \approx 4\mu_1$.

The decay rate of the survival probability is set by ν_0 , not by the reflecting gap. For $d = 2$, $N = 35$, $q = 0.8$ the reflecting gap $q\mu_1$ is 1.610×10^{-3} per step, whereas the survival probability decays at $q\nu_0 = 7.731 \times 10^{-5}$, close to $1/T = 7.092 \times 10^{-5}$; for corner-to-corner transport $\nu_0 qT = 1.090$ ($d = 2$, $N = 35$) and 1.009 ($d = 3$, $N = 40$).

S1.4 Proof of Proposition 2.4 and the zero mode

The argument of Section 2.2 uses only irreducibility: $I - Q$ is invertible and $h(\mathbf{x}) = [(I - Q)^{-1} \mathbf{1}](\mathbf{x}) < \infty$ for $\mathbf{x} \neq \mathbf{a}$, $h(\mathbf{a}) = 0$. The conditions “ $h(\mathbf{a}) = 0$ and $(Lh)(\mathbf{x}) = 1$ for $\mathbf{x} \neq \mathbf{a}$ ” are the linear system $(I - Q)h|_{\mathbf{x} \neq \mathbf{a}} = \mathbf{1}$, whose solution is unique. Next, L is symmetric and positive semi-definite and its kernel is spanned by $\mathbf{1}$, because the eigenvalue 1 of an irreducible stochastic matrix is simple. Hence $LL^+ = I - \frac{1}{n} \mathbf{1}\mathbf{1}^\top$. Put $u = n(L_{\mathbf{aa}}^+ \mathbf{1} - L^+ \mathbf{e}_{\mathbf{a}})$. Then $u(\mathbf{a}) = 0$ and

$$Lu = -nLL^+ \mathbf{e}_{\mathbf{a}} = -n\left(\mathbf{e}_{\mathbf{a}} - \frac{1}{n} \mathbf{1}\right) = \mathbf{1} - n\mathbf{e}_{\mathbf{a}},$$

so $(Lu)(\mathbf{x}) = 1$ for $\mathbf{x} \neq \mathbf{a}$. By uniqueness $u = h$, and $u(\mathbf{x}) = n(L_{\mathbf{aa}}^+ - L_{\mathbf{xa}}^+)$ because L^+ is symmetric. $\square$

By Proposition 2.2(a), $L^+ = q^{-1}(I - P_1)^+$, which is (4) again. Since $1 - P(\mathbf{x}, \mathbf{x}) = \frac{q}{2d} (2d - \eta(\mathbf{x}))$ and $2d - \eta(\mathbf{x})$ is the number of neighbours of $\mathbf{x}$, (1) reads $I - P = \frac{q}{2d} \mathcal{L}$, which gives (13), with $2dn/q$ in the place of the total conductance. When a symmetry of the lattice exchanges $\mathbf{x}$ and $\mathbf{a}$, as for opposite corners, the two directions are equal and $T = (dn/q) R$ (Corollary 4.4).

Generating function and the zero mode. Since $\tilde{F}(1) = 1$ and $\tilde{F}$ is analytic in a disc of radius greater than one,

$$T = \tilde{F}'(1) = \lim_{z \rightarrow 1^-} \frac{1 - \tilde{F}(z)}{1 - z}; \quad (\text{S1.3})$$

the reciprocal $(1 - z)/(1 - \tilde{F}(z))$ tends to $1/T$. The discrete propagator is $\tilde{P}(\mathbf{x}, z \mid \mathbf{y}) = \frac{1}{n(1-z)} + \mathcal{R}_{\mathbf{xy}}(z)$, where $\mathcal{R}(z)$ collects the modes $\mathbf{k} \neq \mathbf{0}$ and $\mathcal{R}(1) = L^+$; its $\mathbf{k} = \mathbf{0}$ pole lies at $z = 1$, on the unit circle. In the renewal ratio this pole cancels,

$$1 - \tilde{F}(z) = \frac{n(1 - z) [\mathcal{R}_{\mathbf{aa}}(z) - \mathcal{R}_{\mathbf{oa}}(z)]}{1 + n(1 - z) \mathcal{R}_{\mathbf{aa}}(z)},$$

and (S1.3) returns (12).

S2 Numerical methods and their validation

Two qualifications of “exact” apply in addition to those of Section 3 of the main text: where a maximiser is needed to a fraction of a step (the offsets of Section S2.2 and the single-defect responses of Section 8) it is refined by a parabola through three exact values, and the one-dimensional modes at $N = 10^5$ and 10^6 were located as local maxima from the sign of the exact increment (their global maximality follows from Theorem 7.13(c)).

S2.1 The routes in detail

Route A: time stepping. We read $f(t + 1)$ as the probability of landing on the target in that step, $f(t + 1) = \sum_{\mathbf{x}} \rho_t(\mathbf{x}) P(\mathbf{x}, \mathbf{a})$; the difference $S(t) - S(t + 1)$ would lose digits by cancellation. The 193 runs reach 1 248 768 steps for the chain with $N = 1000$, 517 376 steps for $d = 2$, $N = 401$, and 85 248 steps for $d = 3$, $N = 100$. Each run continues until t exceeds 1.5 times the running argmax (1.35 for the six largest lattices, 3 for the chain, 4 for EXIT); 35 further runs at smaller sizes continue until $S(t) < 10^{-12}$ and so cover the whole support. No exact tie for the maximum occurred in any run.

Route B: one dimension. For the chain the PMF is the closed form (29) of Proposition 5.1. Double precision suffices if the logarithm of each eigenvalue $\lambda_m = 1 - q\nu_m$ is formed as $\log_{1p}(-q\nu_m)$: taking the logarithm of the rounded difference $1 - q\nu_m$ loses the digits that decide the integer and displaces the located maximum by 353 steps at $N = 10^5$ and by 385 703 steps at $N = 10^6$. The values found in this way by two codes were confirmed as local maxima in 50-digit arithmetic.

Route C: Laplace space. The d -fold spectral sum for $\hat{P}$ is reduced to a $(d - 1)$ -fold sum by carrying out one index in closed form.

Lemma S2.1 (Resolvent of the one-dimensional reflecting chain). *Let $\mathbf{T}$ be the $N \times N$ matrix of Lemma S3.1, $\beta > 0$, and let $\zeta \in \mathbb{C}$ be such that $\zeta I + \beta(I - \mathbf{T})$ is invertible, that is, $\zeta \neq -\beta(1 - \cos(\pi k/N))$ for $k = 0, \dots, N - 1$. Let $\varphi \in \mathbb{C}$ be any solution of $\cosh \varphi = 1 + \zeta/\beta$. Then, for $0 \leq m, m' \leq N - 1$,*

$$G_1(\zeta; m, m') := [(\zeta I + \beta(I - \mathbf{T}))^{-1}]_{m, m'} = \frac{\cosh(\varphi(m_{<} + \frac{1}{2})) \cosh(\varphi(N - \frac{1}{2} - m_{>}))}{\frac{\beta}{2} \sinh \varphi \sinh N\varphi}, \quad (\text{S2.1})$$

where $m_{<} = \min(m, m')$ and $m_{>} = \max(m, m')$. At $\zeta = -2\beta$, where numerator and denominator both vanish, the right-hand side is understood as its limit.

Proof. Put $u(m) = \cosh(\varphi(m + \frac{1}{2}))$ and $v(m) = \cosh(\varphi(N - \frac{1}{2} - m))$ for all integers m . Since $\cosh \varphi(x + 1) + \cosh \varphi(x - 1) = 2 \cosh \varphi \cosh \varphi x$ and $\beta \cosh \varphi = \zeta + \beta$, both satisfy $(\zeta + \beta)w(m) - \frac{\beta}{2}[w(m - 1) + w(m + 1)] = 0$ for every m ; moreover $u(-1) = u(0)$ and $v(N) = v(N - 1)$. The matrix $M = \zeta I + \beta(I - \mathbf{T})$ acts on a vector w by the same three-term expression with the conventions $w(-1) := w(0)$ and $w(N) := w(N - 1)$, because $\mathbf{T}(0, 0) = \mathbf{T}(N - 1, N - 1) = \frac{1}{2}$.

Fix m' and let $g(m) = u(m_{<})v(m_{>})$. For $m < m'$, $(Mg)(m) = v(m') \cdot 0$, and for $m > m'$, $(Mg)(m) = u(m') \cdot 0$; the boundary conventions hold for g because they hold for u at the left end and for v at the right end. At $m = m'$ the recurrence for u gives

$$\begin{aligned} (Mg)(m') &= \frac{\beta}{2} [u(m' + 1)v(m') - u(m')v(m' + 1)] \\ &= \frac{\beta}{4} [\cosh(N + 1)\varphi - \cosh(N - 1)\varphi] = \frac{\beta}{2} \sinh \varphi \sinh N\varphi, \end{aligned}$$

by the product formula for hyperbolic cosines. Hence $Mg = \frac{\beta}{2} \sinh \varphi \sinh N\varphi \mathbf{e}_{m'}$. The factor $\sinh \varphi \sinh N\varphi$ vanishes only if $\cosh \varphi = \cos(\pi j/N)$ for an integer j , i.e. at the excluded values of ζ or at $\zeta = -2\beta$; elsewhere division gives the m' -th column of M^{-1} . The right-hand side does not depend on the choice of φ , being even and $2\pi i$ -periodic in it. $\square$

For $\zeta > 0$ the lemma is equivalent to the ring identity (S3.6) of Lemma S3.2, and hence to Eq. (E1) of Giuggioli (2020); the direct proof also covers the complex and negative arguments that the inversion and the pole search need. Against dense inverses the formula holds to 2×10^{-13} in 164 020 matrix entries with positive, complex and negative real ζ . By Lemma S3.1, $I - P_1 = \frac{1}{d} \sum_i (I - \mathbf{T}^{(i)})$. Expanding the first $d - 1$ coordinates in the cosine basis and applying (S2.1) with $\beta = 1/d$ to the last one gives

$$\hat{P}(\mathbf{x}, s \mid \mathbf{y}) = \sum_{k_1, \dots, k_{d-1}=0}^{N-1} \left[\prod_{i=1}^{d-1} \frac{\alpha_{k_i}}{N} \cos \vartheta_{k_i}(x_i) \cos \vartheta_{k_i}(y_i) \right] G_1 \left(s + \frac{2}{d} \sum_{i=1}^{d-1} s_{k_i}^2; x_d, y_d \right), \quad (\text{S2.2})$$

a single sum for $d = 2$ and a double sum for $d = 3$. For $d = 2$ and $s = 0$ the angle is $\varphi = \varphi_k$, and (S2.2) is the summation that yields the single-sum mean of Theorem 4.2.

Two devices make large lattices cheap. First, the terms of (S2.2) whose shift $\frac{2}{d} \sum_i s_{k_i}^2$ exceeds $20 s_{\max}$, where $s_{\max}$ is the largest $|s|$ required, are analytic in $|s| < 20 s_{\max}$; their sum is replaced once by its Taylor polynomial of degree 15, obtained from 16 samples on the circle $|s| = s_{\max}$. The number of terms summed for each s then does not grow with N (79 of 3000 at $N = 3000$, $d = 2$), and accelerated and plain sums of $\hat{F}$ agree to 5×10^{-16} ($d = 2$, $N = 3000$) and 2×10^{-13} ($d = 3$, $N = 120$). Second, $g(t)$ is recovered from $\hat{F}$ by the fixed-Talbot rule of Abate and Valkó (2004), a trapezoidal rule with M nodes on Talbot's deformed Bromwich contour (Talbot, 1979; Weideman, 2006). In double precision $M = 20$ is the best choice among $M = 16, 20, 24, 28, 32$: compared with the exact density, written as the Erlang mixture $g(t) = \sum_{n \geq 1} f_1(n) t^{n-1} e^{-t} / (n-1)!$ of the PMF f_1 at $q = 1$, the median relative error is 1.4×10^{-13} and the largest 5.4×10^{-11} over 30 values of t between 0.2 and 20 modal times. The mode t_c^* is the root of g' , which we obtain by inverting $s\hat{F}(s)$ (legitimate because $g(0) = 0$ when the start is not adjacent to the target), bracketing the sign change on a logarithmic grid and subdividing until the bracket has relative width below 10^{-11} . The same code gives T from the limit $s \rightarrow 0$ and the smallest poles $-\nu_j$ of $\hat{F}$ with their residues b_j .

Route P. The poles are found by bracketed root finding (all ν_j below $14\mu_1$ for $d = 2$ and $11\mu_1$ for $d = 3$), and the residues are $b_j = \hat{P}(\mathbf{a}, -\nu_j \mid \mathbf{o}) / \partial_s \hat{P}(\mathbf{a}, s \mid \mathbf{a})|_{s=-\nu_j}$. The truncation matters only for t well below the mode: raising the number of poles from 8–11 to 21–33 changes the mode by less than 10^{-15} in the four cases tested.

Monte Carlo check of the walker rule. As a check that involves no linear algebra, we simulated the rule of Section 2 literally for the corner-to-corner problem at $N = 35$, $q = 0.8$: in every step a uniform number decides whether the walker attempts a move, a second one selects the direction, and a move off the lattice is cancelled. Eight independent batches of 500 000 walkers (fixed seeds, all batches reported) give a sample mean of 14 104.6 steps against the exact $T = 14 100.808$ (standard error 6.5, $z = 0.58$). A χ^2 test of the arrival times on 100 bins that are equiprobable under the exact PMF gives 112.5 on 99 degrees of freedom ($p = 0.17$), the fraction

Table S2: Cross-checks of the numerical routes. Deviations are relative unless marked “abs.” (absolute) or given in steps. “Solve” is a sparse direct solution of $(I - Q)h = \mathbf{1}$, which gives $T = h(\mathbf{o})$; cases are corner-to-corner unless stated.

Check	Cases	Largest deviation
Route A against explicit powers of the sparse matrix Q	$d = 1, 2, 3$; $N = 9, 6, 5$; 399 steps	4×10^{-16} (abs.)
$1 - \sum_t f(t)$ over the full support	6 cases, $d = 1, 2, 3$	2×10^{-13} (abs.)
$\sum_t t f(t)$ against solve	same 6 cases	3×10^{-12}
Route B against route A; argmax	$N = 5-200$, $q = 0.3, 0.8, 1$ (6 cases)	3×10^{-17} (abs.); identical
T from route C ($s \rightarrow 0$) and from (17) against solve	$d = 2$, $N = 2-200$, $q = 0.3, 0.8, 1$ (24 cases)	9×10^{-11}
(17) against the spectral sum, second code	$d = 2$, $N = 2-40, 60, 100, 150, 200$; same q	1.9×10^{-14}
T from route C against solve	CC, C2M, M2C; $d = 2, 3$; $N \leq 25$ (13 cases)	5×10^{-12}
Lemma S2.1 against dense inverses	$N \leq 64$; real and complex ζ	2×10^{-13}
Far-term Taylor acceleration against plain sums	$d = 2$, $N = 3000$; $d = 3$, $N = 120$	5×10^{-16} ; 2×10^{-13}
Talbot ($M = 20$) against the Erlang mixture of the exact $q = 1$ PMF	$d = 1, 2, 3$; three geometries; 30 values of t	5.4×10^{-11}
Talbot ($M = 24$) against the closed-form density, $d = 1$	$N = 20, 200, 2000$; 18 values of t	4×10^{-10}
Identity (5) against $\sum_t f(t)z^t$	15 points, $d = 1, 2, 3$	3×10^{-16} (abs.)
Rule (S2.4) ($\gamma = 10$) against route A (Fig. S1(b))	$d = 2$, $N = 35$, $1000 \leq t \leq 6000$	6.4×10^{-10}
Route P against dense eigen-decomposition: rates; residues	$d = 2$, $N = 8, 13$; $d = 3$, $N = 5$	6×10^{-14} ; 7×10^{-13}
Mode t_c^* : route C against route P	40 sizes ($d = 2$); 24 sizes ($d = 3$)	7×10^{-13} ; 1.1×10^{-10}
Integer mode t^* : two time-stepping codes	148 cases	no difference
Certificate of the global mode (Proposition 3.1), third stepping code	all 193 runs of route A	all certified; modes identical
Interpolated maximiser of route A against the exact three-exponential truncation of (6)	$d = 2, 3$; CC, C2M; $N \geq 30$	0.087 steps
Monte Carlo against the exact PMF (Fig. S1(a))	$d = 2$, $N = 35$, $q = 0.8$; 4×10^6 walkers	$z = 0.58$; $\chi_{99}^2 = 112.5$
Monte Carlo mean against (21)	three pairs, $N = 5, 6, 7$	$ z \leq 0.52$

of walkers that arrive by the exact modal time $t^* = 2493$ is 0.09805 against 0.09806, and the Kolmogorov–Smirnov distance is 3.2×10^{-4} ($0.64/\sqrt{n}$). Batch by batch, $|z| \leq 1.27$ and χ_{99}^2 lies between 77.6 and 134.9. The sample confirms the distribution (Fig. S1(a)) but could not have located the mode: the PMF stays within 1% of its maximum for $2162 \leq t \leq 2913$, that is from $0.87 t^*$ to $1.17 t^*$, while a histogram bin of 200 steps near the peak has a relative standard error of 0.4%. The mode has to come from the exact routes.

S2.2 Discrete and continuous time; parity

In the 54 cases computed by both discrete and continuous routes at $q = 0.8$ ($d = 1, 2, 3$; CC, C2M, M2C), $t^* = t_c^*/q + c$ with $-3.05 \leq c \leq -0.22$ steps, and $-2.81 \leq c \leq -0.52$ in the 17 cases of the second implementation; the relative difference is below 10^{-4} for all $N \geq 100$. The offset of the interpolated maximiser is -0.45 steps for the chain and drifts slowly with N for $d \geq 2$ (from -0.56 to -1.01 for $d = 2$, $N = 5$ to 200, and from -1.05 to -2.59 for $d = 3$, $N = 5$ to 100; Fig. S2(b)); whether it stays bounded as $N \rightarrow \infty$ is not established. Other activities follow from

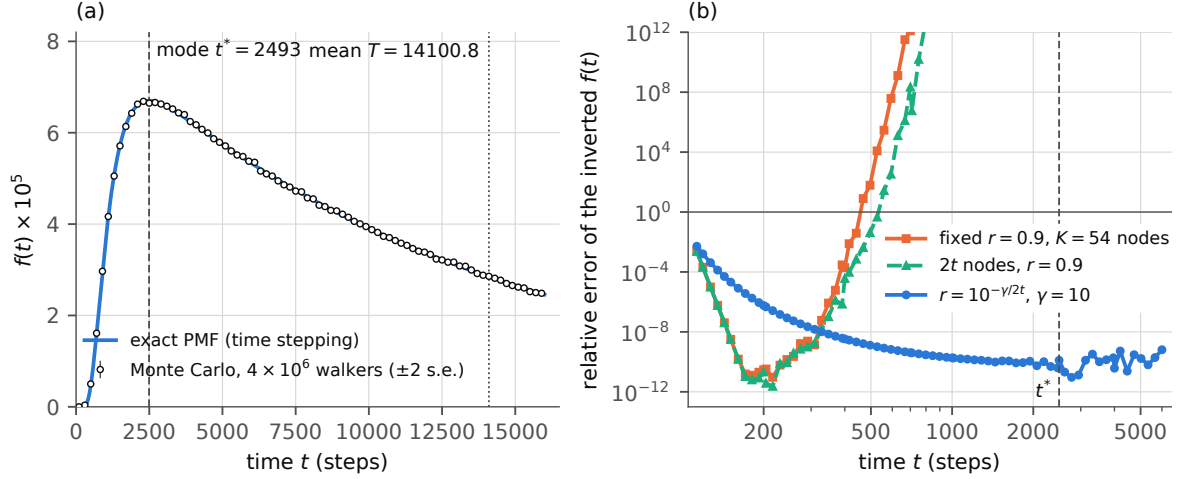


Figure S1: Corner-to-corner first passage on the square lattice, $N = 35$, $q = 0.8$, discrete time. (a) Exact PMF from time stepping (line) and histogram of 4×10^6 simulated walkers in bins of 200 steps (circles; bars are ± 2 standard errors and are smaller than the symbols). Dashed line: mode $t^* = 2493$; dotted line: mean $T = 14100.8$. (b) Relative error of three numerical inversions of the generating function against the exact PMF: the trapezoidal rule on a circle of fixed radius $r = 0.9$ truncated after $K = 54$ nodes, the same rule with all $2t$ nodes, and the rule (S2.4) with $\gamma = 10$. The two fixed-radius curves leave the frame near $t = 700$; at t^* (dashed) the truncated fixed-radius rule returns 3.6×10^{93} .

the scaling with q (Table S18).

If $q \leq \frac{1}{2}$ no component of f alternates in sign, and at $q = 0.8$ alternating components decay at least as fast as 0.6^t (Proposition 2.2(d)). At $q = 1$ parity is visible. On the chain only a cancelled move at the reflecting end changes the parity of position plus time, the values of f at even and at odd t lie on two branches, the maximum of the even branch exceeding that of the odd branch by 4.0% at $N = 51$ (Fig. S2(a)) and by 1.0% at $N = 201$, and the raw argmax is biased: $N = 51$ gives 832 against $t_c^* = 849.9$, and $N = 201$ gives 13332 against 13398.0. The two-step average $\frac{1}{2}[f(t) + f(t+1)]$ is maximal at $t = 848$ and $t = 13398$. For $d = 2$ and $q = 1$ the PMF has a ripple at its peak, with two local maxima two steps apart ($t = 678$ and 680 at $N = 21$; also at $N = 15$ and $N = 51$, but not at $N = 101$); for $d = 3$ none was seen. Exact certificates confirm and complete this: for $2 \leq N \leq 160$ the ripple occurs exactly at 16 sizes, among them 15, 21 and 51 but not 101, and for $d = 3$, $2 \leq N \leq 60$, never (Theorem 7.9, computer-assisted).

S2.3 The certificate of the global mode in practice

In the third time-stepping code the vectors $u_k = Q^k \mathbf{e}_o$ and $v_k = Q^k r$ are advanced alternately; $f(2k+1) = u_k^T v_k$ and $f(2k+2) = u_{k+1}^T v_k$ are sums of non-negative terms; and the run stops at the first k (tested every 16 steps) with $\|u_k\| \|v_k\| < (1 - 10^{-6}) \max_{t \leq 2k} f(t)$, so that $t_{\text{cert}} = 2k+1$. The cost is that of one time-stepping run of length t_{cert} . At the largest sizes $N = 1000$, 401 and 100 for $d = 1, 2, 3$, $t_{\text{cert}} = 584097$, 478369 and 72673 (up to $4.2t^*$ for the smallest lattices). *Round-off.* The certificate has a relative margin of at least 10^{-6} . The integer maximiser on $\{1, \dots, t_{\text{cert}} - 1\}$ is found by comparing computed values. For the 97 corner-to-corner runs at $q \in \{0.5, 0.8, 1\}$ the modes are also certified in exact arithmetic (Section 3.2). For the other 96 runs (the placements C2M, M2C and EXIT, and corner to corner at $q = 0.3$ and 0.9) we use an a-priori bound of the round-off of route A. Each component of ρ_{t+1} is computed as a sum of at most $2d+1$ products of non-negative numbers, each product rounded once and the sum rounded at most $2d$ times, with stored coefficients $q/(2d)$ and $1 - q + \frac{q}{2d}n$ (n the number of cancelled

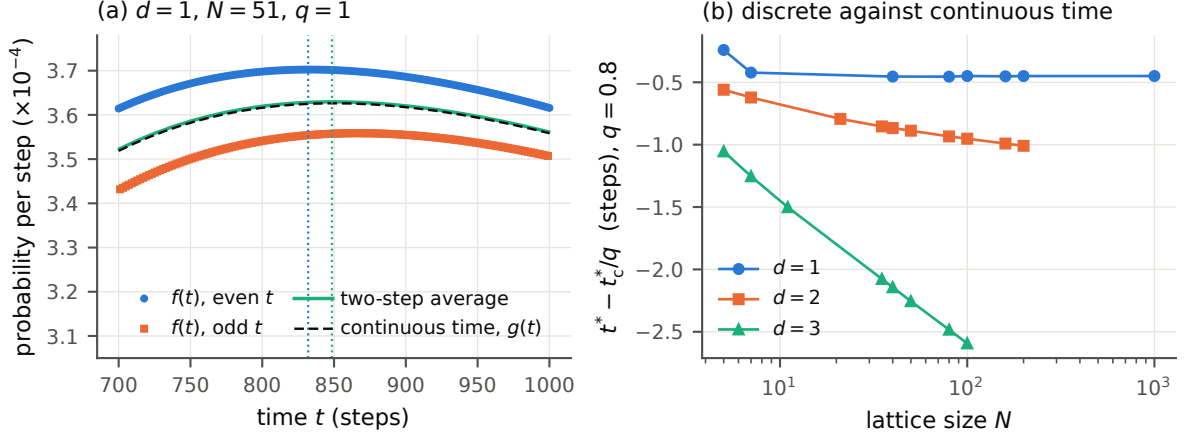


Figure S2: (a) Parity at $q = 1$: exact PMF of the one-dimensional chain (reflecting end to absorbing end, $N = 51$) for $700 \leq t \leq 1000$. Values at even t (circles) and odd t (squares) lie on two branches; the two-step average (solid line) follows the continuous-time density at unit rate (dashed). Dotted vertical lines: raw argmax ($t = 832$) and maximiser of the two-step average. (b) Difference, in steps, between the interpolated maximiser of the discrete-time PMF at $q = 0.8$ (three-point parabola through the exact values) and t_c^*/q , for corner-to-corner transport in $d = 1, 2, 3$ against N .

directions) whose relative error e against the exact values for the exact activity is computed in rational arithmetic. Hence every computed component equals $\sum_{\mathbf{y}} Q(\mathbf{x}, \mathbf{y}) \tilde{\rho}_t(\mathbf{y})(1 + \epsilon_{\mathbf{y}})$ with $|\epsilon_{\mathbf{y}}| \leq \delta := (1 + u)^{2d+1}(1 + e) - 1$, $u = 2^{-53}$, and by induction, since Q is non-negative, every computed $f(t)$ lies between $(1 - \delta)^t$ and $(1 + \delta)^t$ times the exact value (times $(1 \mp u)^{m-1}$ for the sum over the m target sites of EXIT); underflow adds an absolute error below 10^{-300} . In these runs $\delta \leq 1.0 \times 10^{-15}$. The computed maximiser is the exact one if every other computed value, multiplied by the upper factor at the maximiser, stays below the computed maximum multiplied by the lower factor at its own time; this holds in all 96 runs. The smallest relative gap between the maximum and any other computed value is 6.59×10^{-11} ($d = 2$, $N = 141$, centre to corner), the relative allowance $1 - L(t_{\text{cert}} - 1)/U(t^*)$, where $U(t)$ and $L(t)$ denote the upper and lower factors, is at most 9.2×10^{-11} (at $d = 2$, $N = 101$, $q = 0.3$, $t^* = 60\,187$), and in every run the gap exceeds the allowance, by a factor of at least 1.09. The round-off of the tail bound (norms of vectors of up to 10^6 components) is below 2×10^{-10} , far below its margin. As a consistency check, two evaluations of the same $f(t)$ in the certificate code from different pairs of vectors, $u_k^T v_k$ and $u_{k+1}^T v_{k-1}$, differ by at most 1.2×10^{-15} in relative terms; this measures the agreement of two evaluations, not the accumulated round-off, which is larger (see below).

One further size of the chain, $N = 10\,240$, was certified with a compiled stepper: $t_{\text{cert}} = 61\,300\,289$ ($1.40 t^*$) and the maximiser is $t^* = 43\,679\,969$, the value located by route B. At this size the maximum is too flat for the stepping to resolve the integer: neighbouring values of f differ by 1.7×10^{-16} in relative terms, whereas the stepped values, compared with the closed form (29) in multi-precision arithmetic, carry an accumulated round-off of 1.1×10^{-10} at the mode (at most 1.5×10^{-10} over the run). The stepping is therefore used only where its margin exceeds its worst-case round-off. All its operations act on non-negative numbers, so the relative error of every stepped quantity grows by at most five units of round-off per step, which gives the a-priori bound 3.4×10^{-8} for the whole run; underflow affects only components below 10^{-300} . Beyond t_{cert} the bound (b) lies below the maximum by a relative 1.1×10^{-6} , and for $t < t_{\text{cert}}$ outside a window of $\pm 20\,000$ steps around the maximiser the stepped values lie below it by at least 1.5×10^{-7} ; both statements survive the a-priori bound. Inside the window the increments $f(t+1) - f(t)$ of (29), evaluated in 60-digit arithmetic at all 40 001 times (24 slowest terms;

the omitted ones are bounded by 10^{-428} , the smallest increment being 10^{-24} , are positive up to $t^* - 1$ and negative from t^* on. The same comparison shows that accumulated round-off is common to neighbouring times: the relative error of the stepped values varies by 1.6×10^{-15} within 60 steps of the mode and by 3.8×10^{-14} over the window. The a-priori comparison used for the 96 runs above does not rely on this observation.

S2.4 Unimodality on the chain and the observed single maximum

Proposition S2.2 (Unimodality on the chain). *Let $d = 1$, start at the reflecting end 0 and target $N - 1$, and let $\lambda_m = 1 - q + q \cos \omega_m$, $\omega_m = (2m - 1)\pi/(2N - 1)$, be as in Proposition 5.1. Then*

$$\tilde{F}_q(z) = \prod_{m=1}^{N-1} \frac{(1 - \lambda_m)z}{1 - \lambda_m z}, \quad \hat{F}(s) = \prod_{m=1}^{N-1} \frac{1 - \cos \omega_m}{s + 1 - \cos \omega_m}. \quad (\text{S2.3})$$

Consequently (i) the continuous-time first-passage time is a sum of $N - 1$ independent exponential variables and its density g is log-concave, hence unimodal, for every N ; (ii) if $q[1 + \cos(2\pi/(2N - 1))] \leq 1$, in particular if $q \leq \frac{1}{2}$, the discrete first-passage time is a sum of $N - 1$ independent geometric variables and f is log-concave ($f(t)^2 \geq f(t - 1)f(t + 1)$), hence unimodal.

Proof. Here Q is tridiagonal on $\{0, \dots, N - 2\}$ with off-diagonal entries $q/2$, and only site $N - 2$ is adjacent to the target, so $\tilde{F}_q(z) = z \frac{q}{2} [(I - zQ)^{-1}]_{0, N-2}$. By Cramer's rule the corner entry of the inverse of a tridiagonal matrix is the product of its off-diagonal entries, up to sign, divided by the determinant: $[(I - zQ)^{-1}]_{0, N-2} = (zq/2)^{N-2} / \det(I - zQ)$. The eigenvalues of Q are the λ_m (Proposition 5.1), so $\det(I - zQ) = \prod_m (1 - \lambda_m z)$ and $\tilde{F}_q(z) = (zq/2)^{N-1} / \prod_m (1 - \lambda_m z)$. Since $\tilde{F}_q(1) = 1$, $(q/2)^{N-1} = \prod_m (1 - \lambda_m)$, which gives the first product; the second follows from (5). (Numerically, both products agree with direct solves to 6×10^{-13} .) A factor $\nu/(s + \nu)$ is the transform of an exponential density, and a factor $(1 - \lambda)z/(1 - \lambda z)$ with $0 \leq \lambda < 1$ is the generating function of a geometric variable on $\{1, 2, \dots\}$; the smallest λ_m is $1 - q - q \cos(2\pi/(2N - 1))$. Exponential densities and geometric distributions are log-concave, and a density (lattice distribution) is log-concave if and only if it is strongly unimodal, that is, if its convolution with every unimodal density (lattice distribution) is unimodal (Ibragimov, 1956; Keilson and Gerber, 1971). The class of strongly unimodal laws is closed under convolution. Induction over the factors therefore shows that g in case (i) and f in case (ii) are log-concave and unimodal. $\square$

The product form is the classical passage-time representation for birth-and-death processes (Karlin and McGregor, 1959; Keilson, 1971), and in discrete time that of Fill (2009). It does not cover $q = 0.8$ at large N , where some λ_m are negative, nor starts inside the chain, and at $q = 1$ the conclusion is false (Proposition 7.8). Section 7 of the main text extends it.

Numerical observation S2.3 (Single maximum for point targets). In every computation on the clean lattice with a point target and $q \leq 0.9$, the first-passage PMF or the unit-rate density has exactly one local maximum on the range examined:

- (a) 35 full-support runs of route A ($S < 10^{-12}$ at the end; $q = 0.5$ and 0.8 ; CC, C2M, M2C; $N \leq 101$, 35 and 15 for $d = 1, 2, 3$; 27 of them with $d \geq 2$) have one local maximum and no local minimum, as do 26 runs of the second time-stepping code (20 with $d \geq 2$);
- (b) all 151 point-target runs of route A with $q \leq 0.9$ (112 with $d \geq 2$) have one local maximum on $1 \leq t < t_{\text{cert}}$ and decrease monotonically from the mode to t_{cert} , beyond which f stays below its maximum by Proposition 3.1; all 150 point-target runs with $q \leq 0.9$ of the second code have one local maximum and decrease monotonically from the mode to the end of the run (as do its 15 runs of the exit geometry);

- (c) in all 156 two- and three-dimensional cases of route C, g' changes sign exactly once, from positive to negative, on a grid of four points per octave that starts below $0.22 t_c^*$ and ends at $15 qT$ (83 cases) or at 256 times a first estimate of the mode (73 cases), whichever is smaller. The grid always extends beyond $53 t_c^*$, but for the largest three-dimensional lattices that is only $0.03 qT$ (M2C, $N = 1281$) to $0.11 qT$ (CC, $N = 4096$);
- (d) for corner-to-corner transport, the pole sum of route P has $g' > 0$ on $[0.6 t_c^*, t_c^*)$ and $g' < 0$ on $(t_c^*, 15 qT]$ at all 65 sizes ($N \leq 16\,777\,216$ for $d = 2$, $N \leq 4096$ for $d = 3$).

Only (a) covers the whole support; the details are in Table S17. Most of the observation is a theorem: items (c) and (d) follow from Theorem 7.6(i), and items (a) and (b) from Theorems 7.7, 7.9 and 7.13(c) for every run except the two corner-to-corner runs at $q = 0.9$ with $d = 2$, $N = 101$ and $d = 3$, $N = 31$, which lie outside the proved ranges. Extended targets are different: for EXIT in one dimension at $N = 11$ a parity ripple with four local maxima at very short times ($t = 5, 7, 9, 11$) appears already at $q = 0.8$; the certificate applies there as well and returns the global one.

S2.5 Numerical inversion of a generating function

Inverting the generating function $\tilde{F}$ numerically is a tempting alternative to time stepping, and it has a pitfall of general interest. The rule of Abate and Whitt (1992a,b) is the trapezoidal rule on a circle of radius $r < 1$ with $2t$ nodes and a radius that depends on t :

$$f(t) \approx \frac{1}{2t r^t} \sum_{k=1}^{2t} (-1)^k \operatorname{Re} \tilde{F}(r e^{i\pi k/t}), \quad r = 10^{-\gamma/(2t)}. \quad (\text{S2.4})$$

Inserting $\tilde{F}(z) = \sum_m f(m) z^m$ and using that $\sum_{k=1}^{2t} (-1)^k e^{i\pi k m/t}$ equals $2t$ if $m \equiv t \pmod{2t}$ and 0 otherwise shows that the right-hand side equals $\sum_{j \geq 0} f((2j+1)t) r^{2jt}$ exactly. The error of (S2.4) is therefore the aliasing sum $\sum_{j \geq 1} f((2j+1)t) 10^{-\gamma j} \leq 10^{-\gamma}/(1 - 10^{-\gamma})$, and the prefactor $r^{-t} = 10^{\gamma/2}$ that multiplies round-off does not grow with t .

Two simplifications that look harmless are fatal at large t : a radius fixed independently of t , and a sum truncated after a fixed number $K < t - 1$ of nodes. With a fixed radius the bracket has to cancel down to $t r^t f(t)$, with $0.9^{1000} = 1.7 \times 10^{-46}$, from terms of order $\tilde{F}(0.9) = 1.4 \times 10^{-16}$. Figure S1(b) shows the outcome for $N = 35$, $q = 0.8$, for the rule with $r = 0.9$ and the sum truncated after $K = 54$ nodes (with the term $\frac{1}{2} \tilde{F}(r)$ added). These two values are one example of a fixed radius and a fixed truncation, not a tuned choice; the rule with the same radius and all $2t$ nodes, shown as well, separates the effect of the radius from that of the truncation. Its relative error is below 4×10^{-4} for $150 \leq t \leq 400$ (2×10^{-4} at $t = 400$). It passes 1% at $t = 417$ and exceeds 100% at every t from 455 on that we examined (all integers up to 700 and 40 larger values); it returns 4.3×10^{23} at $t = 1000$, where $f = 3.59 \times 10^{-5}$, and 3.6×10^{93} at the modal time $t = 2493$, where $f = 6.70 \times 10^{-5}$. With all $2t$ nodes but the fixed radius the breakdown comes slightly later (error above 100% from $t = 535$ on, 3.2×10^{85} at the modal time). The rule (S2.4) with $\gamma = 10$ has relative error 1.3×10^{-10} at the modal time and at most 6.4×10^{-10} for $1000 \leq t \leq 6000$; on the rising flank, where $f(3t) \gg f(t)$, it is larger (1.5×10^{-8} at $t = 309$), exactly as the aliasing sum predicts. At other sizes, all at $q = 0.8$, the truncated fixed-radius rule is off by 1.6% at the modal time for $d = 2$, $N = 9$ ($t^* = 136$), by 19% for the chain with $N = 21$ ($t^* = 174$), and returns 6.4×10^{19} for $d = 2$, $N = 20$ ($t^* = 765$, $f = 2.42 \times 10^{-4}$). For the sizes of this article, inversion of $\tilde{F}$ is unnecessary: time stepping is simpler and exact, and for larger lattices the Laplace transform is inverted instead (route C).

S3 Proofs of the mean first-passage time formulas

This section proves Theorems 4.1, 4.2 and 4.3, Corollary 4.4 and Proposition 4.7. The inputs are Proposition 2.4 and three elementary lemmas. The symbols $G_k, g_k, H_k, G_{kj}, g_{kj}, \Gamma, O_N$ and u_k are local to this section (G_k is an inner sum, not the Green's function G). Throughout, $N \geq 2$, $q \in (0, 1]$, sites are zero-based, and for $k = 0, \dots, N-1$ and $m \in \mathbb{Z}$

$$\vartheta_k = \frac{\pi k}{N}, \quad c_k = \cos \frac{\pi k}{2N}, \quad s_k = \sin \frac{\pi k}{2N}, \quad \vartheta_k(m) = \frac{\pi k(2m+1)}{2N},$$

so that $1 - \cos \vartheta_k = 2s_k^2$ and $1 + \cos \vartheta_k = 2c_k^2$. For $k \geq 1$, $\sigma_k = 2 - \cos \vartheta_k > 1$ and $\varphi_k = \operatorname{arcosh} \sigma_k > 0$; since $\cosh \varphi = 1 + 2 \sinh^2(\varphi/2)$,

$$\sinh \frac{\varphi_k}{2} = s_k, \quad \coth \frac{\varphi_k}{2} = \frac{\sqrt{1 + s_k^2}}{s_k}. \quad (\text{S3.1})$$

All identities displayed below were evaluated numerically in the form printed here. The formulas, computed to 40 digits, were compared with exact rational solutions of $(I - Q)h = \mathbf{1}$ for the walk built from its rules ($d = 1$: $N \leq 12$; $d = 2$: $N \leq 9$, including all 1848 ordered pairs with $N \leq 5$ at two values of q ; $d = 3$: $N \leq 4$) and with double-precision sparse solves ($d = 2$: $N = 16, 35, 36, 100, 101$; $d = 3$: $N = 5, \dots, 12$): 51 checks, none failed. The largest deviation is 8.5×10^{-36} in the multiprecision comparisons and 1.4×10^{-12} against the sparse solves; the double-precision matrix identities (eigenvectors, pseudo-inverses, resistances) hold to 3×10^{-14} .

S3.1 Spectrum of the walk and the cosine sum

Lemma S3.1 (Spectrum of the reflecting walk). *Let T be the $N \times N$ matrix with $\mathsf{T}(m, m \pm 1) = \frac{1}{2}$ for $0 \leq m, m \pm 1 \leq N-1$, $\mathsf{T}(0, 0) = \mathsf{T}(N-1, N-1) = \frac{1}{2}$ and zero otherwise. Then $\psi_k(m) = \sqrt{\alpha_k/N} \cos \vartheta_k(m)$, $k = 0, \dots, N-1$, is an orthonormal eigenbasis, $\mathsf{T}\psi_k = \cos(\pi k/N) \psi_k$, and $P = (1-q)I + \frac{q}{d} \sum_{i=1}^d \mathsf{T}^{(i)}$, where $\mathsf{T}^{(i)}$ acts as T on the i -th coordinate. Hence the products $\prod_i \psi_{k_i}(x_i)$ diagonalise P with $1 - \lambda_{\mathbf{k}} = \frac{2q}{d} \sum_i s_{k_i}^2$.*

Proof. Put $u_k(m) = \cos \vartheta_k(m)$ for $m \in \mathbb{Z}$. The addition formula gives $u_k(m-1) + u_k(m+1) = 2 \cos(\vartheta_k) u_k(m)$ for every m . Because $\vartheta_k(-1) = -\vartheta_k(0)$ and $\vartheta_k(N) = 2\pi k - \vartheta_k(N-1)$,

$$u_k(-1) = u_k(0), \quad u_k(N) = u_k(N-1). \quad (\text{S3.2})$$

Hence $(\mathsf{T}u_k)(0) = \frac{1}{2}[u_k(0) + u_k(1)] = \frac{1}{2}[u_k(-1) + u_k(1)] = \cos(\vartheta_k) u_k(0)$, likewise at $m = N-1$, and at the interior sites the eigenvalue equation is the addition formula itself. The N eigenvalues $\cos \vartheta_k$ are distinct and T is symmetric, so the u_k are mutually orthogonal. For $1 \leq k \leq N-1$, $\sum_{m=0}^{N-1} \cos^2 \vartheta_k(m) = \frac{N}{2} + \frac{1}{2} \operatorname{Re} \sum_{m=0}^{N-1} e^{i\vartheta_k(2m+1)} = \frac{N}{2}$, because the geometric sum vanishes ($e^{2iN\vartheta_k} = 1$, $e^{2i\vartheta_k} \neq 1$); for $k = 0$ the sum is N . This gives the normalisation $\sqrt{\alpha_k/N}$. Finally, by the rule (1) an attempt along coordinate i has probability q/d ; it changes x_i by ± 1 with probability $\frac{1}{2}$ each, unless the walker would leave $\{0, \dots, N-1\}$, in which case nothing moves. That is T acting on the i -th coordinate, so $P = (1-q)I + \frac{q}{d} \sum_i \mathsf{T}^{(i)}$. The products are therefore eigenvectors with $\lambda_{\mathbf{k}} = 1 - q + \frac{q}{d} \sum_i \cos \vartheta_{k_i}$, and $1 - \cos \vartheta_k = 2s_k^2$. $\square$

As in (7), write $\phi_{\mathbf{k}}(\mathbf{x}) = \prod_i \psi_{k_i}(x_i)$. Since $\sum_i s_{k_i}^2 > 0$ unless $\mathbf{k} = \mathbf{0}$, the eigenvalue 0 of $L = I - P$ is simple, and the pseudo-inverse inverts the other eigenvalues:

$$L_{\mathbf{xy}}^+ = \frac{d}{2q} \sum_{\mathbf{k} \neq \mathbf{0}} \frac{\phi_{\mathbf{k}}(\mathbf{x}) \phi_{\mathbf{k}}(\mathbf{y})}{\sum_i s_{k_i}^2}, \quad \phi_{\mathbf{k}}(\mathbf{x}) \phi_{\mathbf{k}}(\mathbf{y}) = \frac{1}{N^d} \prod_{i=1}^d \alpha_{k_i} \cos \vartheta_{k_i}(x_i) \cos \vartheta_{k_i}(y_i). \quad (\text{S3.3})$$

Proof of Theorem 4.1. P is symmetric, stochastic and irreducible, so Proposition 2.4 applies with $n = N^d$. Inserting (S3.3) into (12) gives (14); the factor $1/q$ is explicit, so $qT_{\mathbf{o} \rightarrow \mathbf{a}}$ does not depend on q . For opposite corners use

$$\cos \vartheta_k(0) = c_k, \quad \cos \vartheta_k(N-1) = \cos\left(\pi k - \frac{\pi k}{2N}\right) = (-1)^k c_k. \quad (\text{S3.4})$$

The numerator of (14) becomes $\prod_i c_{k_i}^2 [1 - (-1)^{k_1 + \dots + k_d}]$, which vanishes when $k_1 + \dots + k_d$ is even and equals $2 \prod_i c_{k_i}^2$ when it is odd. This is (15). For $d = 1$, (14) is the left-hand side of (S3.7) (because $\alpha_k = 2$ and $2s_k^2 = 1 - \cos \vartheta_k$), which equals $h_1(m, m')$ by Lemma S3.4 below; that lemma uses only Proposition 2.4 and Lemma S3.1. In particular $qT_N^{(1)} = h_1(0, N-1) = N(N-1)$. $\square$

S3.2 Two summation lemmas

Lemma S3.2 (Green's function of a ring). *Let $M \geq 1$ be an integer and $\sigma = \cosh \varphi > 1$ with $\varphi > 0$. For every integer $0 \leq m \leq M$,*

$$\sum_{l=0}^{M-1} \frac{\cos(2\pi l m/M)}{\sigma - \cos(2\pi l/M)} = \frac{M \cosh((\frac{M}{2} - m)\varphi)}{\sinh \varphi \sinh(M\varphi/2)}, \quad (\text{S3.5})$$

and consequently, for every integer $0 \leq m \leq 2N$,

$$\sum_{j=0}^{N-1} \frac{\alpha_j \cos(\pi m j/N)}{\sigma - \cos(\pi j/N)} = \frac{2N \cosh((N-m)\varphi)}{\sinh \varphi \sinh N\varphi} - \frac{(-1)^m}{\sigma + 1}. \quad (\text{S3.6})$$

Proof. Let A be the averaging operator on functions on the ring $\mathbb{Z}_M$, $(Af)(m) = \frac{1}{2}[f(m-1) + f(m+1)]$. Its eigenfunctions are $m \mapsto e^{2\pi i l m/M}$ with eigenvalues $\cos(2\pi l/M) \in [-1, 1]$, so $\sigma - A$ is invertible and

$$((\sigma - A)^{-1} \delta_0)(m) = \frac{1}{M} \sum_{l=0}^{M-1} \frac{e^{2\pi i l m/M}}{\sigma - \cos(2\pi l/M)},$$

where δ_0 is the indicator of the site 0. Define $\Gamma(m) = \cosh((\frac{M}{2} - m)\varphi) / (\sinh \varphi \sinh(M\varphi/2))$ for $0 \leq m \leq M$. Then $\Gamma(0) = \Gamma(M)$, so Γ is a function on $\mathbb{Z}_M$. For $1 \leq m \leq M-1$, $\cosh((\frac{M}{2} - m + 1)\varphi) + \cosh((\frac{M}{2} - m - 1)\varphi) = 2 \cosh \varphi \cosh((\frac{M}{2} - m)\varphi)$, that is, $((\sigma - A)\Gamma)(m) = 0$. At $m = 0$ the two neighbours are 1 and $M-1$, $\Gamma(M-1) = \Gamma(1)$, and

$$\sigma \Gamma(0) - \Gamma(1) = \frac{\cosh \varphi \cosh \frac{M\varphi}{2} - \cosh(\frac{M\varphi}{2} - \varphi)}{\sinh \varphi \sinh \frac{M\varphi}{2}} = 1.$$

Hence $(\sigma - A)\Gamma = \delta_0$, so $\Gamma = (\sigma - A)^{-1} \delta_0$, and taking real parts gives (S3.5). For (S3.6) take $M = 2N$: the summand $F(l) = \cos(\pi m l/N) / (\sigma - \cos(\pi l/N))$ satisfies $F(2N-l) = F(l)$, so $\sum_{l=0}^{2N-1} F(l) = F(0) + F(N) + 2 \sum_{l=1}^{N-1} F(l) = \sum_{j=0}^{N-1} \alpha_j F(j) + (-1)^m / (\sigma + 1)$. $\square$

Remark S3.3. Subtracting the term $j = 0$ from (S3.6) and dividing by two gives

$$\sum_{j=1}^{N-1} \frac{\cos(\pi m j/N)}{\sigma - \cos(\pi j/N)} = \frac{N \cosh((N-m)\varphi)}{\sinh \varphi \sinh N\varphi} - \frac{1}{2} \left[\frac{1}{\sigma - 1} + \frac{(-1)^m}{\sigma + 1} \right],$$

which is Eq. (E1) of Giuggioli (2020), written there with Chebyshev polynomials ($\cosh n\varphi$ and $\sinh n\varphi / \sinh \varphi$ are the Chebyshev polynomials of the first and second kind in σ). Essam and Wu (2009) use an equivalent summation lemma, proved by contour integration. So Lemma S3.2 is known; the short proof is included to keep the section self-contained. The range $0 \leq m \leq 2N$ is sharp: the left-hand side of (S3.6) is even and $2N$ -periodic in m , the right-hand side is not, and at $m = -1$ or $m = 2N+1$ the two sides differ by exactly $4N$, because $\cosh(N+1)\varphi - \cosh(N-1)\varphi = 2 \sinh N\varphi \sinh \varphi$. For other m , first reduce m to $[0, 2N]$.

Lemma S3.4 (One-dimensional hitting times). *For the chain T of Lemma S3.1 the mean hitting time from m to m' is $h_1(m, m')$ of (16), and*

$$\sum_{j=1}^{N-1} \alpha_j \frac{\cos^2 \vartheta_j(m') - \cos \vartheta_j(m) \cos \vartheta_j(m')}{1 - \cos(\pi j/N)} = h_1(m, m'). \quad (\text{S3.7})$$

In particular (take $m = 0$, $m' = N - 1$)

$$\sum_{\substack{1 \leq j \leq N-1 \\ j \text{ odd}}} \cot^2 \frac{\pi j}{2N} = \frac{N(N-1)}{2}. \quad (\text{S3.8})$$

Proof. Fix m' and put $h(m) = h_1(m, m')$. For $m \leq m'$ this is $h(m) = m'(m' + 1) - m(m + 1)$. It satisfies $h(m') = 0$; for $1 \leq m \leq m' - 1$, $\frac{1}{2}[h(m-1) + h(m+1)] = h(m) - 1$, because $(m-1)m + (m+1)(m+2) = 2m(m+1) + 2$; and at $m = 0$ (if $m' \geq 1$), $(\mathsf{T}h)(0) = \frac{1}{2}[h(0) + h(1)] = h(0) - 1$, because $h(1) = h(0) - 2$. These equations involve only sites $\leq m'$. For $m \geq m'$ the function is the same expression after the reflection $m \mapsto N - 1 - m$, $m' \mapsto N - 1 - m'$, which maps the chain to itself, so the equations at the sites $m > m'$ hold as well. Thus $h(m') = 0$ and $((I - \mathsf{T})h)(m) = 1$ for all $m \neq m'$, and by the uniqueness statement of Proposition 2.4 (the chain T is symmetric, stochastic and irreducible) h is the mean hitting time. Identity (S3.7) is (12) for T , with $n = N$ and $(I - \mathsf{T})_{mm'}^+ = \sum_{j \geq 1} \psi_j(m) \psi_j(m') / (1 - \cos \vartheta_j)$ from Lemma S3.1. For (S3.8) take $m = 0$, $m' = N - 1$ and use (S3.4): the summand vanishes for even j and equals $2c_j^2/s_j^2$ for odd j , and $h_1(0, N - 1) = N(N - 1)$. $\square$

S3.3 Proof of Theorem 4.2

Step 1 (splitting off the row $k = 0$). For $d = 2$, $s_k^2 + s_j^2 = \frac{1}{2}(2 - \cos \vartheta_k - \cos \vartheta_j)$, and (15) reads $\frac{1}{2} q T_N = \sum_{(k,j) \neq (0,0)} \alpha_k \alpha_j c_k^2 c_j^2 [1 - (-1)^{k+j}] / (2 - \cos \vartheta_k - \cos \vartheta_j)$. In the row $k = 0$ the denominator is $1 - \cos \vartheta_j$ and $\alpha_0 c_0^2 = 1$; in a row $k \geq 1$ it is $\sigma_k - \cos \vartheta_j$, $\alpha_k = 2$, and j runs over $0, \dots, N - 1$. Therefore

$$\frac{q T_N}{2} = \sum_{j=1}^{N-1} \frac{\alpha_j c_j^2 [1 - (-1)^j]}{1 - \cos \vartheta_j} + \sum_{k=1}^{N-1} 2c_k^2 G_k, \quad G_k := \sum_{j=0}^{N-1} \frac{\alpha_j c_j^2 [1 - (-1)^{k+j}]}{\sigma_k - \cos \vartheta_j}. \quad (\text{S3.9})$$

By (S3.7) with $m = 0$, $m' = N - 1$ the first sum equals $h_1(0, N - 1) = N(N - 1)$.

Step 2 (the inner sum). Write $c_j^2 = \frac{1}{2}[(1 + \sigma_k) - (\sigma_k - \cos \vartheta_j)]$ and $(-1)^j = \cos(N\vartheta_j)$. Then

$$G_k = \frac{1 + \sigma_k}{2} \sum_{j=0}^{N-1} \frac{\alpha_j [1 - (-1)^k \cos(N\vartheta_j)]}{\sigma_k - \cos \vartheta_j} - \frac{1}{2} \sum_{j=0}^{N-1} \alpha_j [1 - (-1)^{k+j}].$$

By (S3.6) with $m = 0$ and with $m = N$,

$$\begin{aligned} \sum_{j=0}^{N-1} \frac{\alpha_j}{\sigma_k - \cos \vartheta_j} &= \frac{2N \cosh N\varphi_k}{\sinh \varphi_k \sinh N\varphi_k} - \frac{1}{\sigma_k + 1}, \\ \sum_{j=0}^{N-1} \frac{\alpha_j \cos(N\vartheta_j)}{\sigma_k - \cos \vartheta_j} &= \frac{2N}{\sinh \varphi_k \sinh N\varphi_k} - \frac{(-1)^N}{\sigma_k + 1}, \end{aligned}$$

and $\sum_{j=0}^{N-1} \alpha_j = 2N - 1$, $\sum_{j=0}^{N-1} \alpha_j (-1)^j = -(-1)^N$ (from $\sum_{j=0}^{2N-1} (-1)^j = 0$ and the symmetry $j \mapsto 2N - j$). The terms containing $(-1)^{k+N}$ cancel, the constants combine to $-N$, and

$$G_k = \frac{N(1 + \sigma_k) [\cosh N\varphi_k - (-1)^k]}{\sinh \varphi_k \sinh N\varphi_k} - N. \quad (\text{S3.10})$$

Step 3 (form (a)). Since $\cosh(N-1)\varphi = \cosh N\varphi \cosh \varphi - \sinh N\varphi \sinh \varphi$ and $\cosh \varphi_k = \sigma_k$,

$$\mathcal{B}_k = (1 + \sigma_k) [\cosh N\varphi_k - (-1)^k] - \sinh \varphi_k \sinh N\varphi_k, \quad G_k = \frac{N \mathcal{B}_k}{\sinh \varphi_k \sinh N\varphi_k}, \quad (\text{S3.11})$$

the second relation being (S3.10) rewritten. Inserting it and $N(N-1)$ into (S3.9) gives (17). The rows $k \geq 1$ were summed over all j , including $j = 0$; no cancellation between different sectors of the double sum is needed.

Step 4 (form (b)). Because $(1 + \cosh \varphi)/\sinh \varphi = \coth(\varphi/2)$, $(\cosh N\varphi - 1)/\sinh N\varphi = \tanh(N\varphi/2)$ and $(\cosh N\varphi + 1)/\sinh N\varphi = \coth(N\varphi/2)$, (S3.10) reads

$$G_k = N \left[\coth \frac{\varphi_k}{2} g_k - 1 \right], \quad g_k = \coth \frac{N\varphi_k}{2} \quad (k \text{ odd}), \quad g_k = \tanh \frac{N\varphi_k}{2} \quad (k \text{ even}). \quad (\text{S3.12})$$

Moreover

$$\sum_{k=1}^{N-1} c_k^2 = \frac{N-1}{2} + \frac{1}{2} \sum_{k=1}^{N-1} \cos \vartheta_k = \frac{N-1}{2}, \quad (\text{S3.13})$$

since the terms k and $N-k$ of the cosine sum cancel. Hence, by (S3.9),

$$q T_N = 2N(N-1) + 4N \sum_{k=1}^{N-1} c_k^2 \coth \frac{\varphi_k}{2} g_k - 4N \cdot \frac{N-1}{2} = 4N \sum_{k=1}^{N-1} c_k^2 \coth \frac{\varphi_k}{2} g_k,$$

which is (18) by (S3.1); note $N\varphi_k/2 = N \operatorname{arsinh} s_k$.

Step 5 (form (c)). The summand of (15) is symmetric in (k, j) , and each pair with $k+j$ odd has exactly one odd index. With $1/(s_k^2 + s_j^2) = 2/(\sigma_k - \cos \vartheta_j)$,

$$q T_N = 16 \sum_{\substack{1 \leq k \leq N-1 \\ k \text{ odd}}} c_k^2 H_k, \quad H_k := \sum_{\substack{0 \leq j \leq N-1 \\ j \text{ even}}} \frac{\alpha_j c_j^2}{\sigma_k - \cos \vartheta_j}. \quad (\text{S3.14})$$

For $j = 2l$ we have $\vartheta_j = 2\pi l/N$ and $c_j^2 = \cos^2(\pi l/N)$. The function $l \mapsto \cos^2(\pi l/N)/(\sigma_k - \cos(2\pi l/N))$ on $\mathbb{Z}_N$ is invariant under $l \mapsto N-l$ and vanishes at $l = N/2$ when N is even; hence H_k equals its sum over all $l \in \mathbb{Z}_N$, for both parities of N . Writing $\cos^2(\pi l/N) = \frac{1}{2}[(1 + \sigma_k) - (\sigma_k - \cos(2\pi l/N))]$ and using (S3.5) with $M = N$, $m = 0$,

$$H_k = \frac{1 + \sigma_k}{2} \cdot \frac{N \cosh(N\varphi_k/2)}{\sinh \varphi_k \sinh(N\varphi_k/2)} - \frac{N}{2} = \frac{N}{2} \left[\coth \frac{\varphi_k}{2} \coth \frac{N\varphi_k}{2} - 1 \right]. \quad (\text{S3.15})$$

Next,

$$\sum_{\substack{1 \leq k \leq N-1 \\ k \text{ odd}}} c_k^2 = \frac{N}{4} \quad \text{for every } N \geq 2. \quad (\text{S3.16})$$

Indeed the left-hand side is half the number of odd $k < N$ plus $\frac{1}{2} O_N$, where O_N denotes the sum of $\cos \vartheta_k$ over the odd $k < N$. For even N the number is $N/2$ and $O_N = 0$, since $k \mapsto N-k$ maps odd indices to odd indices and changes the sign of the cosine. For odd N the number is $(N-1)/2$ and $O_N = \frac{1}{2}$. To see this, note that for odd N the alternating sum

$$\sum_{k=0}^{N-1} (-1)^k e^{i\vartheta_k} = \frac{1 - (-1)^N e^{i\pi}}{1 + e^{i\pi/N}}$$

vanishes, so the sum of $\cos \vartheta_k$ over the even k (which includes the term $k = 0$, equal to 1) equals O_N ; on the other hand $k \mapsto N-k$ maps the even indices $k \geq 2$ onto the odd ones and changes

the sign of the cosine, so the same sum equals $1 - O_N$. In both cases the left-hand side is $N/4$. From (S3.14)–(S3.16),

$$q T_N = 8N \sum_{k \text{ odd}} c_k^2 \left[\coth \frac{\varphi_k}{2} \coth \frac{N\varphi_k}{2} - 1 \right] = 8N S_{\text{odd}} - 2N^2.$$

Form (b) is $q T_N = 4N (S_{\text{odd}} + S_{\text{even}})$. Comparing the two gives $S_{\text{odd}} - S_{\text{even}} = N/2$ and then $q T_N = 2N^2 + 8N S_{\text{even}}$, which completes (20). $\square$

Remark S3.5. Form (a) contains $\cosh N\varphi_k$ and $\sinh N\varphi_k$, which are of size $e^{N\varphi_k}/2$. Since $\varphi_{N-1} \rightarrow \text{arcosh } 3 = 1.7627$, the exponent $N\varphi_{N-1}$ passes the logarithm of the largest double-precision number, 709.78, between $N = 402$ (708.62) and $N = 403$ (710.38). Accordingly a direct double-precision evaluation of (17) is finite for every $N \leq 402$ (at $N = 402$ it agrees with form (b) to 5.7×10^{-13}) and returns no finite value at any size tested from $N = 403$ on. Forms (b) and (c) contain only $\tanh$ and $\coth$ and have no such limit.

S3.4 Arbitrary start and target, resistance, and three dimensions

Proof of Theorem 4.3. For $d = 2$, (14) reads $q T_{\mathbf{o} \rightarrow \mathbf{a}} = \sum_{(k,j) \neq (0,0)} 2\alpha_k \alpha_j [\mathcal{C}_k(a_1, a_1) \mathcal{C}_j(a_2, a_2) - \mathcal{C}_k(o_1, a_1) \mathcal{C}_j(o_2, a_2)] / (2 - \cos \vartheta_k - \cos \vartheta_j)$. In the row $k = 0$, $\alpha_0 = 1$, $\mathcal{C}_0 \equiv 1$ and the denominator is $1 - \cos \vartheta_j$, so by (S3.7) this row contributes $2h_1(o_2, a_2)$. In the rows $k \geq 1$ the sums over j have the form $\sum_{j=0}^{N-1} \alpha_j \cos \vartheta_j(m) \cos \vartheta_j(m') / (\sigma_k - \cos \vartheta_j)$. Now

$$\cos \vartheta_j(m) \cos \vartheta_j(m') = \frac{1}{2} [\cos(|m - m'| \vartheta_j) + \cos((m + m' + 1) \vartheta_j)], \quad (\text{S3.17})$$

where $m_1 = |m - m'| \in [0, N - 1]$ and $m_2 = m + m' + 1 \in [1, 2N - 1]$ have opposite parity. Both lie in the range of (S3.6); applying it to the two terms, the contributions $-(-1)^{m_1}/(\sigma_k + 1)$ and $-(-1)^{m_2}/(\sigma_k + 1)$ cancel and

$$\sum_{j=0}^{N-1} \frac{\alpha_j \cos \vartheta_j(m) \cos \vartheta_j(m')}{\sigma_k - \cos \vartheta_j} = N \Psi_k(m, m'), \quad (\text{S3.18})$$

with Ψ_k as in (22). Since $\alpha_k = 2$ for $k \geq 1$, this gives (21). $\square$

For $\mathbf{o} = (0, 0)$ and $\mathbf{a} = (N - 1, N - 1)$, (S3.4) gives $\mathcal{C}_k(a_1, a_1) = c_k^2$ and $\mathcal{C}_k(o_1, a_1) = (-1)^k c_k^2$, the numerators of $\Psi_k(N - 1, N - 1)$ and $\Psi_k(0, N - 1)$ are $\cosh N\varphi_k + \cosh(N - 1)\varphi_k$ and $\cosh \varphi_k + 1$, and (21) reduces to (17). For numerical work one writes $\cosh((N - m)\varphi) / \sinh N\varphi = (e^{-m\varphi} + e^{-(2N-m)\varphi}) / (1 - e^{-2N\varphi})$, valid for $0 \leq m \leq 2N$, which removes the overflow.

Proof of Corollary 4.4. Let $\mathcal{L} = D - A$ be the Laplacian of the $N \times N$ grid graph, with D its degree matrix and A its adjacency matrix. By (1), $P(\mathbf{x}, \mathbf{y}) = q/4$ for neighbours and $1 - P(\mathbf{x}, \mathbf{x}) = \frac{q}{4} \deg(\mathbf{x})$, so

$$I - P = \frac{q}{4} \mathcal{L}, \quad L^+ = \frac{4}{q} \mathcal{L}^+. \quad (\text{S3.19})$$

Proposition 2.4, applied to both directions, gives $T_{\mathbf{o} \rightarrow \mathbf{a}} + T_{\mathbf{a} \rightarrow \mathbf{o}} = n (L_{\mathbf{aa}}^+ + L_{\mathbf{oo}}^+ - 2L_{\mathbf{oa}}^+) = \frac{4N^2}{q} (\mathbf{e}_{\mathbf{a}} - \mathbf{e}_{\mathbf{o}})^T \mathcal{L}^+ (\mathbf{e}_{\mathbf{a}} - \mathbf{e}_{\mathbf{o}})$, with $\mathbf{e}_{\mathbf{x}}$ the unit vector of site $\mathbf{x}$, and the last quadratic form is the effective resistance $R(\mathbf{o}, \mathbf{a})$ (Klein and Randić, 1993). This is (13) for $d = 2$, the commute-time identity of Chandra et al. (1989); see also Tetali (1991). The point reflection $\mathbf{x} \mapsto (N - 1 - x_1, N - 1 - x_2)$ is an automorphism of the walk that exchanges the two corners, so for them $T_{\mathbf{o} \rightarrow \mathbf{a}} = T_{\mathbf{a} \rightarrow \mathbf{o}}$ and $q T_N = 2N^2 R_N$. The two expressions for R_N follow from (20). For example $q T_4 = 416/7$ gives $R_4 = 13/7$, the value that Example 4 of Wu (2004) gives for unit resistors. $\square$

Proof of Proposition 4.7. Put $\sigma_{kj} = 3 - \cos \vartheta_k - \cos \vartheta_j = 1 + 2s_k^2 + 2s_j^2$, so that $s_k^2 + s_j^2 + s_l^2 = \frac{1}{2}(\sigma_{kj} - \cos \vartheta_l)$ and, for $(k, j) \neq (0, 0)$, $\sigma_{kj} = \cosh \varphi_{kj} > 1$. Summing (15) for $d = 3$ over the third index first,

$$qT_N^{(3)} = 3 \sum_{k,j=0}^{N-1} \alpha_k \alpha_j c_k^2 c_j^2 G_{kj}, \quad G_{kj} := \sum_l \frac{\alpha_l c_l^2 [1 - (-1)^{k+j+l}]}{\sigma_{kj} - \cos \vartheta_l}, \quad (\text{S3.20})$$

where l runs over $0, \dots, N-1$, except for $(k, j) = (0, 0)$, where $l \geq 1$. As in Step 1, $G_{00} = N(N-1)$ by (S3.7). For $(k, j) \neq (0, 0)$, Steps 2 and 4 apply verbatim with σ_k , φ_k and $(-1)^k$ replaced by σ_{kj} , φ_{kj} and $(-1)^{k+j}$: $G_{kj} = N[\coth(\varphi_{kj}/2) g_{kj} - 1]$. Finally $\sum_{k=0}^{N-1} \alpha_k c_k^2 = 1 + 2 \cdot \frac{N-1}{2} = N$ by (S3.13), so $\sum_{(k,j) \neq (0,0)} \alpha_k \alpha_j c_k^2 c_j^2 = N^2 - 1$ and

$$qT_N^{(3)} = 3 \left[N(N-1) + N \sum_{(k,j) \neq (0,0)} \alpha_k \alpha_j c_k^2 c_j^2 \coth \frac{\varphi_{kj}}{2} g_{kj} - N(N^2 - 1) \right],$$

which is (27), because $N(N-1) - N(N^2 - 1) = -N^2(N-1)$. $\square$

S4 Proof of the large- N expansion

This section proves Theorem 4.5. The route is that of Essam and Wu (2009), who expanded the corner-to-corner resistance $R_N = qT_N/(2N^2)$ for even N up to the term in N^{-4} , with coefficients given by rapidly convergent series and evaluated numerically: the Euler–Maclaurin formula for the algebraic part of the single sum and a Taylor expansion of its exponentially small part. We add remainder bounds at every order, a finite formula for every coefficient (Proposition S4.3) and the evaluation of C_2 , C_0 and C_{-2} in closed form through Eisenstein series; coefficients for a general aspect ratio are given, in terms of theta functions and Kronecker double series, by Izmailian and Huang (2010). The symbols w , r , c , U , v_k , y_k , ϵ , δ , K , κ_0 , $\mathfrak{p}$, $\mathfrak{D}$, $\mathcal{A}$, $\mathcal{E}$, $\mathcal{L}$, Λ , J , I_r , $e_{n,m}$, $G_k(Y)$, $g_{k,m}$, $\mathfrak{a}_m$ and $\mathfrak{b}_m$ are local to this section; in particular r , G_k and $g_{k,m}$ here are not the absorption vector, the inner sum and the factor g_k of Section 4 and Section S3.

S4.1 Splitting the single sum

Let $N \geq 2$, $\epsilon = \pi/(2N)$, $y_k = k\epsilon$, and

$$w(y) = \frac{\cos^2 y \sqrt{1 + \sin^2 y}}{\sin y}, \quad U(v) = \frac{v}{1+v}, \quad (\text{S4.1})$$

$$v_k = (-1)^k e^{-N\varphi_k} = (-1)^k \exp[-2N \operatorname{arsinh}(\sin y_k)].$$

In (18), $N \operatorname{arsinh} s_k = N\varphi_k/2$, so that $g_k = (1 - v_k)/(1 + v_k) = 1 - 2U(v_k)$ for both parities of k , and the single sum becomes

$$qT_N = 4N \mathcal{S}_N^{\text{alg}} - 8N \mathcal{S}_N^{\text{exp}}, \quad \mathcal{S}_N^{\text{alg}} = \sum_{k=1}^{N-1} w(y_k), \quad \mathcal{S}_N^{\text{exp}} = \sum_{k=1}^{N-1} w(y_k) U(v_k). \quad (\text{S4.2})$$

This is exact for every $N \geq 2$.

S4.2 The algebraic part

Lemma S4.1. *The function $r(y) = w(y) - 1/y$ extends to an odd real-analytic function on $(-\pi, \pi)$. With $I_r = \int_0^{\pi/2} r(y) dy$ and the Bernoulli numbers B_{2m} , for every integer $M \geq 1$, as $N \rightarrow \infty$,*

$$4N \mathcal{S}_N^{\text{alg}} = \frac{8N^2}{\pi} (\ln N + \gamma_E + I_r) + \sum_{m=1}^M \mathfrak{a}_m N^{2-2m} + O(N^{-2M}), \quad (\text{S4.3})$$

$$\mathfrak{a}_m = -\frac{4B_{2m}}{(2m)!} \left(\frac{\pi}{2}\right)^{2m-1} r^{(2m-1)}(0),$$

and

$$I_r = \frac{3}{2} \ln 2 - \ln \pi - \frac{1}{2}. \quad (\text{S4.4})$$

Proof. Regularity. For real y the function $c(y) = \cos^2 y \sqrt{1 + \sin^2 y}$ is real-analytic and even, with $c(0) = 1$, and $1/\sin y - 1/y$ is real-analytic and odd on $(-\pi, \pi)$. Hence $r(y) = c(y)(1/\sin y - 1/y) + (c(y) - 1)/y$ is real-analytic and odd on $(-\pi, \pi)$; in particular $r(0) = 0$ and every derivative of r is bounded on $[0, \pi/2]$. Moreover $w(\pi - y) = w(y)$, so $w(\pi/2) = 0$ and the odd derivatives of w vanish at $\pi/2$; therefore

$$r(\pi/2) = -\frac{2}{\pi}, \quad r^{(2j-1)}(\pi/2) = (2j-1)! \left(\frac{2}{\pi}\right)^{2j} \quad (j \geq 1). \quad (\text{S4.5})$$

Euler–Maclaurin. Apply the Euler–Maclaurin formula with its integral remainder (DLMF, 2026, Eq. 2.10.1, with $m = M + 1$) to $x \mapsto r(\epsilon x)$ on $[0, N]$:

$$\sum_{k=0}^N r(k\epsilon) = \frac{I_r}{\epsilon} + \frac{r(0) + r(\pi/2)}{2} + \sum_{j=1}^M \frac{B_{2j}}{(2j)!} \epsilon^{2j-1} [r^{(2j-1)}(\pi/2) - r^{(2j-1)}(0)] + \rho_M,$$

$$\rho_M = \epsilon^{2M+2} \int_0^N \frac{B_{2M+2} - \tilde{B}_{2M+2}(x)}{(2M+2)!} r^{(2M+2)}(\epsilon x) dx,$$

where $\tilde{B}_{2M+2}$ is the periodic Bernoulli function, which is bounded. Since $N\epsilon = \pi/2$,

$$|\rho_M| \leq \frac{\pi}{2} \frac{\sup_x |B_{2M+2} - \tilde{B}_{2M+2}(x)|}{(2M+2)!} \max_{[0, \pi/2]} |r^{(2M+2)}| \epsilon^{2M+1} = O(N^{-2M-1}). \quad (\text{S4.6})$$

Removing the terms $k = 0$ and $k = N$ and inserting (S4.5),

$$\sum_{k=1}^{N-1} r(k\epsilon) = \frac{2N}{\pi} I_r + \frac{1}{\pi} + \sum_{j=1}^M \left[\frac{2N}{\pi} \frac{B_{2j}}{2j N^{2j}} - \frac{B_{2j}}{(2j)!} \epsilon^{2j-1} r^{(2j-1)}(0) \right] + \rho_M.$$

The pole. The remaining part of $\mathcal{S}_N^{\text{alg}}$ is $\sum_{k=1}^{N-1} 1/(k\epsilon) = (2N/\pi) H_{N-1}$, and the harmonic number is $H_{N-1} = \psi(N) + \gamma_E$ with the digamma function ψ , whose expansion (DLMF, 2026, Eqs. 5.4.14 and 5.11.2) gives

$$H_{N-1} = \ln N + \gamma_E - \frac{1}{2N} - \sum_{j=1}^M \frac{B_{2j}}{2j N^{2j}} + O(N^{-2M-2}).$$

In the sum of the two contributions the upper-endpoint terms of the Euler–Maclaurin formula cancel the Bernoulli terms of H_{N-1} exactly, and $1/\pi$ cancels $(2N/\pi) \cdot (-1/2N)$. What remains is

$$\mathcal{S}_N^{\text{alg}} = \frac{2N}{\pi} (\ln N + \gamma_E + I_r) - \sum_{j=1}^M \frac{B_{2j}}{(2j)!} \left(\frac{\pi}{2N}\right)^{2j-1} r^{(2j-1)}(0) + O(N^{-2M-1}),$$

which is (S4.3) after multiplication by $4N$.

The integral. First, $\int_0^{\pi/2} (1/\sin y - 1/y) dy = [\ln \tan(y/2) - \ln y]_0^{\pi/2} = \ln(4/\pi)$. Second, with $t = \sin y$,

$$\int_0^{\pi/2} \frac{c(y) - 1}{\sin y} dy = \int_0^1 \frac{\sqrt{1-t^4} - 1}{t} dt + \int_0^1 \frac{1 - (1-t^2)^{-1/2}}{t} dt = -\frac{1}{4} [\psi(\frac{3}{2}) + \gamma_E] + \frac{1}{2} [\psi(\frac{1}{2}) + \gamma_E],$$

by the substitutions $u = t^4$ and $u = t^2$ and $\int_0^1 [1 - (1-u)^a] u^{-1} du = \psi(a+1) + \gamma_E$ for $a > -1$ (DLMF, 2026, Eq. 5.9.16). Since $\psi(\frac{1}{2}) + \gamma_E = -2 \ln 2$ and $\psi(\frac{3}{2}) + \gamma_E = 2 - 2 \ln 2$ (DLMF, 2026, Eqs. 5.4.13, 5.4.15), the second integral equals $-\frac{1}{2} - \frac{1}{2} \ln 2$, and (S4.4) follows. $\square$

The Taylor series $r(y) = -\frac{1}{3}y - \frac{47}{90}y^3 + \frac{941}{1890}y^5 - \dots$ gives $\mathbf{a}_1 = \pi/18$ and $\mathbf{a}_2 = -47\pi^3/21600$.

S4.3 The exponentially small part

Let $\kappa_0 = 2 \operatorname{arsinh} 1 = \ln(3 + 2\sqrt{2}) = 1.7627\dots$ and $\mathbf{p} = -e^{-\pi}$. For complex $|y| < \operatorname{arsinh} 1$ one has $|\sin y| \leq \sinh |y| < 1$, and $\sqrt{1+u^2}$ and $\operatorname{arsinh} u$ are analytic for $|u| < 1$; hence $y w(y)$ and $\operatorname{arsinh}(\sin y)/y$ are analytic and even in that disc, and

$$\mathcal{A}(y^2) = y w(y), \quad 1 + \mathcal{E}(y^2) = \frac{\operatorname{arsinh}(\sin y)}{y} \quad (\text{S4.7})$$

define functions $\mathcal{A}(Y) = 1 - \frac{1}{3}Y - \frac{47}{90}Y^2 + \dots$ and $\mathcal{E}(Y) = -\frac{1}{3}Y + \frac{1}{6}Y^2 - \dots$ that are analytic for $|Y| < (\operatorname{arsinh} 1)^2$. We write $\mathbf{D}$ for the Euler operator $z d/dz$, whatever the name of the variable, and, for integers $0 \leq n \leq m$,

$$e_{n,m} = \text{coefficient of } Y^m \text{ in } \mathcal{A}(Y) \frac{(-\mathcal{E}(Y))^n}{n!} \quad (\text{S4.8})$$

(for $n > m$ this coefficient vanishes because $\mathcal{E}(0) = 0$).

Lemma S4.2. *For every integer $M \geq 0$, as $N \rightarrow \infty$,*

$$\begin{aligned} -8N \mathcal{S}_N^{\exp} &= \sum_{m=0}^M \mathbf{b}_m N^{2-2m} + O(N^{-2M}), \\ \mathbf{b}_m &= -\frac{16}{4^m} \sum_{n=0}^m e_{n,m} \pi^{2m-1+n} \sum_{k \geq 1} k^{2m-1+n} (\mathbf{D}^n U)(\mathbf{p}^k). \end{aligned} \quad (\text{S4.9})$$

Proof. (i) Elementary bounds. On $(0, \pi/2]$, $\sin y \geq 2y/\pi$, and arsinh is concave on $[0, 1]$ with $\operatorname{arsinh} 0 = 0$, so $\operatorname{arsinh}(\sin y) \geq (2y/\pi) \operatorname{arsinh} 1$. Hence, for $1 \leq k \leq N-1$,

$$|v_k| \leq e^{-\kappa_0 k}, \quad 0 < w(y_k) \leq \frac{\sqrt{2}}{\sin y_k} \leq \frac{\sqrt{2} N}{k}, \quad |U(v)| \leq \frac{|v|}{1-|v|} \quad (|v| < 1). \quad (\text{S4.10})$$

(ii) Cut-off and tail. Fix $\delta \in (0, \operatorname{arsinh} 1)$ such that $|\mathcal{E}(Y)| \leq \frac{1}{2}$ for $|Y| \leq \delta^2$; this is possible because $\mathcal{E}$ is continuous and $\mathcal{E}(0) = 0$ (for instance $\delta = \frac{1}{2}$, where $|\mathcal{E}| \leq 0.097$). Let $A_\delta = \max_{|Y| \leq \delta^2} |\mathcal{A}(Y)|$ and $K = \lfloor \sqrt{2} \delta N / \pi \rfloor$, so that $y_k^2 \leq \delta^2/2$ for $k \leq K$. By (S4.10) and $(1 - e^{-\kappa_0})^{-2} < 3/\sqrt{2}$,

$$\sum_{k=K+1}^{N-1} w(y_k) |U(v_k)| \leq \frac{\sqrt{2} N}{1 - e^{-\kappa_0}} \sum_{k > K} e^{-\kappa_0 k} \leq 3N e^{-\kappa_0(K+1)} \leq 3N e^{-\sqrt{2} \kappa_0 \delta N / \pi}, \quad (\text{S4.11})$$

which is exponentially small in N .

(iii) Uniform Taylor remainder. For $k \geq 1$ put $G_k(Y) = \mathcal{A}(Y) U(\mathbf{p}^k e^{-\pi k \mathcal{E}(Y)})$. By (S4.7), $2N \operatorname{arsinh}(\sin y_k) = \pi k [1 + \mathcal{E}(y_k^2)]$, so

$$v_k = \mathbf{p}^k e^{-\pi k \mathcal{E}(y_k^2)}, \quad w(y_k) U(v_k) = \frac{G_k(y_k^2)}{y_k}. \quad (\text{S4.12})$$

For $|Y| \leq \delta^2$ we have $|\mathbf{p}^k e^{-\pi k \mathcal{E}(Y)}| \leq e^{-\pi k/2} \leq e^{-\pi/2} < \frac{1}{4}$. Therefore G_k is analytic in $|Y| < \delta^2$, continuous on the closed disc, and bounded there by $\frac{4}{3} A_\delta e^{-\pi k/2}$. Cauchy's inequalities give, for its Taylor coefficients $g_{k,m}$ and its Taylor remainder,

$$|g_{k,m}| \leq \frac{4A_\delta}{3} \frac{e^{-\pi k/2}}{\delta^{2m}}, \quad \left| G_k(Y) - \sum_{m=0}^M g_{k,m} Y^m \right| \leq \frac{8A_\delta}{3} e^{-\pi k/2} \left(\frac{|Y|}{\delta^2} \right)^{M+1} \quad (|Y| \leq \frac{1}{2} \delta^2), \quad (\text{S4.13})$$

uniformly in $k \geq 1$.

(iv) *Summation.* For $k \leq K$ take $Y = y_k^2 = (k\epsilon)^2$ in (S4.13) and divide by y_k ; by (S4.12),

$$\sum_{k=1}^K \left| w(y_k)U(v_k) - \sum_{m=0}^M g_{k,m} (k\epsilon)^{2m-1} \right| \leq \frac{8A_\delta}{3\delta^{2M+2}} \epsilon^{2M+1} \sum_{k \geq 1} k^{2M+1} e^{-\pi k/2} = O(N^{-2M-1}).$$

By the first bound in (S4.13) and $(k\epsilon)^{2m-1} \leq N(\pi k)^{2m}$, the sums $\sum_{k > K} |g_{k,m}| (k\epsilon)^{2m-1}$, $m \leq M$, are exponentially small in N , because $K+1 > \sqrt{2}\delta N/\pi$. With (S4.11) this yields

$$\mathcal{S}_N^{\text{exp}} = \sum_{m=0}^M \epsilon^{2m-1} \sum_{k \geq 1} k^{2m-1} g_{k,m} + O(N^{-2M-1}),$$

and $-8N\epsilon^{2m-1} = -16\pi^{2m-1}4^{-m}N^{2-2m}$.

(v) *The coefficients $g_{k,m}$.* For $|t| < \pi k$ the function $t \mapsto U(\mathbf{p}^k e^{-t})$ is analytic, and d/dt acts on it as $-D$; hence $U(\mathbf{p}^k e^{-t}) = \sum_{n \geq 0} \frac{(-t)^n}{n!} (D^n U)(\mathbf{p}^k)$. Substituting $t = \pi k \mathcal{E}(Y)$, which vanishes at $Y = 0$, and multiplying by $\mathcal{A}(Y)$ gives $g_{k,m} = \sum_{n=0}^m e_{n,m} (\pi k)^n (D^n U)(\mathbf{p}^k)$ with $e_{n,m}$ as in (S4.8). Inserting this in (iv) gives (S4.9); the sums over k converge absolutely because $(D^n U)(z) = O(|z|)$. $\square$

Proposition S4.3 (Form of the expansion). *For every integer $M \geq 1$ the expansion (24) holds, with*

$$C_2 = \frac{8}{\pi}(\gamma_E + I_r) + \mathbf{b}_0, \quad C_{2-2m} = \mathbf{a}_m + \mathbf{b}_m \quad (m \geq 1), \quad (\text{S4.14})$$

where $\mathbf{a}_m$, I_r and $\mathbf{b}_m$ are given by (S4.3), (S4.4) and (S4.9). In particular only even powers of N occur besides $N^2 \ln N$.

Proof. Add (S4.3) and (S4.9) in (S4.2). $\square$

S4.4 Lambert series and Eisenstein series

For $|z| < 1$ let $\Lambda_j(z) = \sum_{k \geq 1} k^j U(z^k)$. Since $D[U(z^k)] = k(DU)(z^k)$ and the series converges locally uniformly, the inner sums in (S4.9) are

$$\sum_{k \geq 1} k^{2m-1+n} (D^n U)(\mathbf{p}^k) = (D^n \Lambda_{2m-1})(\mathbf{p}). \quad (\text{S4.15})$$

The identity $U(z) = z/(1-z) - 2z^2/(1-z^2)$ gives, for $m \geq 1$,

$$\Lambda_{2m-1}(z) = \mathcal{L}_{2m-1}(z) - 2\mathcal{L}_{2m-1}(z^2), \quad \mathcal{L}_{2m-1}(z) = \sum_{k \geq 1} \frac{k^{2m-1} z^k}{1-z^k} = \frac{B_{2m}}{4m} (1 - E_{2m}), \quad (\text{S4.16})$$

where E_{2m} is the normalised Eisenstein series of weight $2m$, regarded as a function of $z = e^{2\pi i \tau}$, $\text{Im } \tau > 0$ (Zagier, 2008, Proposition 5 and Eq. (17)); in this variable $D = (2\pi i)^{-1} d/d\tau$, and $D^n[\mathcal{L}(z^2)] = 2^n (D^n \mathcal{L})(z^2)$. Our point is $\mathbf{p} = e^{2\pi i \tau_0}$ with $\tau_0 = (1+i)/2$, and $\mathbf{p}^2 = e^{-2\pi}$ corresponds to $\tau = i$. Hence

$$(D^n \Lambda_{2m-1})(\mathbf{p}) = (D^n \mathcal{L}_{2m-1})|_{\tau_0} - 2^{n+1} (D^n \mathcal{L}_{2m-1})|_{\tau=i}. \quad (\text{S4.17})$$

We use the following classical facts, all in Zagier (2008).

(F1) $E_2(-1/\tau) = \tau^2 E_2(\tau) + 6\tau/(\pi i)$, $E_4(-1/\tau) = \tau^4 E_4(\tau)$, $E_6(-1/\tau) = \tau^6 E_6(\tau)$, and all three have period 1 (his Proposition 6 and Section 2.1).

(F2) $E_2(i) = 3/\pi$ and $E_6(i) = 0$ (put $\tau = i$ in (F1)), and $E_4(i) = 3\Gamma(1/4)^8/(2\pi)^6 = 3\Xi/\pi^2$ (his Section 6.3).

(F3) Ramanujan's system $\mathcal{D}E_2 = (E_2^2 - E_4)/12$, $\mathcal{D}E_4 = (E_2E_4 - E_6)/3$, $\mathcal{D}E_6 = (E_2E_6 - E_4^2)/2$ (his Proposition 15); every E_{2m} with $m \geq 2$ is a polynomial in E_4 and E_6 (his Proposition 4), with rational coefficients because all Fourier coefficients involved are rational; for instance $E_8 = E_4^2$ and $E_{10} = E_4E_6$.

(F4) The discriminant function, written here as the 24th power of the Dedekind function η , $\eta(\tau)^{24} = e^{2\pi i \tau} \prod_{n \geq 1} (1 - e^{2\pi i n \tau})^{24} = (E_4^3 - E_6^2)/1728$, has period 1 and satisfies $\eta(-1/\tau)^{24} = \tau^{12} \eta(\tau)^{24}$ (his Eqs. (22), (23), (34) and Proposition 7).

Since $-1/\tau_0 = -1 + i$ and $\tau_0^2 = i/2$, (F1) and (F2) give

$$E_2(\tau_0) = \frac{6}{\pi}, \quad E_4(\tau_0) = -4E_4(i), \quad E_6(\tau_0) = 0, \quad \eta(\tau_0)^{24} = -64 \eta(i)^{24}. \quad (\text{S4.18})$$

By (S4.16)–(S4.18) and (F3), every $\mathbf{b}_m$ with $m \geq 1$ is a polynomial in π , $1/\pi$ and $E_4(i)$ with rational coefficients, obtained by finitely many differentiations.

S4.5 The constant C_2

For $m = 0$, $e_{0,0} = 1$ and $\mathbf{b}_0 = (8/\pi)J$ with $J = -2 \sum_{k \geq 1} U(\mathbf{p}^k)/k$. Expanding $U(z) = \sum_{j \geq 1} (-1)^{j-1} z^j$ and summing over k first (the double series converges absolutely),

$$J = 2 \sum_{j \geq 1} (-1)^{j-1} \ln(1 - \mathbf{p}^j) = 2 \ln \prod_{j \geq 1} (1 - \mathbf{p}^j) - 4 \ln \prod_{j \geq 1} (1 - \mathbf{p}^{2j}),$$

where all factors are positive. By (F4) and (S4.18), $\prod_j (1 - \mathbf{p}^j)^{24} = \eta(\tau_0)^{24}/\mathbf{p} = 64 e^\pi \eta(i)^{24}$ and $\prod_j (1 - \mathbf{p}^{2j})^{24} = e^{2\pi} \eta(i)^{24}$, while $\eta(i)^{24} = E_4(i)^3/1728 = \Xi^3/(64\pi^6) = \varpi^{12}/(64\pi^{12})$. Therefore

$$J = \frac{1}{2} \ln 2 - \frac{\pi}{4} - \frac{1}{12} \ln \eta(i)^{24} = \ln \frac{2\pi}{\varpi} - \frac{\pi}{4}, \quad (\text{S4.19})$$

and (S4.14), (S4.4) give $C_2 = \frac{8}{\pi} [\gamma_E + \frac{5}{2} \ln 2 - \ln \varpi - \frac{1}{2} - \frac{\pi}{4}]$, the value stated in (25). (Equivalently, by Jacobi's triple product (DLMF, 2026, Eq. 20.5.9), $J = \ln \theta_3(e^{-\pi}) - 3 \ln \prod_{n \geq 1} (1 - e^{-2\pi n})$ with the classical value $\theta_3(e^{-\pi}) = \pi^{1/4}/\Gamma(3/4)$.)

S4.6 The coefficients C_0 and C_{-2}

Order $m = 1$. Here $e_{0,1} = -\frac{1}{3}$ and $e_{1,1} = \frac{1}{3}$, so $\mathbf{b}_1 = \frac{4}{3}\pi \Lambda_1(\mathbf{p}) - \frac{4}{3}\pi^2(\mathcal{D}\Lambda_1)(\mathbf{p})$. With $\mathcal{L}_1 = (1 - E_2)/24$ and $\mathcal{D}\mathcal{L}_1 = -(E_2^2 - E_4)/288$, (S4.17) and (S4.18) give

$$\Lambda_1(\mathbf{p}) = \sum_{k \geq 1} \frac{k \mathbf{p}^k}{1 + \mathbf{p}^k} = -\frac{1}{24}, \quad (\mathcal{D}\Lambda_1)(\mathbf{p}) = \sum_{k \geq 1} \frac{k^2 \mathbf{p}^k}{(1 + \mathbf{p}^k)^2} = -\frac{E_4(i)}{36}. \quad (\text{S4.20})$$

Hence $\mathbf{b}_1 = -\pi/18 + \pi^2 E_4(i)/27$, and with $\mathbf{a}_1 = \pi/18$,

$$C_0 = \frac{\pi^2 E_4(i)}{27} = \frac{\Xi}{9}.$$

Order $m = 2$. Here $e_{0,2} = -\frac{47}{90}$, $e_{1,2} = -\frac{5}{18}$, $e_{2,2} = \frac{1}{18}$, so $\mathbf{b}_2 = \frac{47}{90}\pi^3 \Lambda_3(\mathbf{p}) + \frac{5}{18}\pi^4(\mathcal{D}\Lambda_3)(\mathbf{p}) - \frac{1}{18}\pi^5(\mathcal{D}^2\Lambda_3)(\mathbf{p})$. With $\mathcal{L}_3 = (E_4 - 1)/240$, $\mathcal{D}\mathcal{L}_3 = (E_2E_4 - E_6)/720$ and, where $E_6 = 0$, $\mathcal{D}^2\mathcal{L}_3 = (E_2^2E_4 + E_4^2)/1728$, we find

$$\Lambda_3(\mathbf{p}) = \frac{1 - 6E_4(i)}{240}, \quad (\mathcal{D}\Lambda_3)(\mathbf{p}) = -\frac{E_4(i)}{20\pi}, \quad (\mathcal{D}^2\Lambda_3)(\mathbf{p}) = -\frac{E_4(i)}{8\pi^2} + \frac{E_4(i)^2}{216}. \quad (\text{S4.21})$$

With $\mathbf{a}_2 = -47\pi^3/21600$ the terms free of $E_4(i)$ cancel again, and

$$C_{-2} = -\frac{\pi^3 E_4(i)}{50} - \frac{\pi^5 E_4(i)^2}{3888} = -\frac{\pi \Xi (25 \Xi + 648)}{10800}.$$

This completes the proof of Theorem 4.5: the expansion and the absence of other powers are Proposition S4.3, and the three closed forms have just been derived. $\square$

S4.7 Higher coefficients and numerical confirmation

Computer algebra. The recipe (S4.14), (S4.15)–(S4.18) is finite at every order. Carried out in exact rational arithmetic by a computer-algebra script for $m = 3, 4, 5$, it returns

$$\begin{aligned} C_{-4} &= \frac{\pi^2 \Xi^2 (1225 \Xi + 12879)}{1905120}, & C_{-6} &= -\frac{\pi^3 \Xi^2 (79625 \Xi^2 + 1215000 \Xi + 960498)}{435456000}, \\ C_{-8} &= \frac{\pi^4 \Xi^3 (7703465 \Xi^2 + 92129400 \Xi + 118943883)}{63228211200}, \end{aligned} \quad (\text{S4.22})$$

that is $2.227516\dots$, $-14.055169\dots$ and $124.706087\dots$ (Table 1). These three closed forms rest on Proposition S4.3 and on the correctness of the symbolic computation; they were not derived by hand.

Numerical confirmation. Three checks that do not use the modular evaluation support every step. (a) The coefficient formula (S4.14), with the Lambert series (S4.15) summed numerically, agrees with the closed forms of $C_2, \dots, C_{-8}$ to better than 10^{-48} in two separately written implementations. (b) A linear solve for the coefficients of even powers of N from 70-digit values of qT_N at $N = 2^j$, which uses no closed form, returns $C_2, C_0, \dots, C_{-8}$ to 3×10^{-67} , 2×10^{-57} , 2×10^{-48} , 7×10^{-40} , 4×10^{-32} and 8×10^{-25} ; when $N \ln N$ and odd powers are admitted, their fitted coefficients are below 10^{-28} for $N \ln N$ and N . (c) For $N = 16, \dots, 4096$ and $M = 1, \dots, 4$ the remainders of (S4.3), of (S4.9) and of their sum, multiplied by N^{2M} , converge to $|\mathbf{a}_{M+1}|$, $|\mathbf{b}_{M+1}|$ and $|C_{-2M}|$ respectively (relative deviation at $N = 4096$ at most 2.2×10^{-6}), so the two lemmas are confirmed separately. Truncation errors at fixed N are in Table S4.

Remark S4.4 (What is not proved). Proposition S4.3 is an asymptotic statement for each fixed M ; the constant in $O(N^{-2M})$ depends on M , and nothing is proved about the convergence of the series in N^{-2} . The computed coefficients grow in modulus (0.53, 1.07, 2.23, 14.1, 124.7, and N^{10} times the remainder after $C_{-8}N^{-8}$ settles at 1685.5 in modulus for $N \geq 1024$), which suggests a divergent asymptotic series; we do not claim this.

S5 Mean first-passage times: verification and three dimensions

S5.1 Exact values and verification of the single sums

Table S3: Exact corner-to-corner mean first-passage times for $d = 2$: qT_N from exact rational solves of $(I - Q)h = \mathbf{1}$, and T_N in steps at $q = 0.8$. Decimals are truncated.

N	qT_N (exact)	decimal value	T_N at $q = 0.8$
2	8	8	10
3	27	27	33.75
4	416/7	59.428571...	74.285714...
5	1175/11	106.818181...	133.522727...
6	9368/55	170.327272...	212.909090...
7	312865/1247	250.894145...	313.617682...
8	125807488/360161	349.309025...	436.636282...
9	35727723/76627	466.255014...	582.818768...
10	85570839000/142065451	602.333912...	752.917391...
11	577168419861/761351911	758.083629...	947.604536...
12	52566792400224/56281934513	933.990504...	1167.488130...

Table S3 lists exact values. Exact rational solves of $(I - Q)h = \mathbf{1}$ for $N = 2, \dots, 40$ (84 cases: $q = 1$ for every N , and $q = 4/5, 3/10, 1/7$ for $N \leq 16$) agree with the double sum and with (a), (b) and both sums of (c) of Theorem 4.2, evaluated with 60 digits, to a relative deviation of at

most 6.6×10^{-60} ; for $N \leq 16$, qT_N is the same rational number at all four values of q . A second, separately written solver gives the same fractions and the same agreement for $N = 2, \dots, 32$ and $N = 35, 38, 41, 44$ (3.3×10^{-50} with 50 digits for $N \leq 22$, 8.4×10^{-60} with 60 digits beyond). As an external anchor, Example 4 of Wu (2004), evaluated for unit resistors, gives $R = 13/7$ between opposite corners of the 4×4 grid, and $2N^2R = 416/7$ (Corollary 4.4). The zero-based formulas exactly as printed in the main text were checked once more against exact solves for $N = 2, \dots, 12$ (4.5×10^{-51}).

Floating point. Evaluated in double precision, form (a) returns NaN for every $N \geq 403$: $\cosh N\varphi_k$ overflows once $N\varphi_k$ exceeds 709.78, and $\varphi_{N-1} \rightarrow \operatorname{arcosh} 3 = 1.7627$. At $N = 402$ it is still correct to 5.7×10^{-13} . Form (b) contains only $\tanh$ and $\coth$ and has no such limit: in double precision it agrees with 40-digit values to 1.4×10^{-16} or better at the five sizes tested between $N = 10$ and $N = 6000$ (Fig. S3). For small k/N , $\varphi_k = \pi k/N + O((k/N)^3)$.

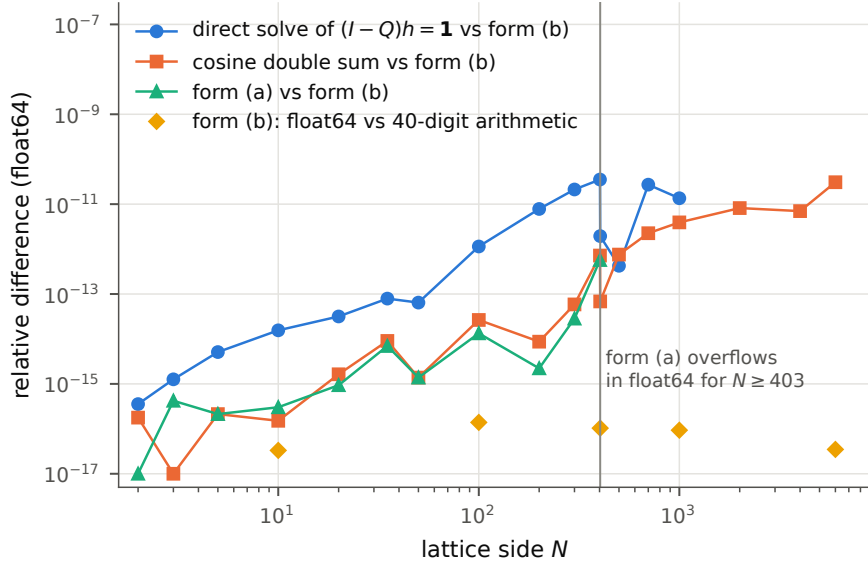


Figure S3: Double-precision evaluation of qT_N ($d = 2$, corner to corner) by different routes, as relative difference from form (b) of Theorem 4.2: sparse solve of $(I - Q)h = \mathbf{1}$ at $q = 0.8$ (direct factorisation up to $N = 300$, conjugate gradients from $N = 402$, up to $N = 1000$), cosine double sum (15) (up to $N = 6000$), and form (a), which is not finite for $N \geq 403$ (vertical line). Diamonds: form (b) in double precision against 40-digit arithmetic. Differences that are exactly zero are drawn at 10^{-17} . The growth of the first two curves is rounding in the linear solve and in the $O(N^2)$ -term sum, not in form (b).

Arbitrary pairs. A second implementation confirms (21) to 1.7×10^{-40} on 1956 cases (the 1848 ordered pairs with $N \leq 5$ at two values of q and 108 random pairs with $6 \leq N \leq 17$). In double precision, with the hyperbolic ratios written through decaying exponentials, the formula reproduces a sparse direct solve for every start site at $N = 35$, $q = 0.8$, to 7.9×10^{-14} for the two targets of Fig. S4 and to 8.6×10^{-14} in the second implementation (three targets). Numerically, $qT_N = 2N^2R_N$ holds to 1.2×10^{-14} for $N = 2, \dots, 20$ when R_N is computed from the pseudo-inverse of the grid Laplacian, and Eq. (57) of Tan and Tan (2020) agrees with the pseudo-inverse of the grid Laplacian to 4.3×10^{-15} for all 4536 ordered pairs with $N \leq 7$. The one-directional time of Theorem 4.3 needs, besides the resistance, the potential of a single node; Tan and Tan (2020) give a single sum for the node potentials as well, which we have not compared.

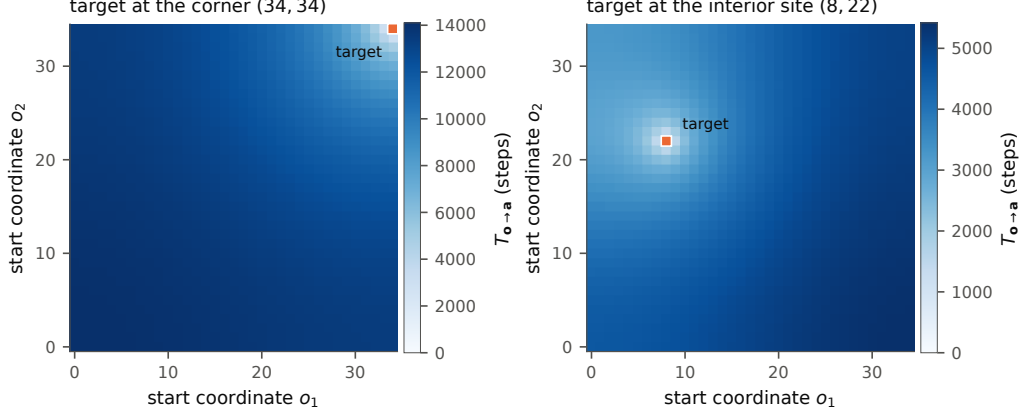


Figure S4: Mean first-passage time $T_{\mathbf{o} \rightarrow \mathbf{a}}$ (steps) from every start site $\mathbf{o} = (o_1, o_2)$ to a fixed target $\mathbf{a}$ (square marker), computed with the single sum (21); $d = 2$, $N = 35$, $q = 0.8$, discrete time, zero-based coordinates. Left: corner target (34, 34); the value at (0, 0) is $T_N = 14\,100.81$. Right: interior target (8, 22).

Table S4: Relative error of the expansion (24) truncated after the stated number of terms (two terms: $(8/\pi)N^2 \ln N + C_2 N^2$; five terms: up to $C_{-4} N^{-4}$), against the exact qT_N .

N	1 term	2 terms	3 terms	4 terms	5 terms
10	2.7×10^{-2}	8.7×10^{-4}	1.7×10^{-5}	3.5×10^{-7}	2.2×10^{-8}
35	1.7×10^{-2}	4.7×10^{-5}	7.7×10^{-8}	1.3×10^{-10}	6.7×10^{-13}
100	1.3×10^{-2}	4.5×10^{-6}	9.0×10^{-10}	1.9×10^{-13}	1.2×10^{-16}
1000	8.7×10^{-3}	3.0×10^{-8}	6.0×10^{-14}	1.3×10^{-19}	7.9×10^{-25}
4001	7.3×10^{-3}	1.6×10^{-9}	2.0×10^{-16}	2.6×10^{-23}	1.0×10^{-29}

S5.2 The expansion in numbers

Table S4 and Fig. S5 give the truncation errors. When terms in $N \ln N$ and in odd powers of N are admitted in the blind solve of Table 1, column (ii), the coefficients found for $N \ln N$ and N are below 10^{-28} , and a free leading coefficient returns $8/\pi$ to 3×10^{-58} ; for the odd sizes $N = 3^j$, $3 \leq j \leq 10$, the two forbidden coefficients are below 2×10^{-15} . The coefficients grow quickly (the same solve gives $C_{-10} \approx -1685.5$), which suggests that the series is asymptotic rather than convergent; we have not proved this. By Theorem 4.5, $r_N = C_0 + C_{-2}N^{-2} + O(N^{-4})$ with $C_{-2} < 0$, so the remainder of Theorem 4.6 approaches C_0 from below, as the certified finite range shows; the limit $r_N \rightarrow C_0$ is also proved, by a different route that evaluates C_0 as a series, in [R2, Thm 9.8].

S5.3 Three dimensions

In double precision (27) agrees with sparse solves of $(I - Q)h = \mathbf{1}$ to 5.4×10^{-13} or better at the sizes tested ($N = 3, 5, 8, 12$). Richardson extrapolation of the exact values gives 4.3204601701 and -3.887257 for the first two constants of Theorem 4.8, and the remainder $qT_N^{(3)} - K_3 N^3 - K'_3 N^2$ equals 1.44, 1.457 and 1.458 at $N = 10, 35$ and 100 , in agreement with the limit K''_3 ; beyond $N \approx 400$ a ten-digit value of K_3 does not resolve the remainder. In Theorem 4.8, $\mathcal{G}_\infty$ is $2d = 6$ times the Green's function of the unit-conductance lattice Laplacian, the normalisation of G elsewhere in this article. The identity for K_3 is elementary (Section S8); the enclosure, the bounds on $R(N)$ and the values of K'_3 and K''_3 rest on ball arithmetic. In Observation 4.9, $\theta_3(x) = \sum_{n \in \mathbb{Z}} x^{n^2}$ and $\theta_4(x) = \sum_{n \in \mathbb{Z}} (-1)^n x^{n^2}$ are theta functions of the nome x , and the observation gives $K'_3 = -\frac{6}{\pi} \xi_H + \frac{6}{\pi^2} \varkappa$ to the precision stated. The constant ξ_H is not ours: it is

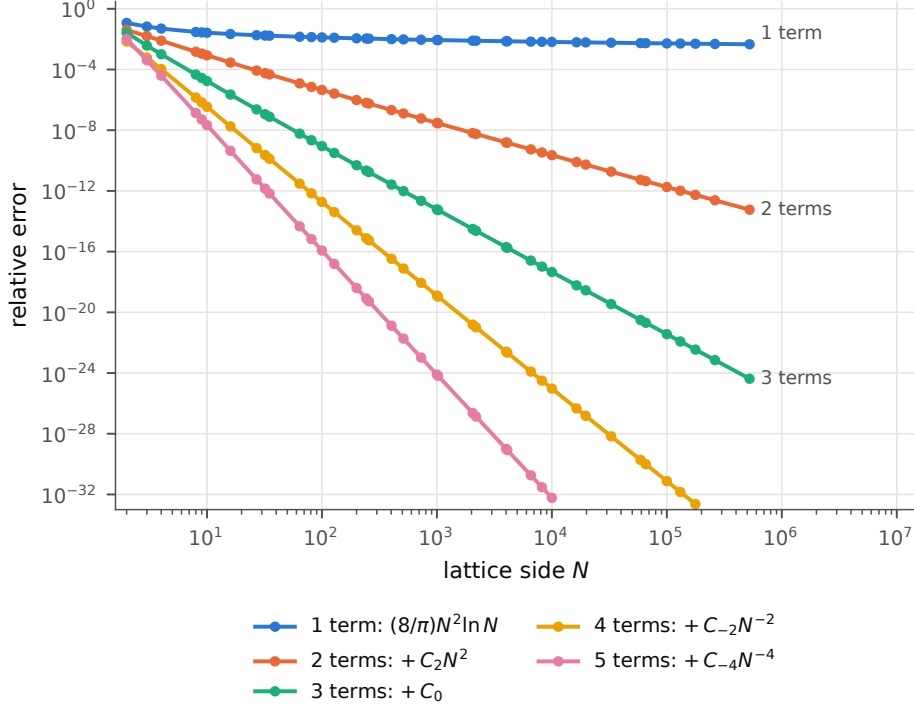


Figure S5: Relative error of the expansion (24) truncated after one to five terms, against exact values of qT_N computed with form (b) to 50 digits ($d = 2$, corner to corner; $N = 2^j, 3^j, 10^j$ and a few other sizes up to $N = 524288$). Curves end where the error reaches the precision of the stored values.

the lattice constant of the periodic simple-cubic lattice that appears in the periodic fundamental solution of Hasimoto (1959) and, with the value 2.837297, in the finite-size correction of diffusion coefficients in periodic boxes (Yeh and Hummer, 2004); the integral above is the representation we used to evaluate it.

S6 Exact modes: proofs, further values and fits

S6.1 Proofs for one dimension

Proof of Proposition 5.1. Let K be the $(N-1) \times (N-1)$ matrix of the $q = 1$ chain restricted to the sites $0, \dots, N-2$: $K(j, j \pm 1) = \frac{1}{2}$ whenever both indices lie in $\{0, \dots, N-2\}$, $K(0, 0) = \frac{1}{2}$ (the cancelled move at the reflecting end), and zero otherwise; the move from $N-2$ to the target is not part of K . Then $Q = (1-q)I + qK$, the vector of one-step absorption probabilities is $r = (q/2) \mathbf{e}_{N-2}$, and $f(t) = \mathbf{e}_0^T Q^{t-1} r$ by (2). Let $v_m(j) = \cos(\omega_m(j + \frac{1}{2}))$ for every integer j . The addition formula gives $\frac{1}{2}[v_m(j-1) + v_m(j+1)] = \cos \omega_m v_m(j)$ for all j . Since $v_m(-1) = v_m(0)$, the left-hand side equals $(Kv_m)(0)$ at $j = 0$; since $v_m(N-1) = \cos((2m-1)\pi/2) = 0$, it equals $(Kv_m)(N-2)$ at $j = N-2$. Hence $Kv_m = \cos \omega_m v_m$ on $\{0, \dots, N-2\}$ for $m = 1, \dots, N-1$. The ω_m are distinct points of $(0, \pi)$, so these are $N-1$ distinct eigenvalues of the symmetric matrix K and the v_m form an orthogonal basis. Their norms follow from $\sum_{j=0}^{N-1} \cos((2j+1)\omega) = \sin(2N\omega)/(2\sin \omega)$ with $n = N-1$ and $2(N-1)\omega_m = (2m-1)\pi - \omega_m$:

$$\|v_m\|^2 = \frac{N-1}{2} + \frac{\sin(2(N-1)\omega_m)}{4\sin \omega_m} = \frac{2N-1}{4}.$$

Expanding Q^{t-1} in this basis, $f(t) = \frac{q}{2} \sum_m \lambda_m^{t-1} v_m(0) v_m(N-2) / \|v_m\|^2$, with $v_m(0) = \cos(\omega_m/2)$ and $v_m(N-2) = \cos((2m-1)\pi/2 - \omega_m) = (-1)^{m+1} \sin \omega_m$, which is (29). For the mean,

$h(j) = [N(N-1) - j(j+1)]/q$ satisfies $h(N-1) = 0$ and $\frac{q}{2}[2h(j) - h(j-1) - h(j+1)] = 1$ for $0 \leq j \leq N-2$, with $h(-1) := h(0)$; these are the equations that characterise the mean hitting time in Proposition 2.4, so $T = h(0) = N(N-1)/q$, in agreement with Theorem 4.1. $\square$

Equation (29) agrees with explicit powers of Q to 4.2×10^{-17} in absolute value ($N = 5, 8, 20, 50$; $q = 0.3, 0.8, 1$) and gives the same maximiser as time stepping at every size compared.

Proof of Proposition 5.2. Rescaling. By Proposition 5.1 and (6) the unit-rate density is $g(t) = \sum_m b_m e^{-\nu_m t}$ with $\nu_m = 1 - \cos \omega_m$ and $b_m = \ell^{-1}(-1)^{m+1} \cos(\omega_m/2) \sin \omega_m$. Put $\Phi_N(\xi) = 2\ell^2 g(2\ell^2 \xi) = \sum_{m=1}^{N-1} c_{N,m} e^{-a_{N,m} \xi}$ with $c_{N,m} = 2\ell(-1)^{m+1} \cos(\omega_m/2) \sin \omega_m$ and $a_{N,m} = 2\ell^2(1 - \cos \omega_m)$. For fixed m , $c_{N,m} \rightarrow (-1)^{m+1}(2m-1)\pi$ and $a_{N,m} \rightarrow (2m-1)^2 \pi^2/4$. Since $|\sin \omega| \leq \omega$ and $1 - \cos \omega \geq 2\omega^2/\pi^2$ on $[0, \pi]$, we have $|c_{N,m}| \leq (2m-1)\pi$ and $a_{N,m} \geq (2m-1)^2$ for all N and m , so the m -th term is bounded by $(2m-1)\pi e^{-(2m-1)^2 \xi_0}$ on $[\xi_0, \infty)$, uniformly in N . Each term converges uniformly on $[\xi_0, \infty)$, and the bounds are summable; hence $\Phi_N \rightarrow \Phi$ uniformly on $[\xi_0, \infty)$ for every $\xi_0 > 0$.

(a) Each Φ_N is log-concave (Proposition S2.2(i)), and the inequality $\Phi_N(\lambda\xi + (1-\lambda)\xi') \geq \Phi_N(\xi)^\lambda \Phi_N(\xi')^{1-\lambda}$ passes to the pointwise limit, so Φ is log-concave. The series (30) converges uniformly on compact subsets of $\{\operatorname{Re} \xi > 0\}$, so Φ is analytic. For $\xi \geq 0.05$ the moduli of the terms decrease from the second term on, so consecutive partial sums bracket Φ : $\Phi(0.05) \leq \pi(e^{-\pi^2/80} - 3e^{-9\pi^2/80} + 5e^{-25\pi^2/80}) < 0.40$, $\Phi(\frac{1}{6}) \geq \pi(e^{-\pi^2/24} - 3e^{-9\pi^2/24}) > 1.84$ and $\Phi(\frac{1}{2}) \leq \pi e^{-\pi^2/8} < 0.92$. For $\xi < 0.05$ write $0.05 = \lambda\xi + (1-\lambda)\frac{1}{6}$ with $0 < \lambda < 1$; log-concavity gives $\Phi(0.05) \geq \Phi(\xi)^\lambda \Phi(\frac{1}{6})^{1-\lambda} \geq \Phi(\xi)^\lambda \Phi(0.05)^{1-\lambda}$, hence $\Phi(\xi) \leq \Phi(0.05) < \Phi(\frac{1}{6})$ (if $\Phi(0.05) = 0$ the first inequality gives $\Phi(\xi) = 0$); in the same way $\Phi(\xi) \leq \Phi(\frac{1}{2}) < \Phi(\frac{1}{6})$ for $\xi > \frac{1}{2}$. Hence Φ attains its supremum on $[0.05, 0.5]$, and not at the end points. The set of maximisers of a log-concave function is an interval, and an analytic function that is constant on an interval is constant; Φ is not constant, so the maximiser ξ^* is unique.

(b) Fix $\varepsilon > 0$ with $[\xi^* - \varepsilon, \xi^* + \varepsilon] \subset (0, \infty)$. By (a), $\Phi(\xi^* \pm \varepsilon) < \Phi(\xi^*)$, so by the convergence just shown $\Phi_N(\xi^* \pm \varepsilon) < \Phi_N(\xi^*)$ for all large N . Φ_N is unimodal, so all its maximisers then lie in $(\xi^* - \varepsilon, \xi^* + \varepsilon)$. The maximiser of g is $t_c^* = 2\ell^2 \xi_N$ with ξ_N a maximiser of Φ_N , and $qT = N(N-1)$; therefore $t_c^*/(qT) = 2\xi_N \ell^2/(N(N-1)) \rightarrow 2\xi^*$. At activity q the mode is t_c^*/q and the mean is T (Proposition 2.2(d)), so the ratio is the same.

(c) For $q \leq \frac{1}{2}$ all $\lambda_m = 1 - q\nu_m$ lie in $[0, 1)$. With $t = 2\ell^2 \xi/q$,

$$\frac{2\ell^2}{q} f(t) = \sum_{m=1}^{N-1} c_{N,m} \lambda_m^{t-1}, \quad 0 \leq \lambda_m^{t-1} \leq e^{-q\nu_m(t-1)} \leq e^{2q} e^{-a_{N,m}\xi},$$

because $\ln(1-x) \leq -x$ and $\nu_m \leq 2$. For fixed m , $(t-1) \ln \lambda_m \rightarrow -(2m-1)^2 \pi^2 \xi/4$ uniformly for ξ in compact subsets of $(0, \infty)$, and the same domination as above gives $(2\ell^2/q) f(2\ell^2 \xi/q) \rightarrow \Phi(\xi)$ uniformly on such subsets, for ξ restricted to the values at which t is an integer. These values have spacing $q/(2\ell^2) \rightarrow 0$, so for large N there are admissible points $\xi_- < \xi_0 < \xi_+$ in $(\xi^* - \varepsilon, \xi^* + \varepsilon)$ with $\xi_\pm$ within $\varepsilon/2$ of $\xi^* \pm \varepsilon$ and ξ_0 so close to ξ^* that the rescaled PMF is larger at ξ_0 than at $\xi_\pm$ (the maximum of Φ over the two closed intervals of length $\varepsilon/2$ at the ends is below $\Phi(\xi^*)$). The PMF is unimodal on the integers (Proposition S2.2(ii)), so every maximiser lies between ξ_- and ξ_+ , and $t^*/T = 2\xi_N \ell^2/(N(N-1)) \rightarrow 2\xi^*$ as in (b). $\square$

Numerical observation S6.1 (One dimension). At $q = 0.8$ the four modes at $N = 50, 100, 200$ and 1000 quoted in Section 5.1 are certified global maximisers. At $N = 10^5$ and 10^6 the local maxima of (29) nearest the continuum prediction are 4 166 011 658 and 416 604 915 263 (route B); they equal $\lceil \tau \rceil$ of Theorem 7.13(c), so they are the global maximisers. In continuous time $t_c^*/(qT) = 0.3332842653$ at $N = 3620$ and 0.33328426549 at $N = 10^6$, and $t_c^* = 0.33328427(N - \frac{1}{2})^2$, to the digits shown, at $N = 10^5$ and 10^6 , in agreement with Proposition 5.2(b).

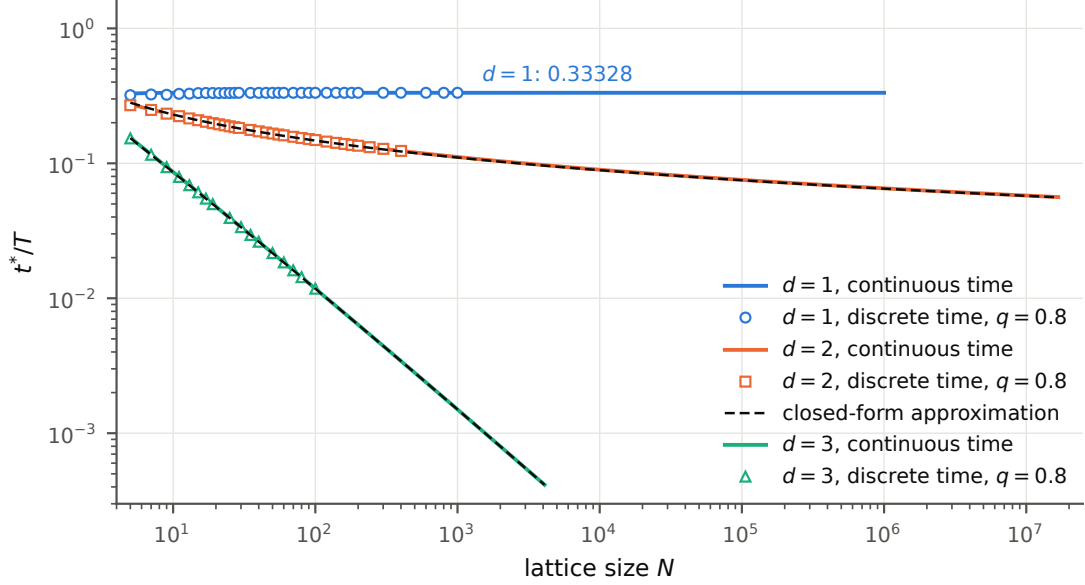


Figure S6: Mode-to-mean ratio t^*/T against lattice size, corner to opposite corner. Lines: exact continuous-time values (independent of q). Symbols: exact discrete-time PMF at $q = 0.8$. Dashed: closed-form approximation (34).

S6.2 Two and three dimensions: further values and figures

At $q = 0.8$ the exact two-dimensional modes are $t^* = 2493, 22\,110, 92\,116$ and $383\,013$ for $N = 35, 100, 200$ and 401 , and the three-dimensional ones $6644, 8863$ and $62\,998$ for $N = 35, 40$ and 100 ; all of them were re-stepped once more with separately written code. In two dimensions the ratio t^*/T is $0.2696, 0.1768, 0.1489, 0.1350$ and 0.1236 at $N = 5, 35, 100, 200$ and 401 , and 0.0651 and 0.0562 at $N = 1\,048\,576$ and $16\,777\,216$. In three dimensions it is 4.2×10^{-3} at $N = 320$. In three dimensions the local exponent of the mode falls from 2.33 ($N \approx 6$) to 2.12 ($N \approx 134$) and 2.085 ($N \approx 3200$), and the two continuous-time routes agree on the position of the maximum to 1.1×10^{-10} , against 7×10^{-13} in two dimensions.

S6.3 What scaling fits can and cannot show

We record what least-squares fits over moderate ranges give when applied to the exact modes. The data are exact, so the intervals quoted below are least-squares 95% intervals computed from the residuals about each functional form: they measure misfit, not noise, and are conditional on the form and on the window. For the exact discrete-time modes at $q = 0.8$, on the computed sizes in $10 \leq N \leq 200$ for $d = 1, 2$ (25 sizes each) and in $10 \leq N \leq 100$ for $d = 3$ (14 sizes), the smallest of them being $N = 11$, a pure power law AN^a gives $a = 2.031 \pm 0.006$ ($d = 1$), 2.090 ± 0.006 ($d = 2$) and 2.169 ± 0.009 ($d = 3$); none of these exponents is a property of the model, since each drifts with the window: for $d = 2$ in continuous time (13 sizes in the first window), 2.093 ± 0.011 on $10 \leq N \leq 200$, 2.033 ± 0.004 on $10 \leq N \leq 1.7 \times 10^7$ and 2.019 ± 0.001 on $10^3 \leq N \leq 1.7 \times 10^7$. Inside the fitting window the residuals do not single out the correct law: on the corrected Akaike criterion the three-parameter form $AN^a(\ln N)^b$ is preferred to $N^2(A \ln \ln N + B)$ in every window examined, with a fitted a below 2. Extrapolation discriminates (Table S5); the complete fits for $d = 2, 3$ are in Table S16, and those for $d = 1$ in the file `data/msc_modes/fit_results.json` of the repository.

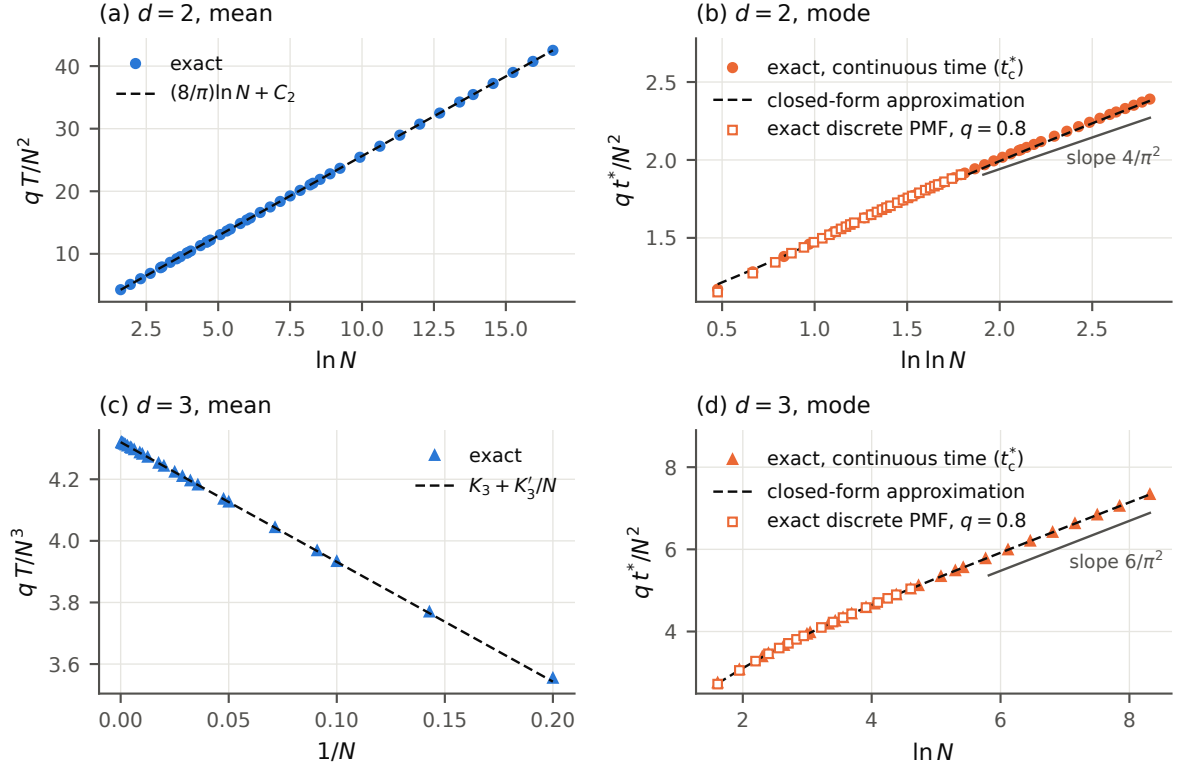


Figure S7: Compensated plots, corner to opposite corner. (a) $d = 2$: qT/N^2 against $\ln N$ with the two-term law of Theorem 4.5. (b) $d = 2$: qt^*/N^2 against $\ln \ln N$. (c) $d = 3$: qT/N^3 against $1/N$ with $K_3 + K'_3/N$. (d) $d = 3$: qt^*/N^2 against $\ln N$. Filled symbols: exact continuous-time values; open squares: exact discrete-time PMF at $q = 0.8$; dashed lines in (b) and (d): the closed-form approximation (34); grey segments: the slopes $4/\pi^2$ and $6/\pi^2$ of the leading laws (Theorem 7.19).

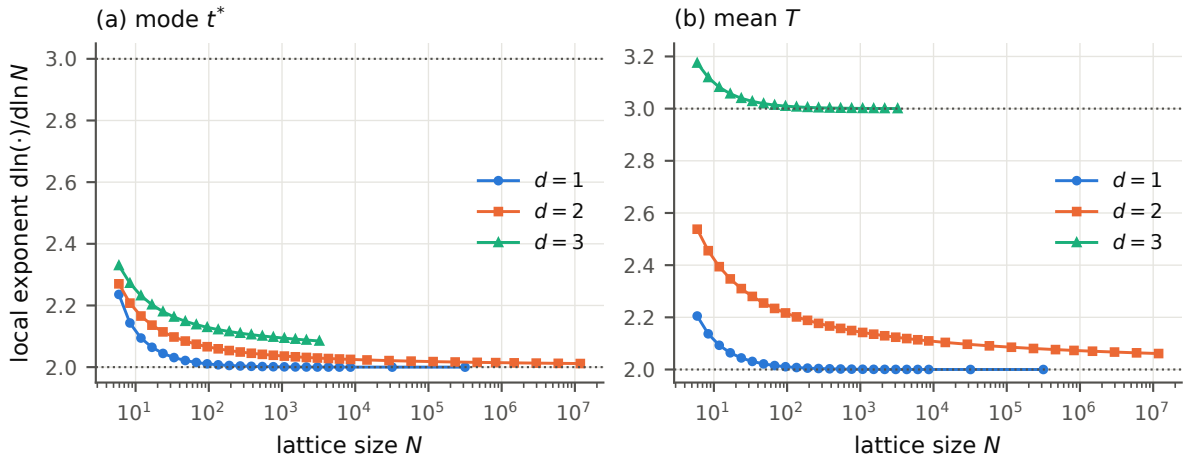


Figure S8: Local exponents $d \ln(\cdot)/d \ln N$ of (a) the mode and (b) the mean, corner to opposite corner, from exact continuous-time values at successive sizes (finite differences, plotted at the geometric mean of the two sizes). The mode laws (Proposition 5.2, Theorem 7.19) give the limit 2 for the mode exponent in every dimension, approached slowly for $d = 2, 3$ because of the logarithmic factors.

Table S5: Least-squares fits of each form to $\ln t_c^*$ (exact continuous-time modes, unit rate) and their extrapolation error, (prediction – exact)/exact, at the largest computed size. Half-widths are 95% least-squares intervals; on exact data they measure misfit, not noise. Last row of each block: closed-form approximation (34), no fitted parameter.

model	fitted exponents or coefficients	rms ln-residual	extrapolation error
$d = 2$: fitted on $10 \leq N \leq 200$ (13 sizes), evaluated at $N = 16\,777\,216$			
AN^2	–	8.3×10^{-2}	–30.8%
AN^a	$a = 2.093 \pm 0.011$	1.5×10^{-2}	+124.9%
$AN^2(\ln N)^b$	$b = 0.343 \pm 0.014$	5.2×10^{-3}	+15.0%
$AN^a(\ln N)^b$	$a = 1.953 \pm 0.004, b = 0.515 \pm 0.014$	5.7×10^{-4}	–19.0%
$N^2(A \ln \ln N + B)$	$A = 0.555 \pm 0.007, B = 0.922 \pm 0.009$	1.6×10^{-3}	+3.8%
approximation (34)	–	–	–0.45%
$d = 3$: fitted on $10 \leq N \leq 100$ (10 sizes), evaluated at $N = 4096$			
AN^3	–	5.8×10^{-1}	+6799%
AN^2	–	1.2×10^{-1}	–41.6%
AN^a	$a = 2.170 \pm 0.014$	1.2×10^{-2}	+31.7%
$AN^2(\ln N)^b$	$b = 0.572 \pm 0.002$	4.9×10^{-4}	–3.8%
$AN^a(\ln N)^b$	$a = 2.006 \pm 0.003, b = 0.553 \pm 0.011$	2.6×10^{-4}	–2.7%
$N^2(A \ln N + B)$	$A = 0.721 \pm 0.023, B = 1.76 \pm 0.08$	4.9×10^{-3}	+5.7%
approximation (34)	–	–	$+6.0 \times 10^{-7}$

S7 Pole structure and shape: proofs and further results

S7.1 Proof of Proposition 6.1

By (9) the first-passage transform from a start $\mathbf{x} \neq \mathbf{a}$ is $\hat{P}(\mathbf{a}, s | \mathbf{x}) / \hat{P}(\mathbf{a}, s | \mathbf{a})$, and the same expression equals 1 for $\mathbf{x} = \mathbf{a}$, in agreement with $\mathcal{T}_u = 0$ there. Averaging over $\mathbf{x}$ gives $\hat{F}_u(s) = n^{-1} \sum_{\mathbf{x}} \hat{P}(\mathbf{a}, s | \mathbf{x}) / \hat{P}(\mathbf{a}, s | \mathbf{a})$. Since P is symmetric, $\sum_{\mathbf{x}} p(\mathbf{a}, t | \mathbf{x}) = \sum_{\mathbf{x}} p(\mathbf{x}, t | \mathbf{a}) = 1$ for all t , hence $\sum_{\mathbf{x}} \hat{P}(\mathbf{a}, s | \mathbf{x}) = 1/s$; this is the first identity. The second follows because $s \hat{u}(s) \hat{F}_u(s) = \hat{P}(\mathbf{a}, s | \mathbf{o}) / \hat{P}(\mathbf{a}, s | \mathbf{a})$. As $u(0) = n \delta_{\mathbf{o}\mathbf{a}} = 0$, $s \hat{u}(s)$ is the transform of u' , and a product of transforms is the transform of a convolution. For (33): at unit rate the d coordinates are independent one-dimensional walks of rate $1/d$ (Lemma S3.1), so p is a product of one-dimensional propagators $\sum_k \psi_k(m) \psi_k(m') e^{-\varepsilon_k t}$, and $\psi_k(N-1) \psi_k(0) = (\alpha_k/N) (-1)^k c_k^2$ because $\vartheta_k(N-1) = \pi k - \vartheta_k(0)$. Finally $\varepsilon_k/\mu_1 = s_k^2/s_1^2 \rightarrow k^2$ and $c_k \rightarrow 1$ for fixed k , while $s_k \geq k/N$ and $s_1 \leq \pi/(2N)$ give $\varepsilon_k/\mu_1 \geq 4k^2/\pi^2$ for all $1 \leq k \leq N-1$; the k -th term is therefore bounded by $2e^{-4k^2x/\pi^2}$ uniformly in N , and dominated convergence gives the limit. $\square$

Numerically the two identities (32) are reproduced by dense linear algebra to 3.5×10^{-14} on four small lattices ($d = 2, 3$), and $\max |u(x/\mu_1) - \theta_4(e^{-x})^2|$ on $0.05 \leq x \leq 12$ is 1.3×10^{-2} , 1.3×10^{-4} and 1.3×10^{-6} at $N = 10, 100$ and 1000 ($d = 2$).

S7.2 The ingredients of the closed form

In (37) the two smallest zeros $\nu_{(0)}, \nu_{(1)}$ of Proposition 2.3 are written ν_0, ν_1 , which is consistent with (6) whenever both residues are non-zero, as they are for the corner start. The two zero conditions are $1/\nu_0 = W/(\mu - \nu_0) + R(-\nu_0)$ and $W/(\nu_1 - \mu) = R(-\nu_1) - 1/\nu_1$, and the residues are dominated by the terms in $1/s$ at $-\nu_0$ and by the terms in $1/(s + \mu)$ at $-\nu_1$.

Equation (34) corrects the rate difference but not $\nu_0 qT$, the ratio b_1/b_0 or the argument of the logarithm, and the omitted terms partly cancel. At $d = 2$, $N = 100$, replacing the exact argument $-b_1\nu_1/b_0\nu_0$ by AX lowers the prediction by 2.5%, the first-order rate difference raises it by 1.1% and the neglect of the poles $j \geq 2$ raises it by 0.4%; the net error is –0.97%. The ingredients can also be checked one by one against the numerically exact poles: $\nu_0 qT$ falls from 1.164 ($N = 5$) to 1.021 ($N = 16\,777\,216$) for $d = 2$ and from 1.076 to 1.0001 ($N = 4096$) for

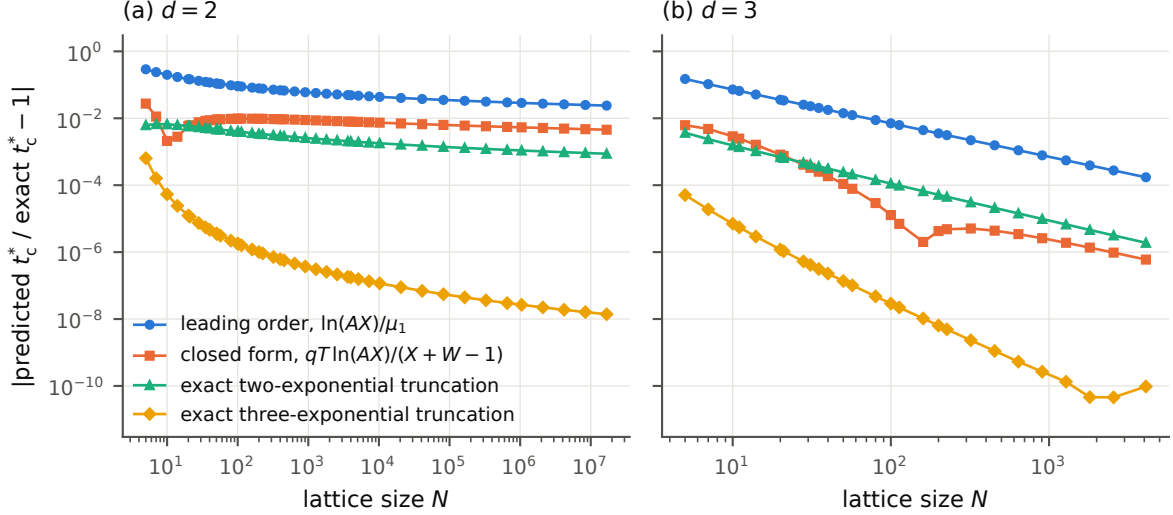


Figure S9: Absolute relative error of the predicted continuous-time mode (unit rate), corner to corner, against the lattice size: (a) $d = 2$, 43 sizes, $5 \leq N \leq 16\,777\,216$; (b) $d = 3$, 27 sizes, $5 \leq N \leq 4096$. Circles: leading order (38); squares: closed-form approximation (34) (for these placements $A = W$); triangles and diamonds: exact two- and three-exponential truncations. Dips of the closed-form curves mark sign changes of the error. The three-exponential curve in (b) levels off near 10^{-10} , the accuracy to which the reference modes are known for $d = 3$.

$d = 3$; $(\nu_1/\mu_1 - 1)X/W$ goes from 1.15 to 1.036 and from 0.911 to 0.9998; and b_1/b_0 reaches -4.087 and -5.9994 at the largest sizes, against $-A = -4$ and -6 . The maximum sits where the two regimes of Fig. 2 meet: $\nu_1 t_c^*$ is 4.81 to 6.13 for $d = 2$ ($N = 35$ to $16\,777\,216$) and 7.44 to 12.07 for $d = 3$ ($N = 40$ to 4096), against $\ln(2dX) = 4.51$ to 6.04 and 7.42 to 12.07 .

Accuracy. The leading-order form (38) errs by $+7.3\%$ at $N = 10$ and $+0.017\%$ at $N = 4096$ for $d = 3$; the closed-form approximation (34) is within 2.6×10^{-4} for $N \geq 35$ in three dimensions, and below $N = 10$ the errors are larger ($+2.7\%$ at $N = 5$ for $d = 2$). The exact two-exponential truncation is within 0.66% ($d = 2$) and 0.15% ($d = 3$), with an error of the opposite sign to that of (34), and the exact three-exponential truncation reaches 1.4×10^{-8} at $N = 16\,777\,216$ (Fig. S9). In discrete time the error of (35) is certified on finite ranges: at most 0.970% for $d = 2$ ($q = 0.8$, $10 \leq N \leq 301$ and five larger sizes) and 0.275% for $d = 3$ ($q = 0.8$, $10 \leq N \leq 120$ and four larger sizes), and for $d = 3$ it stays below 0.3% for every $N \geq 10$ (Theorem 7.21 and Corollary 7.22).

S7.3 Shape in discrete time

The shape statistics of Table 5 are found in discrete time at $q = 0.8$ as well, from distributions summed over their whole support ($\mathbb{P}(\mathcal{T} \leq t^*) = 0.0981$, median $10\,145$ steps against $T = 14\,100.8$ for $d = 2$, $N = 35$), where the coefficient of variation is 0.816 ($d = 1$; the continuum value is $\sqrt{2/3}$), 0.918 ($d = 2$, $N = 35$) and 0.976 ($d = 3$, $N = 15$), increasing towards the value 1 of an exponential law. The density at its maximum is within 5.5% ($d = 2$, $N = 35$) and 1.2% ($d = 3$, $N = 40$) of $1/T$, and for $d = 3$ it stays within 1% of its maximum from 0.62 to 5.6 modal times at $N = 640$. In the variable $x = \mu_1 t$ the density follows $\theta_4(e^{-x})^d$ closely for $d = 3$ and converges slowly for $d = 2$, as expected when the corrections are of order $1/X$.

Table S6: Other placements. (a) Point targets, continuous time at unit rate: corner start and centre target (C2M) and centre start and corner target (M2C), odd N ; the last column is the signed error of (34), which does not apply to M2C ($A = 0$, $b_1 > 0$). (b) Exit from the centre through an absorbing boundary layer (EXIT): continuum limit of t^*/T and exact lattice value at $q = 0.8$ (discrete time).

(a) point targets						
d	placement	N	t_c^*/N^2	t_c^*/qT	sign of b_1	error of (34) (%)
2	C2M	11	0.3838	0.19761	—	−0.1930
2	C2M	161	0.4694	0.12858	—	−0.9491
2	C2M	10 241	0.5369	0.08529	—	−0.7060
2	C2M	655 361	0.5779	0.06463	—	−0.5307
2	M2C	11	0.3122	0.05568	+	—
2	M2C	161	0.4504	0.03623	+	—
2	M2C	10 241	0.5452	0.02370	+	—
2	M2C	655 361	0.5990	0.01783	+	—
3	C2M	11	0.9446	0.06011	—	+0.2055
3	C2M	161	1.3894	0.00571	—	+0.0021
3	C2M	1281	1.7087	0.00088	—	+0.0002
3	M2C	11	0.7765	0.01827	+	—
3	M2C	161	1.3902	0.00201	+	—
3	M2C	1281	1.7779	0.00032	+	—
(b) exit through an absorbing boundary						
d		N	continuum t^*/T	lattice t^*/T	t^* (steps)	T (steps)
1	EXIT	401	0.333284	0.33328	16 664	50 000.0
2	EXIT	201	0.474907	0.47489	6997	14 734.0
3	EXIT	51	0.560839	0.56107	591	1053.3

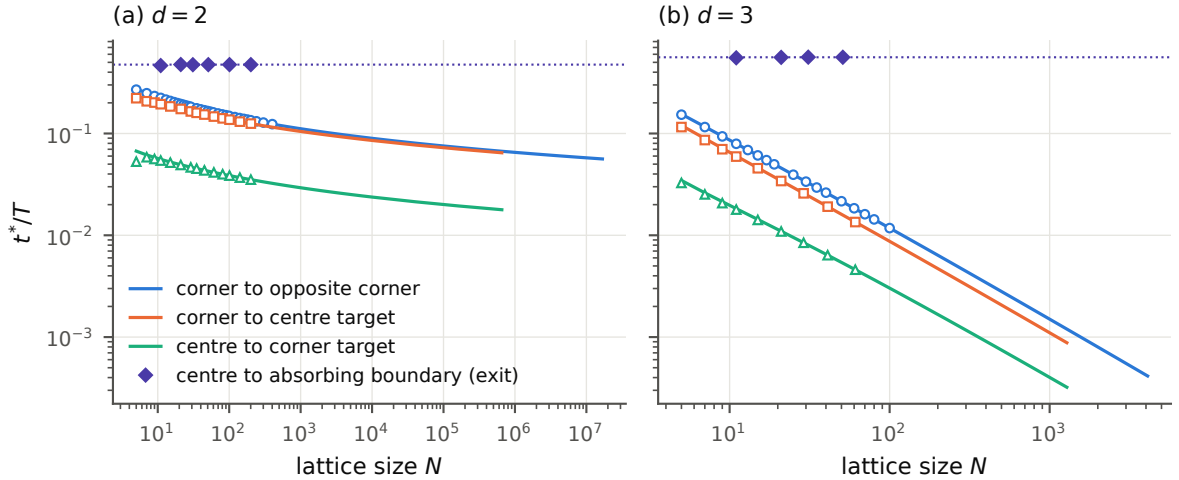


Figure S10: Mode-to-mean ratio t^*/T against the lattice size for four placements, (a) $d = 2$, (b) $d = 3$. Lines: continuous time; open symbols: exact discrete-time modes at $q = 0.8$ (circles: corner to opposite corner; squares: corner to centre; triangles: centre to corner). Diamonds: exit from the centre through an absorbing boundary layer, discrete time, $q = 0.8$; dotted line: its continuum limit, 0.474907 ($d = 2$) and 0.560839 ($d = 3$).

S7.4 Other placements and extended targets

Table S6 and Fig. S10 show how far the results extend. For both centre placements the law is unimodal (Theorem 7.6), but the mode laws are proved only for corner-to-corner transport. A centre target reached from a corner behaves as the corner target does, with $\mu \approx 4\mu_1$. For a centre start and a corner target the density approaches its exponential tail from above, and t_c^*/N^2 again grows slowly with N . Only these placements were examined.

An extended target is different. If the walker starts at the centre and the whole boundary layer absorbs, there is no small parameter: capture and relaxation happen on the same scale. In continuous time the coordinates are independent, so the survival probability is the d -th power of that of a one-dimensional walk of rate $1/d$ between two absorbing ends, and in the diffusive limit t^*/T tends to a constant: 0.333284, 0.474907 and 0.560839 for $d = 1, 2, 3$ (for $d = 1$ this is the constant (31), by symmetry about the centre). The lattice values at $q = 0.8$ agree with these constants to between three and five digits (Table S6b). For exit the ratio therefore tends to a constant that grows with the dimension, whereas for a point target in $d \geq 2$ it decreases with N .

The activity dependence is tabulated in Table S18.

S8 Proofs of the results on the mode

This section gives the short proofs of Section 7 of the main text and of the computer-assisted results of Section 4, and points to the place of every other proof in the companion notes R0–R5.

Throughout, $L_1 = I - P_1$ is the unit-rate generator (with sign reversed) and, for real s outside its spectrum,

$$G(s) = \mathbf{e}_a^\top (L_1 - sI)^{-1} \mathbf{e}_a = \sum_i \frac{w_i}{\mu_{(i)} - s}, \quad G_{\mathbf{x}}(s) = \mathbf{e}_{\mathbf{x}}^\top (L_1 - sI)^{-1} \mathbf{e}_a,$$

with the visible reflecting rates $\mu_{(i)}$ and weights w_i of Proposition 2.3, so that $G(s) = \hat{P}(\mathbf{a}, -s \mid \mathbf{a})$. Its zeros are the rates $\nu_{(i)}$, G increases strictly between consecutive poles, and the residues of the survival function from $\mathbf{x}$ are $r_j(\mathbf{x}) = -G_{\mathbf{x}}(\nu_j)/(\nu_j G'(\nu_j))$ [R2, Thm 3.1].

S8.1 Two lemmas used by the certificates

Lemma S8.1 (Tail lemma). *Let $v_t = Q^{t-1} \mathbf{e}_o$ be the occupation probabilities of the surviving walker. If $v_{T+1} < v_T$ at every site other than the target for some $T \geq 1$, then $v_{t+1} < v_t$ and $f(t+1) < f(t)$ for all $t \geq T$. If moreover $v_{T+1} \leq \gamma v_T$ with $\gamma < 1$, then $f(T+s) \leq \gamma^s f(T)$ for $s \geq 0$. (Source: [R5, Lemma 5], [R4, Lemma 6.1].)*

Proof. $w_t = v_t - v_{t+1}$ satisfies $w_{t+1} = Q w_t$. Every row of Q has a positive entry and $Q \geq 0$, so $w_T > 0$ gives $w_t > 0$ for $t \geq T$, and $f(t) - f(t+1) = \mathbf{r}^\top w_t > 0$ because $f(t) = \mathbf{r}^\top v_t$ by (2) and Q is symmetric. If $v_{T+1} \leq \gamma v_T$, then $v_{T+s+1} = Q^s v_{T+1} \leq \gamma Q^s v_T = \gamma v_{T+s}$ entrywise, since $Q \geq 0$. $\square$

The exact certificates of R4 and R5 compute v_t in integer arithmetic (after scaling Q to an integer matrix) until such a time T ; Proposition 3.1 is a coarser bound of the same kind.

Lemma S8.2 (Rational enclosure of the mean). *Let $h = (I - Q_1)^{-1} \mathbf{1}$ be the vector of unit-rate mean hitting times of $\mathbf{a}$. For every vector z with $R := (I - Q_1)z > 0$ entrywise, $z/\max R \leq h \leq z/\min R$ entrywise. (Source: [R5, Lemma 19].)*

Proof. The spectral radius of Q_1 is below one (Section 2.2), so $(I - Q_1)^{-1} = \sum_{k \geq 0} Q_1^k \geq 0$ entrywise. Apply it to $z/\min R - h$, whose image $R/\min R - \mathbf{1}$ is non-negative, and to $h - z/\max R$. $\square$

A floating-point guess z rounded to integers gives intervals of relative width below 10^{-28} , which is how the means behind Theorem 7.21 are enclosed [R5, Section 5].

S8.2 Structure and signs (Section 7.2)

Proof of Theorem 7.1. (a) By Proposition 2.2, $\sum_j r_j e^{-\nu_j t} = \mathbf{e}_o^\top e^{-(I-Q_1)t} \mathbf{1}$; differentiating i times at $t = 0$ gives $\sum_j r_j \nu_j^i = \mathbf{e}_o^\top (I - Q_1)^i \mathbf{1} = \mathbf{e}_o^\top (I - Q_1)^{i-1} r_1$ for $i \geq 1$. The vector r_1 is supported on the neighbours of $\mathbf{a}$, which are at distance at least $D - 1$ from $\mathbf{o}$, and the entry $(\mathbf{o}, \mathbf{y})$ of $(I - Q_1)^{i-1}$ vanishes if $i - 1 < |\mathbf{o} - \mathbf{y}|_1$; for $i = D$ only the term $(-Q_1)^{D-1}$ of the binomial expansion reaches the neighbours at distance $D - 1$, and it does so with the sign $(-1)^{D-1}$ and a positive weight (a shortest path). Suppose the non-zero r_j had $k \leq D - 2$ changes of sign; choose $\theta_1 < \dots < \theta_k$ separating the blocks of equal sign and put $P(\nu) = \nu \prod_m (\nu - \theta_m)$, of degree $k + 1 \leq D - 1$. Since all $\nu_j > 0$, $r_j P(\nu_j)$ has the same sign for every j with $r_j \neq 0$, so $\sum_j r_j P(\nu_j) \neq 0$; but it is a combination of $\sum_j r_j \nu_j^i$ with $1 \leq i \leq D - 1$, which vanish. On the chain started at the reflecting end, the $N - 1$ rates of Proposition 5.1 all carry a non-zero residue, $D = N - 1$, so the $N - 2$ sign changes force alternation, and $r_0 > 0$.

(b) By Proposition 6.1, $\widehat{F}_u(s) = 1/(nsG(-s))$, which tends to $\mathbb{P}(\mathcal{T}_u = 0) = 1/n$ as $s \rightarrow \infty$ and is finite at $s = 0$ (the term $w_0/(-s)$, $w_0 = 1/n$, cancels the factor ns). Its poles are the zeros $-\nu_j$ of $G(-s)$, simple, with residues $1/(n\nu_j G'(\nu_j)) > 0$ because $G' > 0$. Hence the law of $\mathcal{T}_u$ is the atom $1/n$ at 0 plus the density $\sum_j c_j e^{-\nu_j t}$ with all $c_j > 0$; in discrete time the same partial fractions give geometric terms with ratio $1 - q\nu_j \geq 0$ when $q \leq \frac{1}{2}$.

(c) If $v > 0$ is the Perron eigenvector of Q_1 normalised to a probability, then $\mathbb{P}_v(\mathcal{T} > t) = v^\top e^{-(I-Q_1)t} \mathbf{1} = e^{-\nu_0 t}$, and in discrete time $v^\top Q^t \mathbf{1} = (1 - q\nu_0)^t$. $\square$

Proof of Theorem 7.2. Since $D \geq 2$, $f(1) = r(\mathbf{o}) = 0$ and $g(0) = 0$, whereas $f(D) \geq (q/2d)^D > 0$ along a shortest path, and $g(t) > 0$ for $t > 0$ (by Proposition 2.2(b), $e^t g(t) = \sum_{n \geq 0} t^n f_1(n+1)/n!$ with the PMF f_1 at $q = 1$, a series of non-negative terms one of which is positive). A mixture $\int \lambda^{t-1} m(d\lambda)$ with $m \geq 0$ on $[0, 1)$ equals $m([0, 1))$ at $t = 1$, so $m = 0$ and $f \equiv 0$, a contradiction; in continuous time $m([0, \infty)) = \lim_{t \downarrow 0} g(t) = 0$ by monotone convergence. Some residue is negative by Theorem 7.1(a), since $D - 1 \geq 1$. $\square$

For the proof of Theorem 7.3, let R be the reflection $x_i \mapsto N - 1 - x_i$ in every coordinate. It commutes with P_1 and exchanges $\mathbf{o}$ and $\mathbf{a}$. With $v_\pm = \frac{1}{2}(\mathbf{e}_a \pm \mathbf{e}_o)$ and $G^\pm(s) = v_\pm^\top (L_1 - sI)^{-1} v_\pm$ one has $G = G^+ + G^-$ and $G_o = G^+ - G^-$, because $G_{oo} = G_{aa}$; at a zero ν_j of G , therefore,

$$r_j = \frac{2G^-(\nu_j)}{\nu_j G'(\nu_j)} = -\frac{2G^+(\nu_j)}{\nu_j G'(\nu_j)}, \quad \text{sign } r_j = \text{sign } G^-(\nu_j) = -\text{sign } G^+(\nu_j) \quad (\text{S8.1})$$

[R2, Thm 5.6]. Call a visible rate $\mu_{(i)}$ purely even if its spectral projector E_i satisfies $E_i v_- = 0$, and purely odd if $E_i v_+ = 0$.

Lemma S8.3. *Let $1 \leq i \leq m - 1$ (m as in Proposition 2.3). If $\mu_{(i)}$ is purely even, the residues (r_{i-1}, r_i) do not have the signs $(+, -)$; if it is purely odd, they do not have the signs $(-, +)$. (Source: [R2, Prop 5.9].)*

Proof. R commutes with every E_i and $Rv_\pm = \pm v_\pm$, so $E_i v_+ \perp E_i v_-$ and every pole of $G^\pm$ is a visible rate. If $E_i v_- = 0$, the only visible rate in $I = (\mu_{(i-1)}, \mu_{(i+1)})$ is not a pole of G^- , so G^- is smooth on I with derivative $\sum_l \|E_l v_-\|^2 (\lambda_l - s)^{-2} > 0$ (λ_l the eigenvalues of L_1 ; some weight is positive since $\sum_l \|E_l v_-\|^2 = \frac{1}{2}$). By interlacing, $\nu_{i-1} < \mu_{(i)} < \nu_i$ both lie in I , so $G^-(\nu_{i-1}) < G^-(\nu_i)$, and by (S8.1) $r_{i-1} > 0$ implies $r_i > 0$. The purely odd case is the same with G^+ ($0 \notin I$). $\square$

Proof of Theorem 7.3. (a) is [R2, Cor 5.7]: the first visible rate μ_1 is purely odd (its eigenfunctions $\phi_{\mathbf{e}_i}$ change sign under R), at least two eigenvalues of L_1 (counted with multiplicity) lie in $(0, \mu_1]$ for $d \geq 2$, and the sign rule (S8.1) then gives $r_1 < 0$ [R2, Thm 5.6]; $r_0 > 0$ by Proposition 2.3(iii).

Put $u = \pi/(2N)$, $x = \sin^2 u$ and $\varsigma_j = s_j^2/s_1^2$, strictly increasing in j , $\varsigma_1 = 1$, $\varsigma_2 = 4(1 - x) =: t$, $\varsigma_3 = (3 - 4x)^2$. Then $\mu_{\mathbf{k}} = \mu_1 V(\mathbf{k})$ with $V(\mathbf{k}) = \sum_i \varsigma_{k_i}$. For opposite corners every $\phi_{\mathbf{k}}(\mathbf{a}) \neq 0$, so

every rate is visible and the $\mu_{(i)}$ are μ_1 times the distinct values of V ; $\phi_{\mathbf{k}} \circ R = (-1)^{|\mathbf{k}|} \phi_{\mathbf{k}}$, so a value of V attained only at $\mathbf{k}$ with $|\mathbf{k}|$ of one parity gives a pure rate (purely even for even $|\mathbf{k}|$).

(b) For $d = 1$ this is Theorem 7.1(a). For $N = 2$, $V(\mathbf{k}) = |\mathbf{k}| \in \{0, \dots, d\}$, so there are $m = d = D$ rates, and $D - 1$ sign changes among D non-zero residues force alternation.

(c) Count from the top. Put $\Delta_l = \zeta_{N-1} - \zeta_{N-1-l}$. Since $\sin^2 A - \sin^2 B = \sin(A+B)\sin(A-B)$, $\Delta_l - \Delta_{l-1} = \sin((2l+1)u)/\sin u > 0$. With $l_i = N - 1 - k_i$ and $\Delta(\mathbf{l}) = \sum_i \Delta_{l_i}$ one has $V(\mathbf{k}) = d\zeta_{N-1} - \Delta(\mathbf{l})$ and $|\mathbf{k}| = D - |\mathbf{l}|$; the j -th largest rate is μ_1 times $d\zeta_{N-1}$ minus the j -th smallest value of Δ . Let $\alpha = \Delta_1 = 3 - 4x$ and $\beta = \Delta_2 - \Delta_1 = 5 - 20x + 16x^2$; for $N \geq 3$, $0 < x \leq \frac{1}{4}$, $2\alpha - \beta > 0$, and $\beta - \alpha = 2(1 - 8x + 8x^2)$ is positive for $N \geq 5$ and zero for $N = 4$. For $N \geq 5$, $0 < \alpha < \beta < 2\alpha$; if $\Delta(\mathbf{l}) \leq \alpha + \beta$, all $l_i \leq 2$, an entry 2 forces $\mathbf{l} = (2, 0, \dots)$ up to order, and otherwise $\Delta = n_1\alpha$ with $n_1 \leq 2$. The four smallest values are $0 < \alpha < 2\alpha < \alpha + \beta$, attained only at $|\mathbf{l}| = 0, 1, 2, 2$. For $N = 3$, $\Delta_1 = 2$, $\Delta_2 = 3$, and the four smallest values $0 < 2 < 3 < 4$ are attained only at $|\mathbf{l}| = 0, 1, 2, 2$. For $N = 4$, $\alpha = \beta = 1 + \sqrt{2}$ and $\Delta_3 - \Delta_2 = 1$, so $\Delta = c\alpha + n_3$ with $c = n_1 + 2n_2 + 2n_3$ and $|\mathbf{l}| = c + n_3$; since α is irrational, Δ determines (c, n_3) and hence $|\mathbf{l}|$, every rate is pure, and the values below 3α are $0 < \alpha < 2\alpha < 2\alpha + 1$, with 3α attained. In each case two consecutive rates near the top, the third and fourth largest ($N \neq 4$), or the fourth and fifth ($N = 4$), are pure of the same parity, and they are not the two smallest, because the smallest value of V belongs to $\Delta = d\Delta_{N-1} \geq 2\Delta_2$, larger than all values listed. By Lemma S8.3 applied to both, the three residues adjacent to them have monotone signs, whereas alternation requires $(+, -, +)$ or $(-, +, -)$. $\square$

Proof of Theorem 7.5. (a) Write $G(s) = -1/(ns) + g(s)$ with $g(s) = \sum_{i \geq 1} w_i/(\mu_{(i)} - s)$ ($w_0 = 1/n$). From $\hat{F}_u(s) = 1/(1 + ns g(-s))$ (proof of Theorem 7.1(b)), $\tau_u = -\hat{F}'_u(0) = ng(0)$. The zero $\nu_0 \in (0, \mu_{(1)})$ of G satisfies $1/\nu_0 = ng(\nu_0)$, and for $0 \leq s < \mu_{(1)} \leq \mu$, $\frac{1}{\mu} \leq \frac{1}{\mu-s} \leq \frac{1}{\mu} \frac{1}{1-s/\mu_{(1)}}$, hence $\tau_u \leq ng(s) \leq \tau_u/(1 - s/\mu_{(1)})$. At $s = \nu_0$ the first inequality gives $1/\nu_0 \geq \tau_u$ and the second $1/\nu_0 - 1/\mu_{(1)} \leq \tau_u$.

(b) *Monotone coupling* [R2, Lemma 7.8]: drive two walkers by the same attempted moves; the maps $x_i \mapsto \min(x_i + 1, N - 1)$ and $x_i \mapsto \max(x_i - 1, 0)$ are non-decreasing, so coordinatewise order is preserved, and when the lower walker reaches the maximal site $\mathbf{a}$ the upper one is there as well. Hence $\mathbb{P}_{\mathbf{x}}(\mathcal{T} > t) \geq \mathbb{P}_{\mathbf{y}}(\mathcal{T} > t)$ for $\mathbf{x} \leq \mathbf{y}$, and the corner $\mathbf{o} = \mathbf{0}$ maximises both the mean and $r_0(\cdot) = \lim_{t \rightarrow \infty} e^{\nu_0 t} \mathbb{P}(\mathcal{T} > t)$. By Theorem 7.1(c) the averages of $r_0(\cdot)$ and of $\nu_0 q T_{\rightarrow \mathbf{a}}$ over the quasi-stationary distribution, which charges every site other than $\mathbf{a}$, are both 1 (the second is (11)). So $r_0 \geq 1$ and $\nu_0 q T \geq 1$, with equality only if the function is constant. If the mean hitting time were constant on the sites other than $\mathbf{a}$, the equation $(L_1 h)(\mathbf{x}) = 1$ of Proposition 2.4 would fail at any site that is not a neighbour of $\mathbf{a}$; if $r_0(\cdot)$, which is proportional to the Perron eigenvector of Q_1 , were constant, all row sums of Q_1 would be equal, which again fails if some site is not a neighbour of $\mathbf{a}$. Every site other than $\mathbf{a}$ is a neighbour of $\mathbf{a}$ only for $(d, N) = (1, 2)$.

(c) is [R2, Thm 11.3]; it combines the explicit two-sided bounds on ν_0, ν_1, r_0, r_1 in terms of three Green's-function functionals [R2, Thms 6.2–6.8, 7.10] with their asymptotics for the box in $d = 2, 3$ [R2, Sections 9–11]. $\square$

S8.3 Unimodality (Section 7.3)

Proof of Theorem 7.6(i), corner to corner. At unit rate the coordinates move independently at rate $1/d$ (Lemma S3.1), so $u(t) = np(\mathbf{a}, t | \mathbf{o}) = h(t)^d$, where h is N times the one-dimensional propagator from 0 to $N - 1$ at rate $1/d$. By Cramer's rule for the corner entry of a tridiagonal inverse, as in the proof of Proposition S2.2, $\int_0^\infty e^{-st} h'(t) dt = \prod_{k=1}^{N-1} \varepsilon_k / (s + \varepsilon_k)$ with the rates ε_k of (33): h' is the density of a sum of $N - 1$ independent exponential variables, hence log-concave (Ibragimov, 1956), and h , its distribution function, is log-concave (Bagnoli and Bergstrom, 2005, Theorem 1). So $\varphi := u' = dh^{d-1}h'$ is positive and log-concave on $(0, \infty)$, and $\varphi(0) = 0$ when $D \geq 2$.

By Proposition 6.1 and the proof of Theorem 7.1(b), $g(t) = \varphi(t)/n + \int_0^t k(\tau)\varphi(t-\tau) d\tau$ with $k(\tau) = \sum_j c_j e^{-\nu_j \tau}$, $c_j > 0$. Differentiating the convolution in the form $\int_0^t k(t-s)\varphi(s) ds$,

$$\frac{g'(t)}{\varphi(t)} = \frac{\varphi'(t)}{n\varphi(t)} + k(0) - \int_0^t |k'(\tau)| \frac{\varphi(t-\tau)}{\varphi(t)} d\tau.$$

The first term is non-increasing (φ is log-concave); for fixed τ the ratio $\varphi(t-\tau)/\varphi(t)$ is non-decreasing in t , and $|k'| > 0$, so the integral is strictly increasing. Hence g'/φ is strictly decreasing and g' changes sign at most once, from positive to negative; since $g(0) = 0$ for $D \geq 2$ and $g(t) \rightarrow 0$, it changes sign exactly once. For $D = 1$ ($N = 2$, $d = 1$) the law is exponential. Rate q only rescales time. This is the argument of [R4, Lemma 4.8, Thm 4.11(a)] and [R3, Thm 1.11], written with the factorisation of Section 6. $\square$

The centre geometries use the folded chain $|X - \frac{N-1}{2}|$, which is again a birth-and-death chain between its two ends, in place of the one-dimensional factor [R4, Thm 4.12]. Part (ii) is the discrete counterpart: for $q \leq q_N^\square$ the one-dimensional propagator and its increments are log-concave sequences (sums of independent geometric variables), the multinomial theorem writes the d -dimensional increment as a binomial convolution of one-dimensional ones, binomial convolution preserves log-concavity (Davenport and Pólya, 1949), and a discrete form of the criterion above applies [R4, Lemmas 4.7–4.10, Thms 4.11(b), 4.12].

Proof of Proposition 7.8 (outline). Count paths from 0 to the first visit of $N-1$, with $\eta = 1-q$ and $\eta_1 = 1 - \frac{q}{2}$ the probabilities of resting in the interior and at the reflecting end: $f(N-1) = (\frac{q}{2})^{N-1}$, $f(N) = (\frac{q}{2})^{N-1}[\eta_1 + (N-2)\eta]$ and $f(N+1) = (\frac{q}{2})^{N-1}[\eta_1^2 + (N-2)\eta_1\eta + \binom{N-1}{2}\eta^2 + (N-2)\frac{q^2}{4}]$. The first relation gives $f(N) < f(N-1)$ if and only if $q > \frac{2N-4}{2N-3}$; $f(N+1) - f(N)$ is a quadratic in η that is positive on the relevant range [R1, Prop 3.9], [R5, Prop 32]. For $N = 4$ and $q = \frac{4}{5}$ the integer vectors $(5Q)^t \mathbf{e}_0$ satisfy $(5Q)^5 \mathbf{e}_0 \leq 5(5Q)^4 \mathbf{e}_0$ entrywise, so by Lemma S8.1 (with $\leq$) the PMF does not increase from $t = 5$ on, while $f(3) = f(4) = \frac{8}{125} < f(5) = \frac{208}{3125}$; by the monotonicity of unimodality in q [R4, Thm 2.5] this gives every $q \leq \frac{4}{5}$ [R4, Prop 4.4]. $\square$

Theorems 7.7, 7.9 and 7.10 are proved in [R4, Sections 7–9]; the exact counter-examples compare rational numbers, the continuous-time ones after truncating the Poisson series $e^t g(t) = \sum_n t^n f_1(n+1)/n!$ with explicit tail bounds.

S8.4 One dimension and the closed form (Sections 7.4 and 7.5)

Theorem 7.13 is proved in [R1, Sections 4–7], Refuted statement 7.14 in [R1, Rem 7.9] and [R5, Thms 21, 43], Theorem 7.15 in [R5, Sections 4–6, 10], Proposition 7.16 and Theorems 7.17–7.19 in [R3, Sections 1–6], cases (b) and (c) of Theorem 7.20 in [R3, Section 7], and Theorem 7.21 in [R5, Section 7]. The two deductions made in R0 follow.

Proof of Theorem 7.20(a). For $q \leq \frac{1}{2}$ all multipliers $1 - q\nu_j$ lie in $(0, 1)$, and the discrete analogue of Theorem 7.17 [R3, Prop 7.1, Lemma 7.6, Thm 7.7] certifies, for every $N \geq N_0$, the sign of the increment $f(t+1) - f(t)$ at two explicit times $t_- < t_+$: positive at t_- , negative at t_+ . The PMF is unimodal by Theorem 7.6(ii), since $q \leq \frac{1}{2} < q_N^\square$. For a unimodal sequence, a strict increase from t_- to $t_- + 1$ excludes maximisers $\leq t_-$, and a strict decrease from t_+ to $t_+ + 1$ excludes maximisers $> t_+$; so every maximiser lies in $(t_-, t_+]$, which is the window stated [R3, Thm 7.8]. R3 used a published closure theorem for ultra-log-concave sequences to obtain unimodality [R3, Thm 7.4]; this proof uses Theorem 7.6(ii) instead. $\square$

Proof of Corollary 7.22. Write $\mu = q\mu_1 \leq \frac{2q}{d}(\frac{\pi}{2N})^2$ and $\tau_{\text{cf}}(X) = X \ln(2dX)/(X + 2d - 1)$. A window $-K^-(X\mu) < t^* - t_{\text{cf}} < K^+(X\mu) + 2$ gives $t^* > (\tau_{\text{cf}}(X) - \max(K^-, 0)/X)/\mu$ and

$|t^* - t_{\text{cf}}| < \max(K^+/X + 2\mu, K^-/X)/\mu$, hence

$$\frac{|t^* - t_{\text{cf}}|}{t^*} < B(N) := \frac{\max(K^+/X + 2\mu, K^-/X)}{\tau_{\text{cf}}(X) - \max(K^-, 0)/X}.$$

τ_{cf} is increasing in X (its derivative is $((2d-1)\ln(2dX) + X + 2d-1)/(X+2d-1)^2 > 0$). Insert the lower bound $X \geq X_{\text{lo}}(N)$ of [R3, Cor 4.7] ($2\pi \ln N - 0.24$ for $d=2$, $7.106860N - 7.63$ for $d=3$; $N \geq 10$) and the upper bound on μ ; then B is decreasing in N on every range where the constants are fixed, and it suffices to evaluate it, in interval arithmetic, at the left end N_0 of a row of the window tables. For $d=3$ the row (50; 1.851, 2.973) of Theorem 7.20(b) ($q_{\text{max}} = 0.8$) gives $B(50) < 0.0013$, and for $q = \frac{1}{2}$ the row (30; 1.962, 2.829) of Theorem 7.20(a) gives $B(30) < 0.0023$; Theorem 7.21(b) covers $10 \leq N \leq 120$ and $10 \leq N \leq 80$. For $d=2$ the rows with $N_0 = 1000$ give $B(N) < 0.0100$ exactly from $N = 1\,761\,450$ ($q = \frac{4}{5}$) and $N = 554\,846$ ($q = \frac{1}{2}$) on, and Theorem 7.21(a) covers the finite ranges. $\square$

S8.5 The additions to Section 4

Proof of Theorem 4.6. [R2, Thm 9.1] proves, for every $N \geq 2$, $qT_N = \frac{8}{\pi}N^2 \ln N + (c_\tau + c_{m\tau})N^2 + R_m(N)$ with $R_m = R_\tau + R_{m\tau}$, where the two parts satisfy $-2.14 \leq R_\tau(N) \leq 3.53$ and $-9.06 \leq R_{m\tau}(N) \leq 7.72$ [R2, Eqs (9.3), (9.4)], so that $-11.2 \leq R_m(N) \leq 11.25$; here $c_\tau = \frac{8}{\pi}(\gamma_E - \frac{1}{2} + \frac{7}{2} \ln 2 + \frac{1}{2} \ln \pi - 2 \ln \Gamma(\frac{1}{4})) - 2$ and $c_{m\tau} = 4 \ln 2/\pi$ (the numerical constants of the proof certified in ball arithmetic). Since $\ln \varpi = 2 \ln \Gamma(\frac{1}{4}) - \frac{3}{2} \ln 2 - \frac{1}{2} \ln \pi$, the sum $c_\tau + c_{m\tau} = \frac{8}{\pi}[\gamma_E - \frac{1}{2} + 4 \ln 2 + \frac{1}{2} \ln \pi - 2 \ln \Gamma(\frac{1}{4})] - 2$ equals the expression of C_2 in (25). For the second part, [R5, Thm 26] encloses qT_N for $2 \leq N \leq 301$ by Lemma S8.2 with integer guesses (base T1), and [R5, Prop 27] treats $301 < N \leq 1500$ in ball arithmetic under the hypothesis that qT_N is given by the single sum (18); that hypothesis is Theorem 4.2(b). The junction $r_{301} < r_{302}$ is an inequality between the outward-rounded bounds 0.532129078393 and 0.532129156439. $\square$

Theorem 4.8 is [R2, Thm 10.1]. The identity $K_3 = \sum_{\mathbf{y} \in \{0,1\}^3} \mathcal{G}_\infty(\mathbf{y})$ is elementary: $K_3 = 3 \int_0^\infty [e^{-s}(I_0(s) + I_1(s))]^3 ds$, and expanding $\prod_i (1 + \cos \theta_i) = \sum_{\mathbf{y} \in \{0,1\}^3} \prod_i \cos(y_i \theta_i)$ in the equivalent Fourier integral gives the Fourier representation of the Green's function; the bounds on the remainders use the closed form of a one-dimensional integral by complete elliptic integrals and a trapezoidal estimate [R2, Section 10].

S8.6 Further remarks on the statements of Section 7

Residues. For a pair exchanged by a symmetry of the lattice, as for opposite corners, the sign of every residue is determined by one of two half-sums of the Green's function at the target [R2, Thm 5.6]; this is the sign rule (S8.1). For $d \geq 2$, $\bar{X} > 1$ and $t \geq 1/\mu_1$, the unit-rate density differs from its two slowest exponential terms by at most $2e c_*^{d/2} \beta \sqrt{Z} \bar{X}^{-1} \mu_1 e^{-2\mu_1 t}$, with $c_* = 1 + \pi^{3/2}/2$, $\beta = \bar{X}/(\bar{X} - 1)$ and a lattice constant Z (for $d \leq 3$ the sum $\sum_{\mathbf{k} \in \mathbb{Z}^d \setminus 0} |\mathbf{k}|^{-4}$) [R2, Thm 12.1, Cor 12.2]. The bound is absolute: where $e^{-\mu_1 t}$ is of order $1/\bar{X}$, near the mode, it is a relative error $O(\bar{X}^{-2})$, whereas at $t = 1/\mu_1$ it is of the same order $\mu_1 \bar{X}^{-1}$ as the two slowest terms [R2, Rem 12.3].

Unimodality. The extension of the statement on g' in Theorem 7.6(i) to $d=1$ and to $N=2$ is also covered: unimodality follows from Proposition S2.2 and from part (ii). A second proof of the one-dimensional case of Theorem 7.7, by total positivity of Q^k for $k \geq 10$, is [R1, Thm 3.15]. The evidence for Conjecture 7.12(a) is exact arithmetic at $q = \frac{1}{2}$ near the mode for $d=2$, $N \leq 20$, and $d=3$, $N \leq 8$; for (b) it is Theorem 7.9(b).

Certified modes. In Theorem 7.15, for $d=2,3$ every occupation probability of the surviving walker is already decreasing at the mode. The ranges $d=1$, $N \leq 600$ ($q = \frac{4}{5}$) and $N \leq 300$ ($q = \frac{1}{2}$), $d=2$, $N \leq 160$ and $N \leq 80$, and $d=3$, $N \leq 60$ and $N \leq 32$ rest on Python integers

alone and were re-derived by a second, separately written program; the others rest in addition on the C program, each certificate computed twice with identical checksums.

One dimension. All poles are of order ℓ^{-2} and all contribute at the mode; there is no small parameter, which is why the closed form fails there (Refuted statement 7.14). For $q = 0.8$, $t^* = \lceil c\ell^2/q - 1.09902\dots \rceil$ whenever $\text{dist}(\tau_\ell, \mathbb{Z}) \geq E$ (Theorem 7.13(c)); in particular the modes 4166011658 and 416604915263 at $N = 10^5$ and 10^6 are the global maximisers.

The closed form. t_{cf} differs from (34) by a relative amount $O(N^{-2})$; the scaled discrete mode is $q\mu_1 t^*$, the scaled continuous mode $\mu_1 t_{\text{cf}}^*$, and $q\mu_1 t_{\text{cf}} = X \ln(2dX)/(X + 2d - 1)$. Theorem 7.18 also gives $|t_{\text{cf}}^* - qt_{\text{cf}}| \leq K_d/(X\mu_1)$. Without the corrections of Theorem 7.17, g' vanishes at an explicit time given by the two slowest poles and their weights.

S9 Inert defects: validation, extended results and proofs

S9.1 Model and exact computation

Configurations with pockets (open sites enclosed by blocked ones) are kept; the pockets are simply not part of the state space $\mathcal{C}$. On $\mathcal{C}$ the definitions of Section 2.2 and Propositions 2.2 and 2.4 carry over with Ω replaced by $\mathcal{C}$. Blocking a site deletes its row and column of the transition matrix and adds $q/4$ to the diagonal entry of each of its open neighbours. A *partial trap* is a site whose diagonal entry is lowered by some $\Delta \leq 1 - q$ while the moves onto and off it are kept: the row sum becomes $1 - \Delta$, so that a walker on the site is killed with probability Δ per step (Section S9.10).

Proof of Proposition 8.1, in full. Two neighbouring sites of $\mathcal{C}$ exchange probability $q/(2d)$ per step in both directions, and every cancelled attempt is added to the diagonal. Hence P is symmetric and stochastic on $\mathcal{C}$, it is irreducible because $\mathcal{C}$ is connected, and $I - P = \frac{q}{2d}\mathcal{L}_{\mathcal{C}}$, where $(\mathcal{L}_{\mathcal{C}}V)(\mathbf{x}) = \sum_{\mathbf{y} \sim \mathbf{x}} [V(\mathbf{x}) - V(\mathbf{y})]$ is the Laplacian of $\mathcal{C}$ with unit conductance on each bond between neighbouring sites of $\mathcal{C}$: both sides have the same off-diagonal entries and vanishing row sums. Therefore $(I - P)^+ = \frac{2d}{q}G$, and (12) gives (41). $\square$

The factor $G_{\text{aa}} - G_{\text{oa}}$ is not an effective resistance and is not monotone in the set of blocked sites. Blocking the single site $(1, 1)$ lowers it by the factor 0.99978 at $N = 35$, and so does the blocking of each of 16 other single sites, all within four lattice spacings of the start.

The certificate of Proposition 3.1 is exact in exact arithmetic; in double precision the peak heights (10^{-5} to 10^{-4} here) exceed the rounding error of f by about ten orders of magnitude. This is an estimate of the margin, not the a-priori round-off bound used for the lattice without defects (Section 3 and Section S2.3): the defect modes are certified up to double-precision round-off. Bound (a) is used with the dense eigendecomposition and bound (b) with time stepping. For $N > 35$ the first of the two codes used a bound of the same type built on the leading eigenpair of Q computed by the Lanczos method; that variant is not covered by the proposition, and its results enter only as a secondary sample in Section S9.4. Figure S11 shows exact PMFs for four configurations.

S9.2 The density sweep in full

Among the 20800 configurations the largest single ratio is 0.48 (0.32 for $p \leq 0.2$) and the smallest 0.062. The least-squares slope of the per-placement ratios against p under uniform placement is $+0.081 \pm 0.009$ for $p \leq 0.2$ but $+0.038 \pm 0.015$ for $p \leq 0.12$: the dependence is not linear (95% intervals from heteroscedasticity-consistent standard errors; 8001 and 4801 points). Under corner-thinned placement the slope is $+0.159 \pm 0.007$ for $p \leq 0.2$ (8001 points). Under corner-thinned placement the coefficient of variation falls from 0.918 to 0.902 at $p = 0.2$. In sample A alone the mean of the per-placement ratios never falls below the clean value 0.1768, whereas under uniform placement the ratio of the ensemble means does (0.1733 to 0.1760 at $p = 0.04, 0.06, 0.10, 0.12$): at low density the sign of the small change depends on the estimator.

Table S7: Validation of the exact machinery ($q = 0.8$). “Code A” and “code B” are the two separately written implementations that produced samples A and B of Section 8.2.

Check	Result
Clean lattice, $N = 35$: T by sparse solve, spectral sum over Q , cosine sum (15) and identity (41), in both codes	14 100.80805 by every route; largest relative spread 1.5×10^{-12}
Clean lattice: mode by eigendecomposition with bound (a) and by time stepping (code A); by time stepping with bound (b) and by eigendecomposition (code B)	$t^* = 2493$ four times; the two PMFs of code B differ by at most 4×10^{-16}
Clean lattice: derived quantities	$\varrho = 0.1768$; s.d. 12 948.4; CV 0.918; median 10 145; $S(t^*) = 0.902$; $S(T) = 0.368$
One uniform placement, $p = 0.15$ ($n = 1041$)	$T = 16\,499.48678$ by solve, spectral sum and identity (spread 2.4×10^{-12}); $t^* = 3590$ by both routes; peak heights agree to 5×10^{-14}
Identity (41) against the sparse solve on every placement of the sweep	largest relative deviation 2.8×10^{-11} (10 402 configurations, sample A) and 3.5×10^{-12} (10 400, sample B)
400 placements of sample A regenerated from their seeds and solved with code B	cluster sizes, rejection counts and modes identical in 400 of 400; T agrees to 1.7×10^{-11}
Monte Carlo simulation of the walker rule, $N = 15$, 34 blocked sites, 2×10^5 walkers	code A: sample mean 2405.6 (s.e. 4.8) against exact 2402.04, $z = 0.74$; binned PMF $\chi^2 = 42.2$ on 39 degrees of freedom. Code B, another placement: 3323.6 (s.e. 6.8) against 3315.48, $z = 1.19$; $\chi^2 = 34.9$ on 39

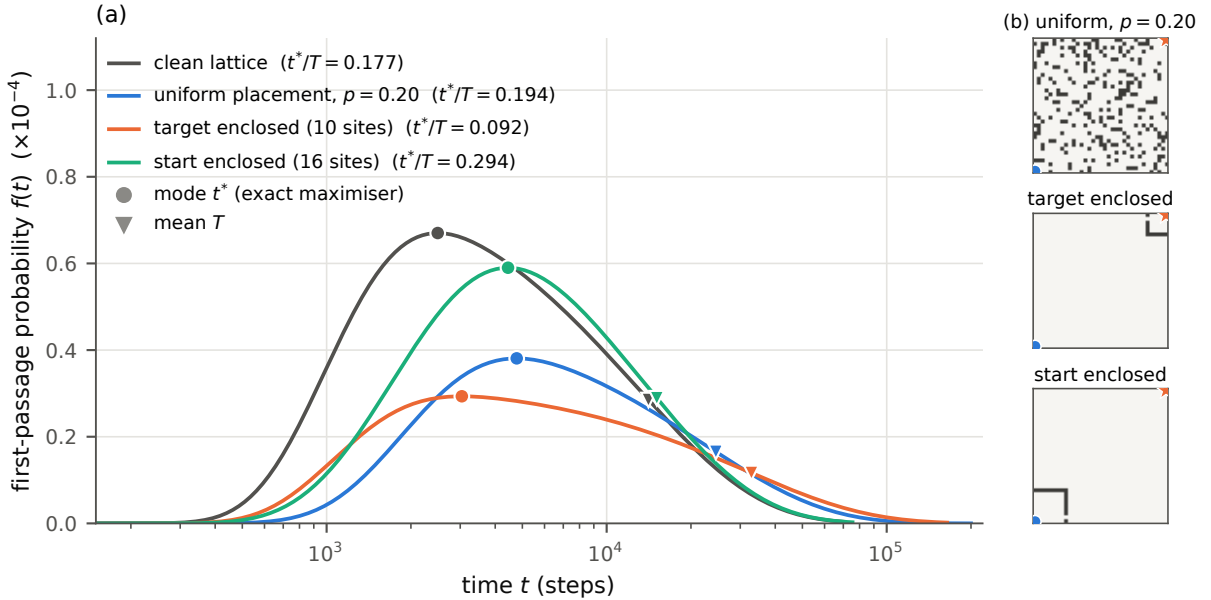


Figure S11: Exact first-passage PMFs in discrete time for $N = 35$, $q = 0.8$, corner to opposite corner (logarithmic time axis). (a) Clean lattice, one uniform placement with $p = 0.20$, and two deterministic geometries of Section 8.4: the target enclosed by 10 blocked sites and the start enclosed by 16, each with a one-site opening. Circles mark the modes (exact maximisers), triangles the means; the legend gives t^*/T . (b) The three configurations with blocked sites (dark; start: dot, bottom left; target: star, top right).

Table S8: Density sweep at $N = 35$, $q = 0.8$: samples A and B pooled (800 placements per row; $\pm$ is a 95% interval over placements). “acc.” is the fraction of drawn placements in which the target is connected to the start. Time factors are ensemble means divided by the clean values $T = 14100.8$ and $t^* = 2493$; $\Theta(p)$ is the homogenised factor (42), estimate Θ_{192} of Table S9. The interval of $\langle t^* \rangle / \langle T \rangle$ is a percentile bootstrap. Clean lattice: $t^*/T = 0.1768$, CV = 0.918.

p	acc.	$\langle T \rangle / T_{\text{clean}}$	$\langle t^* \rangle / t^*_{\text{clean}}$	$\Theta(p)$	$\langle t^* / T \rangle$	$\langle t^* \rangle / \langle T \rangle$	CV
<i>uniform placement</i>							
0.02	1.00	1.046 ± 0.006	1.046 ± 0.002	1.045	0.1775 ± 0.0006	0.1768 [0.1761, 0.1776]	0.918
0.04	1.00	1.106 ± 0.010	1.099 ± 0.003	1.096	0.1774 ± 0.0009	0.1756 [0.1744, 0.1769]	0.918
0.06	0.99	1.163 ± 0.012	1.157 ± 0.003	1.152	0.1782 ± 0.0011	0.1758 [0.1744, 0.1772]	0.918
0.08	0.98	1.221 ± 0.015	1.219 ± 0.004	1.216	0.1795 ± 0.0012	0.1765 [0.1750, 0.1782]	0.917
0.10	0.98	1.290 ± 0.017	1.294 ± 0.005	1.289	0.1810 ± 0.0014	0.1773 [0.1756, 0.1791]	0.916
0.12	0.96	1.396 ± 0.022	1.385 ± 0.006	1.372	0.1804 ± 0.0016	0.1754 [0.1733, 0.1774]	0.916
0.14	0.94	1.488 ± 0.024	1.487 ± 0.007	1.469	0.1825 ± 0.0018	0.1766 [0.1744, 0.1788]	0.915
0.16	0.89	1.569 ± 0.026	1.599 ± 0.009	1.583	0.1863 ± 0.0019	0.1802 [0.1779, 0.1825]	0.913
0.18	0.89	1.688 ± 0.030	1.739 ± 0.010	1.719	0.1891 ± 0.0021	0.1821 [0.1796, 0.1846]	0.912
0.20	0.85	1.825 ± 0.033	1.911 ± 0.013	1.877	0.1926 ± 0.0022	0.1851 [0.1825, 0.1878]	0.910
0.25	0.70	2.303 ± 0.046	2.531 ± 0.028	2.489	0.2035 ± 0.0029	0.1943 [0.1911, 0.1974]	0.905
0.30	0.50	3.075 ± 0.065	3.662 ± 0.060	3.739	0.2214 ± 0.0039	0.2105 [0.2063, 0.2148]	0.897
0.35	0.24	4.205 ± 0.093	5.643 ± 0.136	7.564	0.2456 ± 0.0046	0.2373 [0.2324, 0.2425]	0.885
<i>corner-thinned placement</i>							
0.02	1.00	1.027 ± 0.003	1.042 ± 0.001	1.045	0.1794 ± 0.0003	0.1792 [0.1788, 0.1796]	0.917
0.04	1.00	1.060 ± 0.005	1.087 ± 0.001	1.096	0.1817 ± 0.0006	0.1812 [0.1805, 0.1818]	0.916
0.06	1.00	1.092 ± 0.006	1.137 ± 0.002	1.152	0.1849 ± 0.0007	0.1841 [0.1834, 0.1849]	0.914
0.08	1.00	1.133 ± 0.008	1.195 ± 0.003	1.216	0.1877 ± 0.0009	0.1865 [0.1853, 0.1875]	0.912
0.10	1.00	1.182 ± 0.010	1.262 ± 0.003	1.289	0.1904 ± 0.0010	0.1887 [0.1875, 0.1900]	0.911
0.12	1.00	1.237 ± 0.012	1.342 ± 0.004	1.372	0.1938 ± 0.0011	0.1918 [0.1903, 0.1932]	0.909
0.14	0.99	1.300 ± 0.014	1.436 ± 0.006	1.469	0.1978 ± 0.0013	0.1952 [0.1936, 0.1967]	0.907
0.16	0.99	1.390 ± 0.017	1.540 ± 0.007	1.583	0.1996 ± 0.0016	0.1959 [0.1940, 0.1978]	0.906
0.18	0.99	1.458 ± 0.020	1.665 ± 0.008	1.719	0.2063 ± 0.0017	0.2019 [0.1997, 0.2042]	0.902
0.20	0.98	1.606 ± 0.023	1.829 ± 0.011	1.877	0.2066 ± 0.0020	0.2014 [0.1990, 0.2037]	0.902
0.25	0.94	2.040 ± 0.043	2.436 ± 0.026	2.489	0.2207 ± 0.0029	0.2111 [0.2073, 0.2149]	0.895
0.30	0.81	2.833 ± 0.066	3.625 ± 0.070	3.739	0.2378 ± 0.0041	0.2262 [0.2215, 0.2308]	0.887
0.35	0.49	4.008 ± 0.102	5.516 ± 0.140	7.564	0.2550 ± 0.0050	0.2433 [0.2381, 0.2488]	0.879

For $p \geq 0.30$ many drawn placements are rejected (acceptance 0.496 and 0.241 under uniform placement in sample A), so those rows describe an ensemble conditioned on connectivity. The tables of each sample separately are in the repository. Under corner-thinned placement the mode factor is 0.3–3.1% below Θ for $p \leq 0.2$ and the factor of the mean 1.7–15% below. At $p = 0.35$, where one uniform placement in four connects the corners, neither the mean nor the mode follows Θ (4.2 and 5.6 against 7.6).

S9.3 Effective medium

Here $(\mathcal{L}V)(\mathbf{x}) = \sum_{\mathbf{y} \sim \mathbf{x}} [V(\mathbf{x}) - V(\mathbf{y})]$ is the Laplacian of $\mathbb{Z}^2$ with unit conductances and $\mathcal{K}$ the potential kernel: $\mathcal{L}\mathcal{K} = -4\delta_{\mathbf{0}}$, $\mathcal{K}$ is invariant under the symmetries of the lattice, and $\mathcal{K}(\mathbf{x} + \mathbf{e}_1) - \mathcal{K}(\mathbf{x} - \mathbf{e}_1) \rightarrow 0$ as $|\mathbf{x}| \rightarrow \infty$. Equivalently, the resistance between two sites at distance two along an axis of the infinite lattice is $\mathcal{K}(2\mathbf{e}_1)/2 = 2 - 4/\pi$ (Cserti, 2000). In Proposition 8.2, U satisfies Kirchhoff’s law at every remaining site and $U(\mathbf{x}) + Ex_1 \rightarrow 0$ as $|\mathbf{x}| \rightarrow \infty$; at $\pm \mathbf{e}_2$ the potential is 0; and in (b) U_b^0 and U_b are the potential drops of $U^0(\mathbf{x}) = -Ex_1$ and of U (extended by any value at $\mathbf{0}$) across bond b .

Proof of Proposition 8.2. (a) Put $U(\mathbf{x}) = -Ex_1 + \frac{u}{4}[\mathcal{K}(\mathbf{x} + \mathbf{e}_1) - \mathcal{K}(\mathbf{x} - \mathbf{e}_1)]$ on all of $\mathbb{Z}^2$, with a constant u . Then $\mathcal{L}U = u(\delta_{\mathbf{e}_1} - \delta_{-\mathbf{e}_1})$, because x_1 is harmonic, and $U(\mathbf{x}) + Ex_1 \rightarrow 0$

at infinity. By the symmetries of $\mathcal{K}$, $U(\mathbf{0}) = 0$, $U(\pm\mathbf{e}_2) = 0$ and $U(-\mathbf{e}_1) = -U(\mathbf{e}_1)$, with $U(\mathbf{e}_1) = -E + \frac{u}{4}\mathcal{K}(2\mathbf{e}_1) = -E + u(1 - 2/\pi)$. Kirchhoff's law on the diluted lattice at a site $\mathbf{x} \neq \mathbf{0}$ reads $(\mathcal{L}U)(\mathbf{x}) - [U(\mathbf{x}) - U(\mathbf{0})]\mathbf{1}_{\mathbf{x} \sim \mathbf{0}} = 0$. It holds at every site that is not a neighbour of $\mathbf{0}$, and at $\pm\mathbf{e}_2$, where both terms vanish. At $\pm\mathbf{e}_1$ it reads $u = U(\mathbf{e}_1)$, that is, $u = -E + u(1 - 2/\pi)$, so $u = -\pi E/2$. For uniqueness, the difference of two solutions is harmonic on the diluted lattice, which is connected, and tends to zero at infinity; by the maximum principle on balls of increasing radius, it vanishes.

(b) Since $U^0(\mathbf{0}) = 0$ and $\sum_{\mathbf{y} \sim \mathbf{0}} U^0(\mathbf{y}) = 0$, the sum is $\sum_{\mathbf{y} \sim \mathbf{0}} [U^0(\mathbf{0}) - U^0(\mathbf{y})][U(\mathbf{0}) - U(\mathbf{y})] = \sum_{\mathbf{y} \sim \mathbf{0}} U^0(\mathbf{y})U(\mathbf{y}) = (-E)u + E(-u) = \pi E^2$, whatever value is given to $U(\mathbf{0})$. For the finite statement, let U^0 and U satisfy Kirchhoff's law at every non-boundary node before and after the removal of a set R of bonds (U is constant, with an arbitrary value, on any part cut off from the boundary), and $\mathcal{W}^0 = \sum_b (U_b^0)^2$, $\mathcal{W} = \sum_{b \notin R} U_b^2$ the dissipated powers. For any function V that vanishes on the boundary set, with drop V_b across bond b , Kirchhoff's law gives $\sum_b U_b^0 V_b = 0$ and $\sum_{b \notin R} U_b V_b = 0$. With $V = U - U^0$ these read $\sum_b U_b^0 U_b = \mathcal{W}^0$ and $\sum_{b \notin R} U_b U_b^0 = \mathcal{W}$; subtracting, $\mathcal{W}^0 - \mathcal{W} = \sum_{b \in R} U_b^0 U_b$. $\square$

Derivation. In a large sample of n_s sites under a mean field E the clean lattice dissipates $\mathcal{W}^0 = n_s E^2$ (one bond per site along the field, $\Sigma_0 = 1$) and the diluted one $\mathcal{W} = (\Sigma/\Sigma_0) n_s E^2$. If each of the pn_s blocked sites is treated as isolated in the unperturbed field, Proposition 8.2(b) gives $\mathcal{W}^0 - \mathcal{W} = pn_s \pi E^2$, the interaction between defects entering at order p^2 . An open site is cut off only if its four neighbours are blocked, so $P_\infty = 1 - p + O(p^4)$, and (42) gives the rest. The step from one defect to a density of defects is not proved here, hence the label. The law is not new. [Watson and Leath \(1974\)](#) measured the conductivity of a 137×137 conducting mesh from which sites were removed and gave an effective-medium theory with $\Sigma/\Sigma_0 = 1 - \pi p + \frac{1}{2}\pi p^2$, whose linear term is the first relation of (43). [Nieuwenhuizen et al. \(1986\)](#) calculated the diffusion coefficient and the static conductivity of exactly this walk, a random walk on the square lattice with randomly excluded sites, as an expansion in the fraction of excluded sites that is exact through the terms of order p^2 ; the coefficients are printed in Eqs (4.2) and (4.3) of [Ernst et al. \(1987\)](#); they are reproduced as (44) in the main text. As a check that does not depend on the printed numbers we evaluated the pair term of this expansion ourselves (minus one half of the sum, over all separations on a torus of N_t^2 sites, of the loss of conductivity caused by two removed sites minus twice that of one site; no random sampling): the coefficient of p^2 in Σ/Σ_0 is 1.28614 at $N_t = 256$ and 1.28588 after extrapolation in N_t^{-2} , which gives -0.85571 for the diffusivity; both are within 10^{-4} of the printed values. Numerical conductivities of diluted networks go back to [Kirkpatrick \(1973\)](#).

Three numerical checks support (43). One blocked site on a torus of $N_t \times N_t$ sites gives $(1 - \Sigma/\Sigma_0)N_t^2 = 3.064378, 3.122318, 3.136774, 3.140388, 3.141291, 3.141517$ for $N_t = 8, 16, \dots, 256$ ($\pi = 3.141593$), identically in both codes. In a square of $(2j+1)^2$ sites with the potential fixed on two opposite sides, the potential next to a removed central site and the loss of power are 0.99602, 0.99901, 0.99975, 0.99994, 0.99998 times $-\pi E/2$ and πE^2 for $j = 10, 20, 40, 80, 160$. At finite p , Σ and P_∞ were measured on tori three times (Table S9); we write Θ_{192} , Θ_{160} and Θ_{200} for the three estimates of Θ , after the side of the torus. Θ_{192} (12 placements per density) is 1.045, 1.289, 1.877 and 3.739 at $p = 0.02, 0.1, 0.2$ and 0.3 ; Θ_{160} (10 placements, second code) agrees with it within its intervals; and Θ_{200} (40 placements, second code, computed for $p = 0.1$ and 0.2 only) is the most precise: 1.2877 ± 0.0012 and 1.8783 ± 0.0057 . Beyond first order the measurements can be compared with (44), which has no free parameter. The measured conductivity is $\Sigma/\Sigma_0 = 0.9378, 0.6981$ and 0.4253 at $p = 0.02, 0.10$ and 0.20 , against 0.9377, 0.6987 and 0.4231 from the exact expansion truncated after p^2 and 0.9378, 0.7015 and 0.4345 from the effective-medium expression of [Watson and Leath \(1974\)](#). Measurement and truncated expansion differ by at most 6.2×10^{-4} at the eleven densities with $p \leq 0.18$, which is inside the 95% interval of the measurement at ten of them, and by $+0.0022 (\pm 0.0015)$ at $p = 0.20$. The measured coefficient of p^2 , $(\Sigma/\Sigma_0 - 1 + \pi p)/p^2$, averaged over $p = 0.06\text{--}0.14$ with inverse-variance

Table S9: Homogenised transport coefficients of the site-diluted square lattice with uniformly placed blocked sites. Columns 2–6: 192×192 tori, 12 placements per p ($\pm$: 95% interval; P_∞ is the fraction of sites in the largest cluster); column 6 is the estimate Θ_{192} . Θ_{160} : independent estimate from 160×160 tori, 10 placements. Θ_{200} : 200×200 tori, 40 placements.

p	Σ/Σ_0	P_∞	D_{eff}/D_0	$1 - (\pi - 1)p$	Θ_{192}	Θ_{160}	Θ_{200}
0.02	0.9378 ± 0.0002	0.9800	0.9569	0.9572	1.045	1.0450 ± 0.0003	–
0.04	0.8761 ± 0.0003	0.9600	0.9126	0.9143	1.096	1.0956 ± 0.0008	–
0.06	0.8160 ± 0.0004	0.9400	0.8681	0.8715	1.152	1.1517 ± 0.0018	–
0.08	0.7568 ± 0.0006	0.9200	0.8227	0.8287	1.216	1.2138 ± 0.0017	–
0.10	0.6981 ± 0.0009	0.8999	0.7757	0.7858	1.289	1.2883 ± 0.0033	1.2877 ± 0.0012
0.14	0.5852 ± 0.0011	0.8596	0.6808	0.7002	1.469	1.4685 ± 0.0042	–
0.20	0.4253 ± 0.0015	0.7983	0.5328	0.5717	1.877	1.883 ± 0.013	1.8783 ± 0.0057
0.25	0.2994 ± 0.0022	0.7453	0.4017	0.4646	2.489	2.490 ± 0.042	–
0.30	0.1840 ± 0.0024	0.6878	0.2675	0.3575	3.739	3.79 ± 0.09	–
0.35	0.0815 ± 0.0023	0.6166	0.1322	0.2504	7.564	7.25 ± 0.63	–

weights, is 1.26 ± 0.03 , against the exact 1.2858 and the effective-medium value $\pi/2 = 1.571$. For the diffusivity, D_{eff}/D_0 lies below the linear law by 0.0017, 0.0101 and 0.0389 at $p = 0.04$, 0.10 and 0.20, of which the term $-0.8558p^2$ supplies 0.0014, 0.0086 and 0.0342. The remainder corresponds to a third-order coefficient of about -0.9 (-0.95 ± 0.48 at $p = 0.14$, -0.87 ± 0.28 at $p = 0.18$). That is the value, -0.8558 , which the factor $1/P_\infty = 1/(1-p)$ in (42) generates from (44) when the conductivity has no third-order term; the measured third-order coefficient of the conductivity is indeed compatible with zero (-0.05 ± 0.42 and $+0.04 \pm 0.23$ at the same two densities). The third-order statements are measurements, not derived coefficients.

S9.4 Size dependence and the whole distribution

Larger lattices. For uniform placement at $p = 0.1$ and 0.2 we computed certified modes up to $N = 200$ and means up to $N = 280$ (Table S10, Fig. S12), using the most precise estimate, $\Theta_{200}(0.1) = 1.2877$ and $\Theta_{200}(0.2) = 1.8783$.

- The mode factor equals Θ to within 1.11% at $p = 0.1$ and 1.6% at $p = 0.2$ for every N from 15 to 200, and within its interval for $N \geq 70$.
- The factor of the mean divided by Θ , averaged over $N = 35$ –280 with inverse-variance weights, is 1.004 ± 0.005 at $p = 0.1$ and 0.987 ± 0.007 at $p = 0.2$.
- The shift of $\langle t^*/T \rangle$ relative to the clean lattice of the same size is between $+0.6\%$ and $+1.2\%$ for $N \geq 50$ at $p = 0.1$, with intervals of ± 1.0 to $\pm 1.8\%$. At $p = 0.2$ it falls from $+9.4 \pm 1.8\%$ at $N = 35$ to $+4.5 \pm 1.4\%$ at $N = 100$, $+2.9 \pm 1.8\%$ at $N = 140$ and $+3.9 \pm 2.7\%$ at $N = 200$: it halves but is not shown to vanish. About half of it is the Jensen gap; the ratio of ensemble means is shifted by $+5.3\%$, $+2.2\%$, $+0.5\%$ and $+2.4\%$ at $N = 35$, 100, 140 and 200.

In the smaller sample A (40–100 placements per size for $N \geq 70$; open symbols in Fig. S12(b)) both time factors agree with Θ within their intervals at these sizes; the larger sample B is needed to quantify the residual shift of the ratio at $p = 0.2$.

Whole distribution. Mean and mode are two functionals of the distribution. Comparing each placement’s survival function, in units of its own T , with the clean one in units of its T ($N = 35$, 200 uniform placements per p , $0 \leq t \leq 6T$), the Kolmogorov distance averages 0.008 at $p = 0.1$ and 0.013 at $p = 0.2$ (largest values 0.027 and 0.039). The same distance is 0.023 to 0.072 for five of the deterministic obstacle geometries of Section S9.7 and 0.008 for the most favourable phase of the period-3 array (the phase tabulated in Table S19).

Table S10: Size dependence under uniform placement ($q = 0.8$; $\pm$ is a 95% interval over placements; Θ is the estimate Θ_{200} of Table S9). K : placements with certified modes; “shift” is the relative difference between the ratio with defects and the clean ratio at the same N , for the mean of per-placement ratios and for the ratio of ensemble means (percentile-bootstrap interval). K' : placements of a separate sample for which only the mean was computed. Both samples are from the second code.

N	K	$\langle t^* \rangle / t_{\text{clean}}^*$	shift of $\langle t^* / T \rangle$ (%)	shift of $\langle t^* \rangle / \langle T \rangle$ (%)	K'	$\langle T \rangle / (\Theta T_{\text{clean}})$
$p = 0.1, \Theta = 1.2877 \pm 0.0012$						
15	200	1.283 ± 0.015	$+4.8 \pm 2.0$	$+2.3$ [+0.1, +4.7]	–	–
25	200	1.302 ± 0.011	$+2.0 \pm 1.8$	-0.6 [–2.8, +1.7]	–	–
35	400	1.294 ± 0.007	$+2.9 \pm 1.1$	$+1.1$ [–0.3, +2.3]	800	1.007 ± 0.014
50	200	1.296 ± 0.009	$+1.2 \pm 1.6$	-0.7 [–2.4, +1.1]	800	1.008 ± 0.012
70	400	1.293 ± 0.006	$+0.9 \pm 1.0$	-0.8 [–2.1, +0.4]	800	1.009 ± 0.012
100	300	1.290 ± 0.005	$+1.1 \pm 1.0$	-0.1 [–1.2, +1.0]	600	0.990 ± 0.010
140	200	1.288 ± 0.006	$+1.1 \pm 1.1$	$+0.1$ [–1.2, +1.3]	400	1.008 ± 0.014
200	60	1.290 ± 0.009	$+0.6 \pm 1.8$	-0.1 [–2.0, +1.6]	300	1.017 ± 0.018
280	–	–	–	–	200	1.006 ± 0.018
$p = 0.2, \Theta = 1.8783 \pm 0.0057$						
15	200	1.908 ± 0.054	$+9.8 \pm 3.7$	$+3.7$ [–0.4, +8.1]	–	–
25	200	1.900 ± 0.033	$+8.3 \pm 2.9$	$+3.1$ [–0.2, +6.5]	–	–
35	400	1.905 ± 0.019	$+9.4 \pm 1.8$	$+5.3$ [+3.2, +7.3]	800	0.986 ± 0.018
50	200	1.899 ± 0.021	$+8.1 \pm 2.3$	$+4.3$ [+1.6, +7.2]	800	0.977 ± 0.016
70	400	1.888 ± 0.013	$+5.4 \pm 1.5$	$+1.9$ [+0.2, +3.8]	800	0.988 ± 0.015
100	300	1.886 ± 0.012	$+4.5 \pm 1.4$	$+2.2$ [+0.6, +3.7]	600	0.989 ± 0.016
140	200	1.890 ± 0.014	$+2.9 \pm 1.8$	$+0.5$ [–1.3, +2.6]	400	0.980 ± 0.017
200	60	1.868 ± 0.019	$+3.9 \pm 2.7$	$+2.4$ [–0.6, +5.2]	300	1.001 ± 0.021
280	–	–	–	–	200	0.994 ± 0.022

In summary, for uniformly random blocked sites time dilation by $\Theta(p)$ is accurate at $p = 0.1$ and is a 1–3% approximation at $p = 0.2$ at the sizes computed, where $\langle t^* / T \rangle$ is still 3–4.5% above its clean value at $N = 100$ –200. A slow approach is to be expected: in (41) the constant that accompanies the logarithm of N in G_{aa} is set by the immediate surroundings of the target and need not scale with Σ_0 / Σ . The limit itself is not established.

S9.5 What controls the mean and the mode

Placements with one of the two neighbours of the target blocked have a mean larger by 20–31% and a mode larger by 6–9% than those of the same ensemble with both open (two placement laws, $p = 0.1$ and 0.2, sample A; 21–33% and 8–10% in sample B). Two other explanations are not supported. The shortest open path between the corners keeps its clean length of 68 steps in every placement up to $p = 0.1$ and has mean length 68.06 at $p = 0.2$ (68.04 in sample B): the direct route is not lengthened. Nor do dead ends set the modal time: the relaxation time of the reflecting cluster, whose slowest mode localises on weakly attached pockets, grows by a factor 2.21 at $p = 0.2$ and 6.6 at $p = 0.3$ (2.18 ± 0.04 and 6.3 ± 0.5 in sample B), while the mode grows by 1.92 and 3.70 (sample A), close to $\Theta = 1.88$ and 3.74.

S9.6 Single-defect susceptibilities

Let $\chi_A(\mathbf{x}) = A_{\mathbf{x}} / A_{\text{clean}} - 1$ be the relative change of $A \in \{T, t^*\}$ when the single site $\mathbf{x} \neq \mathbf{o}, \mathbf{a}$ is blocked (Fig. S14). Because one defect shifts the mode by a few steps at most, the position of the maximum is refined to a fraction of a step by a parabola through $f(t^* - 1)$, $f(t^*)$, $f(t^* + 1)$.

Blocking a single site shortens the mean first-passage time for 67.6% of the sites (60.1% at

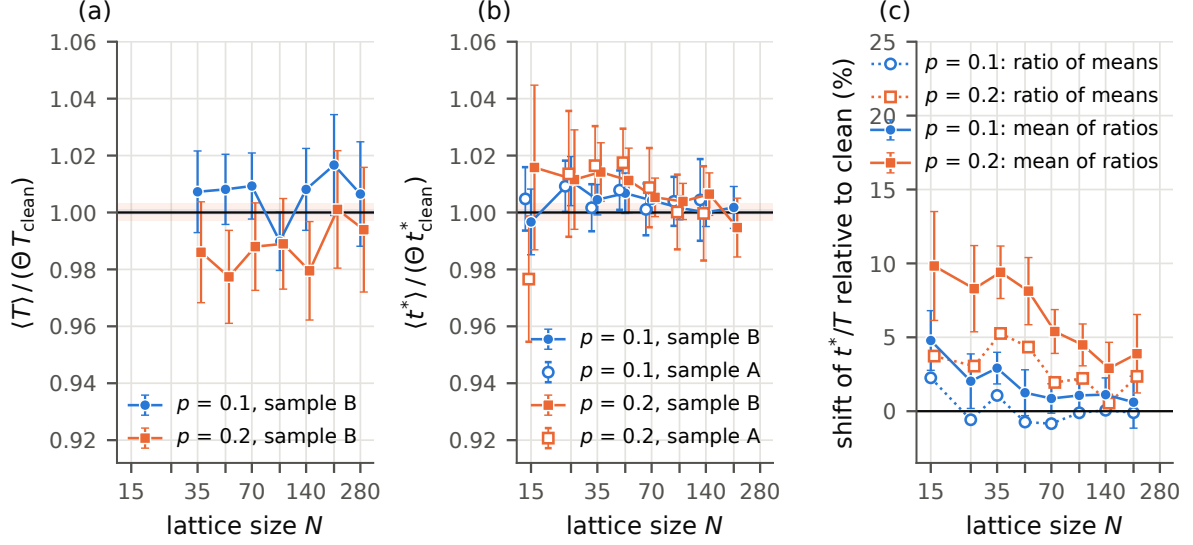


Figure S12: Size dependence of time dilation under uniform placement, $q = 0.8$, corner to opposite corner, discrete time; circles $p = 0.1$, squares $p = 0.2$ (displaced horizontally for clarity); bars are 95% intervals over placements, and the shaded bands the 95% intervals of Θ_{200} . (a) Ensemble mean of T divided by ΘT_{clean} (200–800 placements per point, sample B). (b) Ensemble mean of t^* divided by $\Theta t^*_{\text{clean}}$: filled symbols sample B (60–400 placements), open symbols sample A (40–200 placements). (c) Shift of t^*/T relative to the clean lattice of the same N (sample B): mean of per-placement ratios (filled, solid) and ratio of ensemble means (open, dotted).

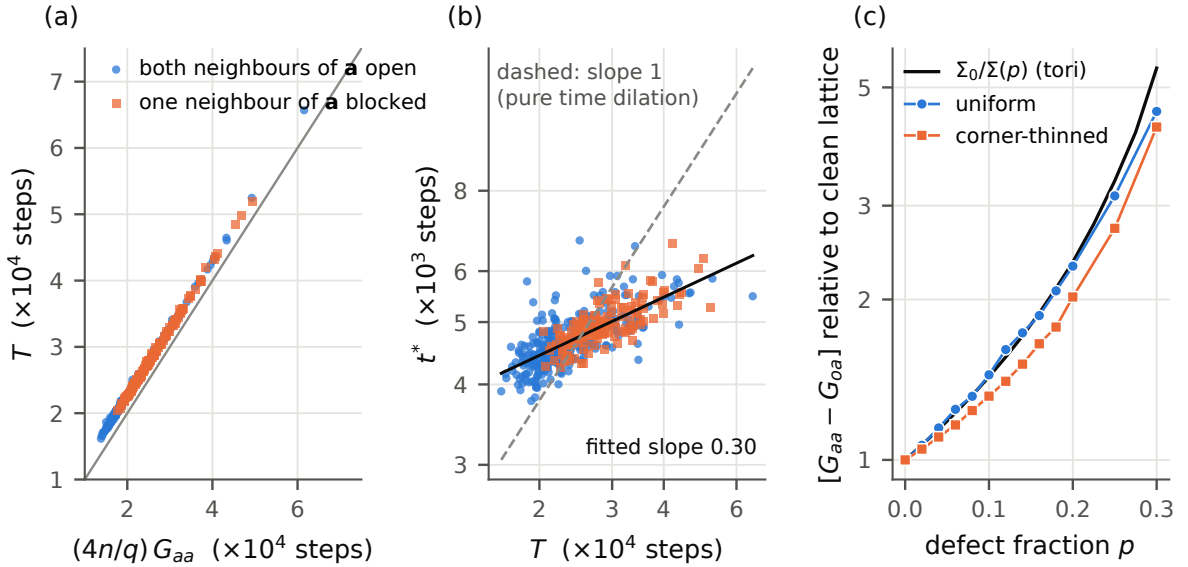


Figure S13: What controls the mean and the mode ($N = 35$, $q = 0.8$, corner to opposite corner, discrete time). (a) Exact mean first-passage time against $(4n/q)G_{aa}$ for the 400 uniform placements of sample A at $p = 0.2$; the line is the diagonal, and the offset from it is the cross term $-(4n/q)G_{0a}$. (b) Mode against mean for the same placements, logarithmic axes; solid line: least-squares fit (slope 0.30); dashed line: slope 1, the behaviour under a pure dilation of time. (c) Ensemble mean of $G_{aa} - G_{0a}$ relative to the clean lattice for the uniform and the corner-thinned placement (400 placements per point), and the homogenised value $\Sigma_0/\Sigma(p)$ of Section 8.3.

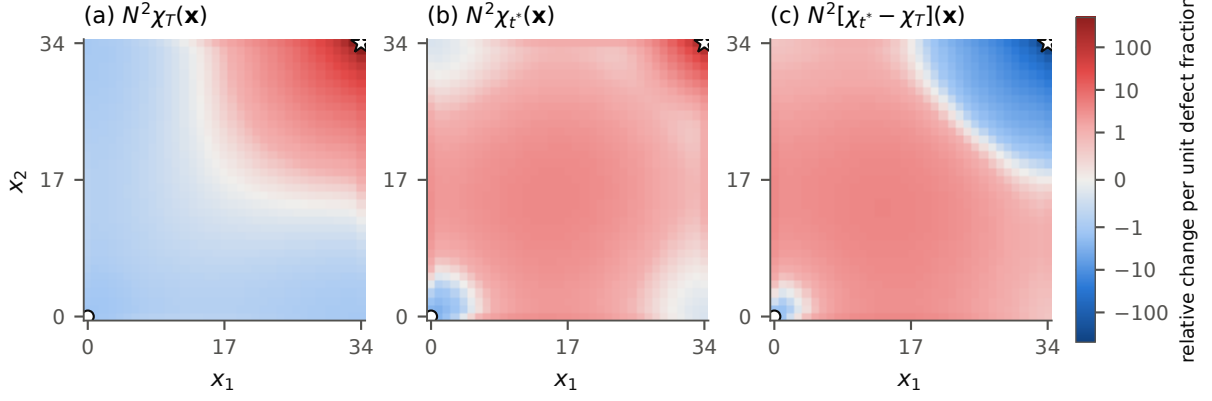


Figure S14: Single-defect susceptibilities at $N = 35$, $q = 0.8$ (discrete time): each site is blocked alone and the exact mean and mode are recomputed. (a) $N^2\chi_T(\mathbf{x})$, (b) $N^2\chi_{t^*}(\mathbf{x})$, (c) their difference. Circle: start; star: target. The colour scale is linear between -1 and 1 and logarithmic beyond. Blue sites shorten the first-passage statistic when blocked.

Table S11: First-order slopes (S9.1) at $N = 35$ and measured secants $(\langle A \rangle / A_{\text{clean}} - 1)/p$ at $p = 25/1225$ (400 placements per sample, $\pm 95\%$ confidence interval). The secants contain the second-order terms, which the estimate omits.

placement	statistic	first order	secant, sample A	secant, sample B
uniform	mean	2.271	2.12 ± 0.41	2.37 ± 0.42
uniform	mode	2.183	2.21 ± 0.12	2.30 ± 0.12
corner-thinned	mean	1.261	1.30 ± 0.17	1.39 ± 0.23
corner-thinned	mode	1.923	2.04 ± 0.05	2.04 ± 0.07

$N = 15$, 65.3% at $N = 25$). Far from the target the only effect is the lost volume, $\chi_T \approx -1/n = -8.2 \times 10^{-4}$ (minimum -1.04×10^{-3}); the positive values are concentrated around the target, with a maximum of $+38.2\%$ for either of its neighbours. Of $\sum_{\mathbf{x}} \chi_T(\mathbf{x})$, 64%, 88% and 105% come from sites within Euclidean distance 3, 6 and 10 of the target. The mode behaves differently: only 5.4% of the sites lower it, its largest response is $+10.5\%$ (the same two sites), and the corresponding shares of $\sum_{\mathbf{x}} \chi_{t^*}(\mathbf{x})$ are 19%, 25% and 28%: its response is spread over the whole region between start and target.

For a placement law with site probabilities $w(\mathbf{x})$, $\sum_{\mathbf{x}} w(\mathbf{x}) = 1$, adding the effects of the pN^2 single defects gives the first-order estimate

$$\left. \frac{d \ln \langle A \rangle}{dp} \right|_{p=0} = N^2 \sum_{\mathbf{x}} w(\mathbf{x}) \chi_A(\mathbf{x}). \quad (\text{S9.1})$$

This is first-order perturbation theory, not a theorem; Table S11 compares it with measured secants. For uniform placement the two slopes nearly coincide, so the ratio hardly moves at first order ($d \ln(t^*/T)/dp = -0.09$), and both approach the bulk value $\pi - 1 = 2.142$ of (43) from above: (2.291, 2.227), (2.278, 2.197) and (2.271, 2.183) for $N = 15, 25$ and 35 .

The same estimate explains why the ratio rises under corner-thinned placement (Section 8.2). That law thins the defects inside the 4×4 boxes at the start and target corners, to 55% of the uniform probability on average. The box at the target holds 71% of $\sum \chi_T$ but only 21% of $\sum \chi_{t^*}$ (the box at the start less than 1% of either, in absolute value). Thinning lowers the slope of the mean from 2.27 to 1.26 and that of the mode only from 2.18 to 1.92, so the ratio must

Table S12: Selected deterministic geometries ($N = 35$, $q = 0.8$; exact values). A wall is a full column of blocked sites with a gap; a serpentine consists of equally spaced walls with three-site gaps on alternating sides; an enclosure is an L-shaped wall at Chebyshev distance R from a corner with a one-site door, the second site from the lattice edge. CV is the coefficient of variation of the first-passage time.

geometry	blocked	T factor	t^* factor	t^*/T	CV
clean lattice	0	1	1	0.1768	0.918
mid-lattice wall, one-site gap	34	1.760	3.139	0.3153	0.855
serpentine, three walls	96	3.537	6.773	0.3385	0.820
wall near the target, three-site gap	32	2.284	1.741	0.1348	0.949
enclosure of the target, $R = 5$	10	2.332	1.219	0.0925	0.963
enclosure of the start, $R = 5$	10	1.018	1.248	0.2168	0.895
solid 12×12 block at the centre	144	0.954	1.233	0.2284	0.890
solid 6×6 block near the start	36	0.971	1.002	0.1825	0.913

rise: $d \ln(t^*/T)/dp = +0.66$, that is $d(t^*/T)/dp = 0.177 \times 0.66 = +0.12$ at $p = 0$. The rise is a consequence of where the placement law puts the defects, not of their density.

S9.7 Localised placements, deterministic geometries and periodic arrays

Forty blocked sites are 3.3% of the lattice. Near the target they multiply the mean by 3.37 ± 0.13 (3.43 ± 0.13 in sample B) and the mode by 1.36, and the ratio falls to 0.0752 ± 0.0022 (0.0741 ± 0.0021), less than half its clean value. Near the start they leave the mean unchanged ($\times 0.999 \pm 0.003$; $\times 1.002 \pm 0.004$), delay the mode by a third, and raise the ratio to 0.2329 ± 0.0040 (0.2368 ± 0.0048). In the centre the ratio is 0.2015; anywhere on the lattice, 0.1771. With ten blocked sites near the start the mean is *below* its clean value in all 200 placements of either sample. Hence t^*/T is not a function of the defect density. It decreases when the defects surround the target, increases when they lie around the start or between start and target, and, when they are spread uniformly, stays within a few per cent of its clean value at low density ($p \leq 0.14$) but rises by 4.7% to 8.9%, depending on the estimator, at $p = 0.2$ (Section S9.2).

Table S12 lists a selection of the 28 deterministic geometries of Table S19 and Fig. 4(a); all were computed twice with separately written programs (identical modes, means equal to 2×10^{-11}).

Walls and corridors push the law towards the one-dimensional shape: the five serpentes have $t^*/T = 0.316$ – 0.339 and a coefficient of variation of 0.820–0.847, against $1/3$ and 0.816 for the clean chain (Section 5); such obstacles lengthen the transit more than the capture. Enclosing the target does the opposite and makes the law more nearly exponential (ratio 0.0925, coefficient of variation 0.963 for ten blocked sites), while the same ten sites around the start change the mean by 1.8%. A solid 12×12 block in the centre *lowers* the mean by 4.6% (13 459 against 14 101 steps) and delays the mode by 23%: it removes 11.8% of the volume and adds only 8.2% to $G_{\text{aa}} - G_{\text{oa}}$.

For a periodic array with one blocked site per $A \times A$ cell the response at $N = 35$ depends on the phase of the array relative to the target. For $A = 3$ the mean factor ranges from 1.08 to 1.47 over the nine phases (average 1.23) and the mode factor from 1.18 to 1.32 (average 1.24); the phases that block a neighbour of the target give 1.47, more than the average 1.30 of uniformly random placements at $p = 0.1$. The periodic rows of Table S19 are single phases: for $A = 3$ and $A = 2$ the phase with the smallest mean factor (1.08 and 1.25; ranges 1.08–1.47 and 1.25–1.54), for $A = 4$ the twelfth smallest of the sixteen (1.16; range 1.03–1.43, average 1.14). In the bulk the comparison is cleaner. The unit-cell problem of the infinite array, solved in rational arithmetic, gives $\Sigma/\Sigma_0 = 1/2$, $5/7$ and $14/17$ for $A = 2$, 3 and 4, so that by (42) the time factors are $3/2$,

$56/45 = 1.244$ and $255/224 = 1.138$, against about 2.49, 1.33 and 1.16 for random placement at the same densities $1/4$, $1/9$ and $1/16$ (interpolated from the values behind Table S9). A periodic array is thus a much weaker obstruction than a random one at $p = 1/4$, but only 7% weaker at $p = 1/9$ and 2% at $p = 1/16$.

S9.8 A two-time-scale heuristic

Why does an obstacle near the target move the mean so much more than the mode? The following argument is a heuristic: it accounts for the sign and the approximate size of the effect and has no controlled error.

Beyond the transit time the first-passage probability is dominated by the two slowest terms $j = 0, 1$ of the spectral sum (6). In the time constants $T_{\text{cap}} = -1/\ln(1 - q\nu_0)$ of the slowest term (the capture time) and $\tau_1 = -1/\ln(1 - q\nu_1)$ of the next,

$$f(t) \approx q b_0 e^{-t/T_{\text{cap}}} [1 - c e^{-t/\tau}], \quad c = -\frac{b_1}{b_0} > 0, \quad \frac{1}{\tau} = \frac{1}{\tau_1} - \frac{1}{T_{\text{cap}}}, \quad (\text{S9.2})$$

where a shift of t by one step is neglected. Its maximiser is the discrete-time form of the two-exponential maximiser $\ln(-b_1\nu_1/b_0\nu_0)/(\nu_1 - \nu_0)$ of Section 6.2, since $1/\tau$ corresponds to $q(\nu_1 - \nu_0)$ and $c(1 + T_{\text{cap}}/\tau)$ to $-b_1\nu_1/(b_0\nu_0)$:

$$t^* \approx \tau \ln \left[c \left(1 + \frac{T_{\text{cap}}}{\tau} \right) \right], \quad \left. \frac{\partial \ln t^*}{\partial \ln T_{\text{cap}}} \right|_{\tau, c} = \frac{\tau}{t^*} \frac{T_{\text{cap}}/\tau}{1 + T_{\text{cap}}/\tau}. \quad (\text{S9.3})$$

What is new here is only the question asked of it: how the maximiser moves when an obstacle changes T_{cap} . On the clean lattice $T_{\text{cap}} = 12935$ steps, $\tau_1 = 518.0$ steps, hence $\tau = 540$, and $c = 4.15$; the two-term sum peaks at $t = 2506$ and three terms give the exact mode 2493 (Fig. S15(a)). The rate ν_1 is the one that (37) places at $\mu_1(1 + W/X)$, a time constant of 528 steps. It must not be confused with the relaxation time of the reflecting lattice, 621 steps: of the two degenerate slowest modes, the combination that is odd under exchange of the coordinates vanishes on the diagonal and has zero weight in the corner-to-corner law (10^{-31} against 8.5×10^{-5} numerically).

A pure dilation multiplies T_{cap} and τ by the same factor and leaves t^*/T unchanged. A perturbation that raises only the capture time moves the mode sub-linearly, with the exponent of (S9.3), 0.21 on the clean lattice when τ and c are held fixed. If instead τ_1 and c are held fixed, τ follows T_{cap} ($d\tau/dT_{\text{cap}} = -\tau^2/T_{\text{cap}}^2$) and the exponent is 0.17. The exact elasticities $\ln(t^* \text{ factor})/\ln(T \text{ factor})$ of obstacles that enclose the target or block one of its neighbours are somewhat larger: 0.310 for one blocked neighbour of the target, 0.255, 0.234 and 0.269 for enclosures with $R = 3, 5$ and 8, 0.241 for $R = 5$ with a three-site door (Fig. S15(b)), and 0.247–0.253 for the near-target ensembles of Table 7 in both samples. With τ and c frozen and T_{cap} multiplied by its exact factor, (S9.3) gives mode factors of 1.072, 1.170, 1.188, 1.202 and 1.102 for these five deterministic cases, against the exact 1.105, 1.215, 1.219, 1.279 and 1.116, that is 69–88% of the logarithmic shift; the rest arises because τ_1 and c change as well (τ_1 rises from 518 to 543 steps for one blocked neighbour). The closed-form approximation (34), evaluated with the clean-lattice μ_1 , A , W and the exact mean of the perturbed lattice, lets ν_1 move with the mean and errs on the other side: 1.107, 1.247, 1.272, 1.294 and 1.150. The heuristic fails for the sixth deterministic case examined, the wall near the target of Table S12 (mean $\times 2.284$, mode $\times 1.741$): that obstacle also lengthens the transit, τ_1 rises from 518 to 1163 steps, the exact elasticity is 0.67, and the frozen- τ formula (1.180) accounts for only 30% of the logarithmic shift of the mode.

In short, a change of the capture time alone moves the mode with an exponent of about $1/4$ at $N = 35$, so that t^*/T falls roughly as $T^{-3/4}$ when defects surround the target without obstructing the transit ($0.1768 \times 3.37^{-3/4} = 0.071$ against the measured 0.075 for forty sites); an

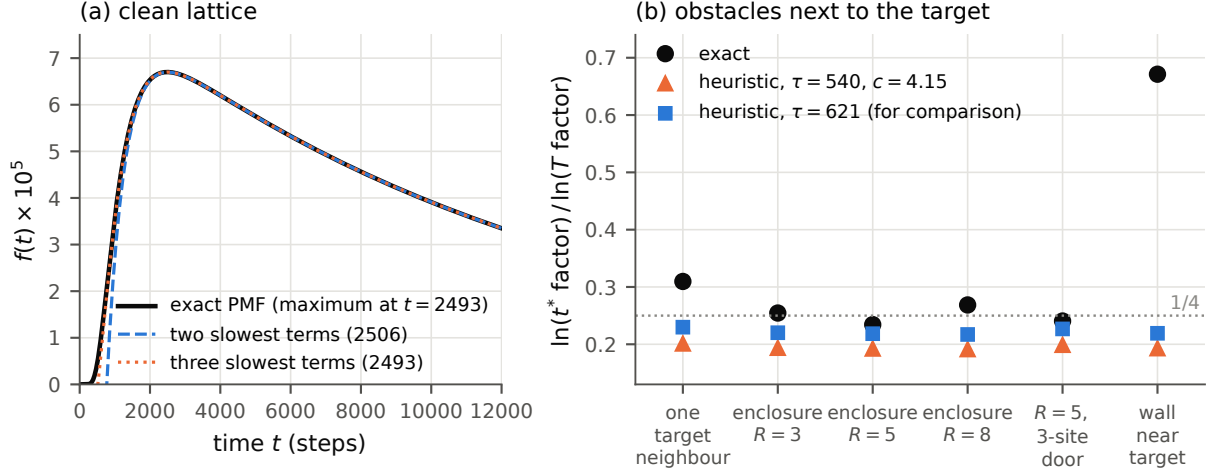


Figure S15: Two time scales ($N = 35$, $q = 0.8$, corner to opposite corner, discrete time). (a) Exact first-passage probability of the clean lattice and its truncations to the two and three slowest terms with non-zero weight (the legend gives the position of each maximum; the truncations are drawn only where they are on scale). (b) Elasticity of the mode with respect to the mean for the six deterministic obstacles examined: exact values (circles) and the heuristic (S9.3) with τ and c frozen, either at the values taken from the spectrum ($\tau = 540$, $c = 4.15$; triangles) or with τ set to the relaxation time 621 of the reflecting lattice and c fitted to the clean mode (squares). The second choice is shown for comparison only: that mode has zero weight in the corner-to-corner law. The heuristic fails for the wall near the target, which also lengthens the transit.

obstacle that also changes τ_1 , such as a wall, is outside this estimate. The fluctuations at fixed density show the same behaviour: across the 400 uniform placements at $p = 0.2$ the regression of $\ln t^*$ on $\ln T$ has slope 0.30 (sample A; 0.31 in sample B; Fig. S13(b)). The exponent is not universal: by (S9.3) it is close to $1/\ln[c(1 + T_{\text{cap}}/\tau)]$, which decreases slowly as N grows.

S9.9 A one-parameter family of shapes

Over all 13 631 exactly solved configurations of sample A (10 402 random placements of the density sweep, 3200 localised placements, the clean lattice and the 28 deterministic geometries) the coefficient of variation is nearly a function of $\varrho = t^*/T$ alone (Fig. S16): the Pearson correlation is -0.981 , and the quadratic $\text{CV} = 1.0073 - 0.4564 \varrho - 0.2461 \varrho^2$ fits with a root-mean-square residual of 0.0045. The 67 configurations (0.5%) with a residual above 0.02, the largest being 0.235, are all random placements with $p \geq 0.25$, towards the percolation threshold; over the whole sample ϱ ranges from 0.047 to 0.469 and the coefficient of variation from 0.729 to 1.125. The 13 629 configurations of sample B give a correlation of -0.983 and, with their own quadratic, a residual of 0.0041 (largest 0.14). Clean lattices with $N = 10$ –200 lie on the same curve (residuals between -0.0033 and $+0.0004$); the clean one-dimensional chain, $(1/3, 0.8165)$, lies 0.011 below it, at the end of the trend rather than on the curve. We record this as an empirical regularity of the model, without a theory: size, defect density and defect placement appear to act mainly by moving the law along one line between a transit-limited shape and a capture-limited, nearly exponential one.

S9.10 Other defect types

One dimension: permeable bonds. A blocked site between start and target disconnects them, so the inert model is empty in one dimension. Its analogue is a bond that is crossed with

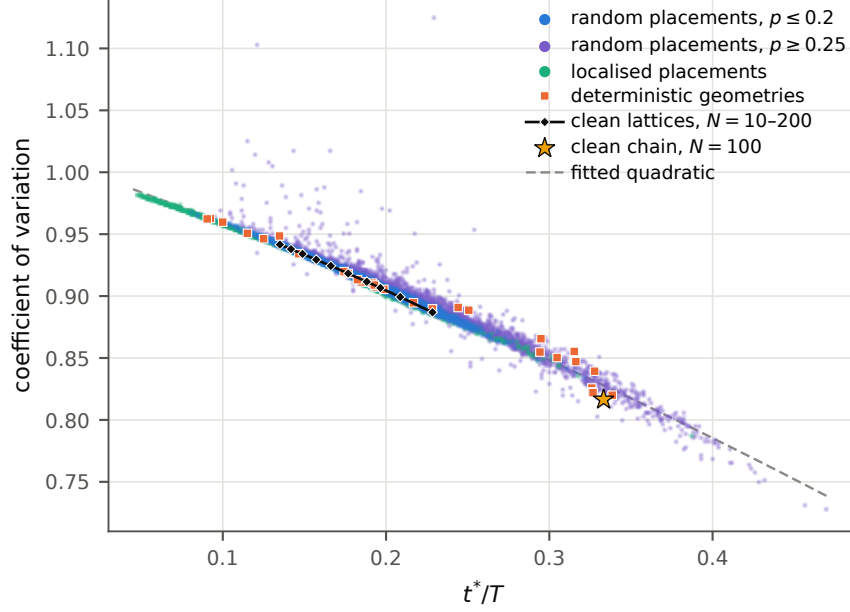


Figure S16: Coefficient of variation against t^*/T for all 13 631 exactly solved configurations of sample A ($N = 35$, $q = 0.8$, corner to opposite corner, discrete time; every configuration lies inside the frame): random placements of the density sweep (both placement laws; $p \leq 0.2$ and $p \geq 0.25$ in different colours), localised placements, deterministic geometries, the clean lattices with $N = 10$ – 200 , and the clean one-dimensional chain with $N = 100$. Dashed: the quadratic of the text.

a reduced probability.

Proposition S9.1 (Chain with barriers). *On $\{0, \dots, N-1\}$ let the walker cross the bond between k and $k+1$ with probability $w_k > 0$ per step in either direction ($k = 0, \dots, N-2$) and stay otherwise, with a reflecting end at 0 and $w_{k-1} + w_k \leq 1$ for $1 \leq k \leq N-2$. Then*

$$T_{0 \rightarrow N-1} = \sum_{k=0}^{N-2} \frac{k+1}{w_k}. \quad (\text{S9.4})$$

Proof. The transition matrix is symmetric and irreducible, so all mean hitting times are finite (Proposition 2.4). Let h_k be the mean time to reach $k+1$ from k . From site 0 the walker moves to 1 with probability w_0 per step and stays otherwise, so $h_0 = 1/w_0$. For $k \geq 1$ condition on the first step: with probability w_k the walker arrives; with probability w_{k-1} it moves to $k-1$, needs h_{k-1} steps on average to return to k and then, by the strong Markov property, h_k more; otherwise it stays at k . Hence $h_k = 1 + w_{k-1}(h_{k-1} + h_k) + (1 - w_{k-1} - w_k)h_k$, that is $w_k h_k = 1 + w_{k-1} h_{k-1}$, and by induction $w_k h_k = k+1$. A nearest-neighbour walk started at 0 reaches $N-1$ by passing from k to $k+1$ for $k = 0, \dots, N-2$ in turn, so $T_{0 \rightarrow N-1} = \sum_k h_k$. $\square$

This is a case of the classical formula for passage times of birth-and-death chains (see, e.g., Levin et al., 2009). For a single barrier it is Eq. (12) of Sarvahrman and Giuggioli (2023): a barrier between start and target raises the mean linearly in its distance from the reflecting end, whereas a barrier behind the start leaves the mean unchanged while it changes the mode and the tail, the ‘disorder indifference’ of these authors (with the start at the reflecting end, as here, every barrier lies between start and target). With $w_k = q/2$ it returns $T = N(N-1)/q$ (16); with barriers, $w_k = \kappa_k q/2$ and $T = (2/q) \sum_k (k+1)/\kappa_k$. In the terms of Section 8.4, $1/w_k$ is the resistance of bond k and $k+1$ the number of sites upstream of it: a barrier costs more the nearer it is to the target. For independent, identically distributed κ_k the disorder average

is $\langle T \rangle = T_{\text{clean}} \langle 1/\kappa \rangle$ exactly, for every N . For $N = 100$ and $q = 0.8$ (clean values $T = 12\,375$, $t^* = 4124$, ratio 0.3333) one barrier with $\kappa = 0.1$ moves t^*/T between 0.313 (bond 77) and 0.353 (bond 21); when each bond is a barrier with $\kappa = 0.2$ with probability p , the ensemble mean of the ratio stays between 0.3317 and 0.3336 for $p = 0.05$ – 0.5 in both samples (400 chains per p , confidence intervals ± 0.0010 to ± 0.0015) (Fig. S17(a)). Uniformly random barriers in one dimension are again a dilation of time.

Two dimensions: permeable obstacles. Let every bond that touches a defect site be crossed with probability $\kappa q/4$ per step ($\kappa = 0$: inert site; $\kappa = 1$: clean lattice), refused attempts being cancelled as before; an analytic framework for lattice walks with inert heterogeneities is given by Sarvaharman and Giuggioli (2023). At $p = 0.1$ (100 uniform placements per sample) and $\kappa = 0.01, 0.1, 0.3, 0.6$ the ratio is $0.1797 \pm 0.0041, 0.1788 \pm 0.0031, 0.1786 \pm 0.0019, 0.1778 \pm 0.0009$ in sample A and $0.1824 \pm 0.0038, 0.1811 \pm 0.0028, 0.1800 \pm 0.0017, 0.1785 \pm 0.0008$ in sample B: within 1.6% and 3.2% of the clean value, an excess of the size found for inert sites at this density, while mean and mode move together (Fig. S17(b)). The limit $\kappa \rightarrow 0^+$ is singular. A site that can be entered at all carries its full share of the uniform stationary distribution, so that n in (41) jumps from $(1-p)N^2$ to N^2 : at $\kappa = 10^{-3}$ the mean is $20\,656 \pm 797$, against $18\,616/0.9 = 20\,685$ from the inert mean (sample B: $20\,113 \pm 825$ against $18\,121/0.9 = 20\,135$). The mode, reached long before the slow sites equilibrate, stays near its inert value (3292 ± 46 against 3245 ± 40), and the ratio drops to 0.1628 ± 0.0037 (0.1648 ± 0.0034 in sample B).

Partial traps. A partial trap removes a walker standing on a defect site with probability Δ per step ($\Delta \leq 1 - q$); it is obtained by subtracting Δ from the diagonal element of the transition matrix at every defect site. The first-passage law is then defective. At $p = 0.1$ (100 placements per sample) and $\Delta = 10^{-5}, 10^{-4}, 10^{-3}, 10^{-2}$ the probability P_{hit} of ever reaching the target is $0.9861, 0.8758, 0.3912, 0.0277$ (sample B: $0.9861, 0.8756, 0.3907, 0.0277$); killing at the uniform rate $p\Delta$ on the clean lattice, for which this probability is $\tilde{F}(1 - p\Delta)$, gives $0.9861, 0.8752, 0.3887, 0.0256$. The mean conditional on arrival falls ($13\,936, 12\,621, 6797, 1896$ steps), the mode moves less ($2486, 2429, 2060, 1210$), and their ratio rises: $0.178, 0.192, 0.303, 0.638$ ($0.178, 0.192, 0.303, 0.639$ in sample B; Fig. S17(c)). Traps remove the long trajectories and act in the opposite sense to obstacles.

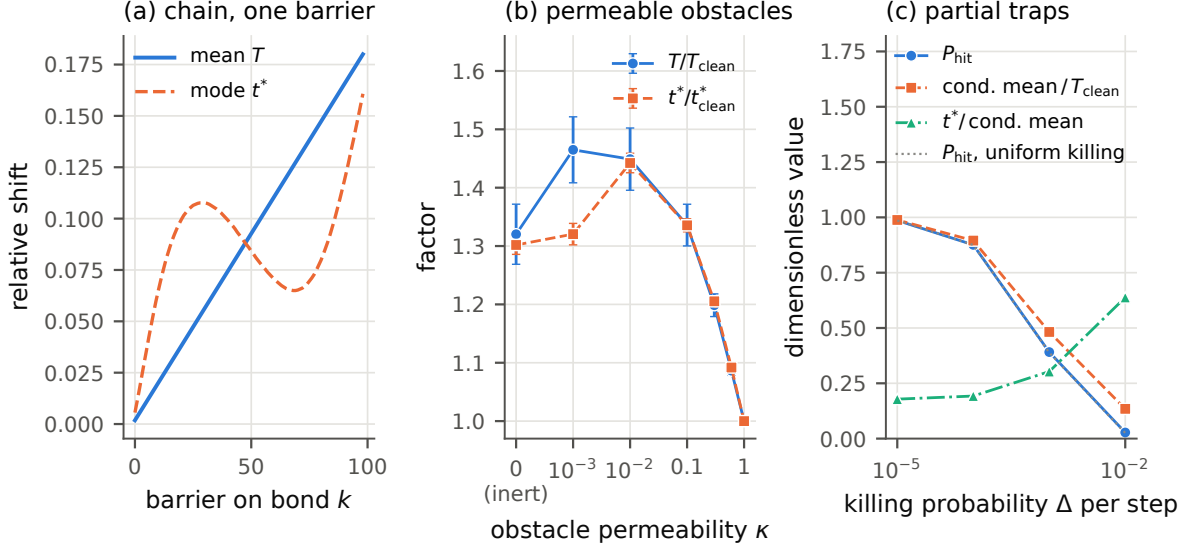


Figure S17: Other defect types (discrete time, $q = 0.8$; sample A). (a) Chain with $N = 100$ and one barrier of permeability $\kappa = 0.1$ on bond k : relative shift of the mean, from (S9.4), and of the mode. (b) $N = 35$, corner to opposite corner, permeable obstacles on 10% of the sites: mean and mode relative to the clean lattice against κ (100 placements; bars: 95% confidence intervals). (c) The same placements with partial traps of killing probability Δ : probability P_{hit} of reaching the target, the same under uniform killing at rate $p\Delta$ (dotted), conditional mean relative to the clean mean, and mode divided by the conditional mean.

S10 Extended tables

This section gives the full tables behind Sections 3–8 of the main text. The numerical rows are copied from the result files by a script, not retyped; only the typography differs from the sources. Unless stated otherwise, times are in steps, $q = 0.8$, and the walk goes from one corner to the opposite corner (CC).

S10.1 Exact modes and means

All rows of Tables S13–S15 up to $N = 10\,240$ are certified global maximisers (Proposition 3.1; for $N = 10\,240$ see Section S2.3). The last two rows of Table S13 are the local maxima of (29) nearest the continuum prediction, located in double precision and confirmed in 50-digit arithmetic; they are the global maximisers by Theorem 7.13(c) (computer-assisted). The ratios of the three large- N rows, 0.33328426, 0.33328427 and 0.33328427, agree with the limit (31) to seven decimals.

Table S13: One dimension, reflecting end to absorbing end, $q = 0.8$, discrete time: exact mode t^* and mean $T = N(N-1)/q$ (15 of the 33 computed sizes, and three larger sizes).

N	t^*	T	t^*/T
5	8	25	0.32000
9	29	90	0.32222
13	64	195	0.32821
17	113	340	0.33235
21	174	525	0.33143
25	249	750	0.33200
29	338	1015	0.33300
35	495	1487.5	0.33277

Table S13 (continued)

N	t^*	T	t^*/T
50	1020	3062.5	0.33306
70	2012	6037.5	0.33325
100	4124	12 375	0.33325
140	8107	24 325	0.33328
200	16 580	49 750	0.33327
400	66 490	199 500	0.33328
1000	416 188	1.24875×10^6	0.33328
10 240	43 679 969	131 059 200	0.33328
100 000	4 166 011 658	12 499 875 000	0.33328
1 000 000	416 604 915 263	1 249 998 750 000	0.33328

Table S14: Two dimensions, CC, $q = 0.8$, discrete time: exact mode and mean (15 of the 31 computed sizes). Means are printed to six significant figures.

N	t^*	T	t^*/T
5	36	133.523	0.2696
9	136	582.819	0.2333
13	304	1413.12	0.2151
17	541	2662.84	0.2032
21	849	4359.64	0.1947
25	1228	6525.22	0.1882
29	1679	9177.43	0.1829
35	2493	14 100.8	0.1768
50	5253	31 614.8	0.1662
70	10 570	67 212.4	0.1573
100	22 110	148 521	0.1489
140	44 248	312 092	0.1418
200	92 116	682 335	0.1350
301	212 975	1.6634×10^6	0.1280
401	383 013	3.09907×10^6	0.1236

Table S15: Three dimensions, CC, $q = 0.8$, discrete time: exact mode and mean (all computed sizes). Means are printed to six significant figures.

N	t^*	T	t^*/T
5	85	555.333	0.1531
7	187	1616.08	0.1157
9	332	3545.23	0.0936
11	523	6602.02	0.0792
13	760	11 045.7	0.0688
15	1044	17 135.5	0.0609
17	1376	25 130.6	0.0548
19	1757	35 290.2	0.0498
25	3202	81 348.9	0.0394
30	4759	141 444	0.0336
35	6644	225 599	0.0295
40	8863	337 864	0.0262
50	14 323	662 926	0.0216
60	21 175	1.14903×10^6	0.0184
70	29 447	1.82859×10^6	0.0161
80	39 161	2.734×10^6	0.0143
100	62 998	5.35199×10^6	0.0118

S10.2 Scaling fits

Each law is fitted by least squares to $\ln t^*$ on $10 \leq N \leq 200$ ($10 \leq N \leq 100$ for $d = 3$). The data are exact, so half-widths and residuals measure misfit, not noise. Inside the window the laws are hard to tell apart; outside it they are not: a pure N^2 law for $d = 2$ is 30.80% too low at $N = 16\,777\,216$, and a pure N^3 law for $d = 3$ is 6798.98% too high at $N = 4096$. On the information criterion $AN^a(\ln N)^b$ is preferred to $N^2(A \ln \ln N + B)$ in both two-dimensional panels, so the fits do not reveal the $\ln \ln N$ law, which is proved, with a slowly decaying remainder, in Theorem 7.19. A second implementation reproduces parameters and extrapolation errors to the printed digits; its ΔAICc values differ by at most 2.8 units, without changing a ranking.

Table S16: Least-squares fits of $\ln t^*$ against N (CC). Parameters are given with the half-width of their 95% least-squares interval in brackets; “rms” and “max” are the root-mean-square and the largest absolute residual of $\ln t^*$, in units of 10^{-2} ; ΔAICc is relative to the best law of the panel. The last columns give the relative error, in per cent, of the fitted law at sizes outside the fitted window ($2^{24} = 16\,777\,216$). The closed-form approximation (34) and the closed form (35), which have no fitted parameter, are assessed in Table 4.

law	parameters (half-width)	rms	max	ΔAICc	extrapolation error (%)			
(a) $d = 2$, continuous time at unit rate, fitted on 13 sizes with $10 \leq N \leq 200$; errors at								
					$N = 1280$	10 240	163 840	2^{24}
AN^2	$\ln A = 0.50339 (0.052)$	8.34	18.1	121.7	−17.04	−21.89	−26.19	−30.80
AN^a	$\ln A = 0.14378 (0.046)$ $a = 2.0925 (0.011)$	1.53	3.42	81.2	+12.21	+28.04	+56.34	+124.92
$AN^2(\ln N)^b$	$\ln A = 0.047535 (0.019)$ $b = 0.34275 (0.014)$	0.519	1.09	53.0	+3.23	+6.08	+9.67	+14.99
$AN^a(\ln N)^b$	$\ln A = 0.0034407 (0.0041)$ $a = 1.9526 (0.0037)$ $b = 0.51462 (0.014)$	0.0572	0.0933	0.0	−1.34	−4.03	−8.99	−18.97
$N^2(A \ln \ln N + B)$	$A = 0.55506 (0.0069)$ $B = 0.92175 (0.009)$	0.160	0.299	22.3	+1.00	+1.78	+2.67	+3.84
$N^2(A \ln N + B)$	$A = 0.1537 (0.015)$ $B = 1.0622 (0.056)$	1.19	2.55	74.7	+8.42	+17.17	+29.72	+51.40
(b) $d = 3$, continuous time at unit rate, fitted on 10 sizes with $10 \leq N \leq 100$; errors at								
					$N = 320$	1280	4096	
AN^3	$\ln A = -2.0909 (0.44)$	58.3	101	143.7	+584.88	+2286.85	+6798.98	
AN^2	$\ln A = 1.4547 (0.091)$	12.0	23.4	112.2	−25.82	−35.37	−41.62	
AN^a	$\ln A = 0.85038 (0.05)$ $a = 2.1704 (0.014)$	1.19	2.25	70.2	+8.35	+19.56	+31.67	
$AN^2(\ln N)^b$	$\ln A = 0.74292 (0.0024)$ $b = 0.57194 (0.0019)$	0.0494	0.105	6.5	−0.81	−2.25	−3.77	
$AN^a(\ln N)^b$	$\ln A = 0.74632 (0.0024)$ $a = 2.0058 (0.0033)$ $b = 0.55273 (0.011)$	0.0264	0.0368	0.0	−0.50	−1.57	−2.72	
$N^2(A \ln N + B)$	$A = 0.72063 (0.023)$ $B = 1.7587 (0.081)$	0.489	0.878	52.4	+2.46	+4.34	+5.67	
(c) $d = 2$, exact discrete modes at $q = 0.8$, 25 sizes with $11 \leq N \leq 200$								
AN^2	$\ln A = 0.72188 (0.033)$	7.77	16.1	242.8				
AN^a	$\ln A = 0.37735 (0.025)$ $a = 2.0901 (0.0064)$	1.27	3.26	154.8				
$AN^2(\ln N)^b$	$\ln A = 0.27394 (0.011)$ $b = 0.34042 (0.0085)$	0.448	1.09	102.8				
$AN^a(\ln N)^b$	$\ln A = 0.22176 (0.0031)$ $a = 1.9508 (0.0026)$ $b = 0.52296 (0.0099)$	0.0542	0.112	0.0				
$N^2(A \ln \ln N + B)$	$A = 0.69628 (0.0061)$ $B = 1.1483 (0.0079)$	0.159	0.296	51.1				

Table S16 (continued)

law	parameters (half-width)	rms	max	ΔAICc	extrapolation error (%)
$N^2(A \ln N + B)$	$A = 0.18788 (0.01)$ $B = 1.346 (0.04)$	0.997	2.50	142.8	
(d) $d = 3$, exact discrete modes at $q = 0.8$, 14 sizes with $11 \leq N \leq 100$					
AN^3	$\ln A = -1.8032 (0.34)$	57.3	96.1	206.3	
AN^2	$\ln A = 1.665 (0.07)$	11.7	20.1	161.8	
AN^a	$\ln A = 1.0788 (0.032)$ $a = 2.169 (0.0091)$	0.999	2.03	96.2	
$AN^2(\ln N)^b$	$\ln A = 0.96098 (0.0012)$ $b = 0.57548 (0.001)$	0.0322	0.0667	0.0	
$AN^a(\ln N)^b$	$\ln A = 0.96208 (0.0026)$ $a = 2.0016 (0.0034)$ $b = 0.56994 (0.012)$	0.0307	0.0684	2.7	
$N^2(A \ln N + B)$	$A = 0.89816 (0.02)$ $B = 2.2067 (0.069)$	0.423	0.877	72.1	

S10.3 Full-support check and dependence on the activity

In 35 runs the exact probability mass function was iterated over its whole support, until $S(t) < 10^{-12}$. In every run it has one local maximum and no local minimum, and the accumulated mass differs from 1 by at most 2.0×10^{-12} ; for $d = 1$ the mean recomputed from the mass function equals $N(N-1)/q$ to the four decimals printed (Table S17). For these placements and activities unimodality is proved (Theorems 7.6, 7.7 and 7.13).

Table S17: Exact discrete first-passage mass functions iterated until $S(t) < 10^{-12}$: number of steps, missing mass $1 - \sum_t f(t)$, mode, number of local maxima and of local minima over the whole support, mean recomputed from the mass function, coefficient of variation, probability of arrival by the mode, and median. Geometry labels as in Section 2.6.

d	geom.	N	q	steps	$1 - \sum f$	t^*	max.	min.	mean	CV	$\mathbb{P}(\mathcal{T} \leq t^*)$	median
1	CC	5	0.8	1024	2.2×10^{-16}	8	1	0	25.0000	0.8124	0.1700	19
1	CC	5	0.5	1024	3.0×10^{-14}	13	1	0	40.0000	0.8216	0.1743	30
1	CC	21	0.8	12 288	3.7×10^{-13}	174	1	0	525.0000	0.8163	0.1655	398
1	CC	21	0.5	19 456	5.0×10^{-13}	279	1	0	840.0000	0.8167	0.1662	636
1	CC	51	0.8	72 704	7.7×10^{-13}	1062	1	0	3187.5000	0.8165	0.1665	2414
1	CC	51	0.5	115 712	8.9×10^{-13}	1699	1	0	5100.0000	0.8165	0.1665	3863
1	CC	101	0.8	285 696	9.7×10^{-13}	4207	1	0	12 625.0000	0.8165	0.1665	9563
1	CC	101	0.5	456 704	9.8×10^{-13}	6732	1	0	20 200.0000	0.8165	0.1665	15 301
2	CC	5	0.8	4096	-4.4×10^{-16}	36	1	0	133.5227	0.8581	0.1413	99
2	CC	5	0.5	5120	8.4×10^{-13}	58	1	0	213.6364	0.8597	0.1430	158
2	CC	9	0.8	14 336	8.6×10^{-13}	136	1	0	582.8188	0.8832	0.1239	426
2	CC	9	0.5	23 552	4.1×10^{-13}	219	1	0	932.5100	0.8836	0.1253	682
2	CC	15	0.8	50 176	6.5×10^{-13}	414	1	0	1983.6493	0.8992	0.1129	1440
2	CC	15	0.5	79 872	7.5×10^{-13}	663	1	0	3173.8388	0.8993	0.1130	2304
2	CC	21	0.8	110 592	7.7×10^{-13}	849	1	0	4359.6443	0.9077	0.1064	3153
2	CC	21	0.5	176 128	8.9×10^{-13}	1359	1	0	6975.4308	0.9078	0.1065	5044
2	CC	29	0.8	233 472	8.7×10^{-13}	1679	1	0	9177.4308	0.9147	0.1009	6614
2	CC	29	0.5	372 736	9.4×10^{-13}	2688	1	0	14 683.8893	0.9147	0.1010	10 583
2	CC	35	0.8	359 424	9.0×10^{-13}	2493	1	0	14 100.8080	0.9183	0.0981	10 145
2	CC	35	0.5	574 464	9.7×10^{-13}	3989	1	0	22 561.2929	0.9183	0.0981	16 233
3	CC	5	0.8	14 336	9.4×10^{-13}	85	1	0	555.3328	0.9290	0.0852	397
3	CC	7	0.8	43 008	8.0×10^{-13}	187	1	0	1616.0828	0.9494	0.0665	1146
3	CC	9	0.8	95 232	9.6×10^{-13}	332	1	0	3545.2319	0.9607	0.0551	2501
3	CC	11	0.8	177 152	1.3×10^{-12}	523	1	0	6602.0241	0.9679	0.0475	4642
3	CC	15	0.8	462 848	2.0×10^{-12}	1044	1	0	17 135.4649	0.9764	0.0375	12 003
2	C2M	5	0.8	2048	-4.4×10^{-16}	10	1	0	45.0000	0.8840	0.1238	33
2	C2M	11	0.8	8192	4.5×10^{-14}	57	1	0	293.7896	0.9063	0.1061	213

Table S17 (continued)

d	geom.	N	q	steps	$1 - \sum f$	t^*	max.	min.	mean	CV	$\mathbb{P}(\mathcal{T} \leq t^*)$	median
2	C2M	21	0.8	33 792	5.3×10^{-13}	226	1	0	1297.6238	0.9197	0.0971	933
2	M2C	5	0.8	4096	3.3×10^{-16}	6	1	0	113.4280	1.0030	0.0319	78
2	M2C	11	0.8	23 552	7.1×10^{-13}	46	1	0	848.1058	0.9913	0.0352	589
2	M2C	21	0.8	109 568	9.1×10^{-13}	196	1	0	3995.3900	0.9886	0.0313	2780
3	C2M	5	0.8	6144	4.2×10^{-14}	24	1	0	207.5000	0.9490	0.0681	147
3	C2M	11	0.8	64 512	1.0×10^{-12}	141	1	0	2376.7869	0.9770	0.0367	1664
3	M2C	5	0.8	14 336	8.8×10^{-13}	17	1	0	520.3252	0.9907	0.0205	362
3	M2C	11	0.8	177 152	1.3×10^{-12}	115	1	0	6429.2117	0.9937	0.0112	4469

For $q \leq 0.9$ the product of q and the discrete mode differs from the continuous-time mode at unit rate by at most 2.16: qt^* is independent of q up to a few steps (Table S18). At $q = 1$ the walk has a parity. For $d = 1$ the raw maximiser (832 at $N = 51$, 13 332 at $N = 201$) lies below the continuous-time mode (849.87, 13 398.02); the maximiser of the two-step average (848, 13 398) removes the bias. For $d = 2$ and $q = 1$ the mass function has two neighbouring local maxima at its peak ($t = 678$ and 680 for $N = 21$), which moves the maximiser by at most two steps.

Table S18: Dependence on the activity q and parity (CC, exact discrete mass functions). “Two-step” is the first index of the maximising pair of $\frac{1}{2}[f(t) + f(t + 1)]$; t_c^* is the mode of the continuous-time density at unit rate.

d	N	q	$\arg \max_t f(t)$	two-step	$q \times \arg \max_t f(t)$	t_c^*
1	51	0.3	2832	2832	849.6	849.87
1	51	0.5	1699	1699	849.5	849.87
1	51	0.9	944	943	849.6	849.87
1	51	1.0	832	848	832.0	849.87
1	201	0.3	44 660	44 659	13 398.0	13 398.02
1	201	0.5	26 796	26 795	13 398.0	13 398.02
1	201	0.9	14 886	14 886	13 397.4	13 398.02
1	201	1.0	13 332	13 398	13 332.0	13 398.02
2	21	0.3	2265	2265	679.5	679.83
2	21	0.5	1359	1358	679.5	679.83
2	21	0.8	849	848	679.2	679.83
2	21	0.9	755	754	679.5	679.83
2	21	1.0	678	678	678.0	679.83
2	51	0.3	14 599	14 598	4379.7	4379.94
2	51	0.5	8759	8758	4379.5	4379.94
2	51	0.9	4866	4865	4379.4	4379.94
2	51	1.0	4380	4379	4380.0	4379.94
2	101	0.3	60 187	60 186	18 056.1	18 056.39
2	101	0.5	36 112	36 111	18 056.0	18 056.39
2	101	0.9	20 062	20 061	18 055.8	18 056.39
2	101	1.0	18 056	18 055	18 056.0	18 056.39
3	11	0.3	1397	1397	419.1	419.61
3	11	0.5	838	837	419.0	419.61
3	11	0.8	523	522	418.4	419.61
3	11	0.9	465	464	418.5	419.61
3	11	1.0	418	418	418.0	419.61
3	21	0.3	5839	5839	1751.7	1752.30
3	21	0.5	3503	3502	1751.5	1752.30
3	21	0.9	1945	1945	1750.5	1752.30
3	21	1.0	1750	1750	1750.0	1752.30
3	31	0.3	13 628	13 628	4088.4	4089.06
3	31	0.5	8176	8176	4088.0	4089.06
3	31	0.9	4541	4541	4086.9	4089.06
3	31	1.0	4087	4086	4087.0	4089.06

S10.4 Deterministic geometries

Table S19 lists the clean lattice and the 28 deterministic placements of Section S9.7 under the identifiers of the data file; the key is in the caption. The ratio t^*/T ranges from 0.0905 (target enclosed) to 0.3385 (serpentine), against 0.1768 on the clean lattice. A second implementation reproduces every row, with identical modes. The periodic rows are for one phase of each array, and the phase matters: for $A = 3$ and $A = 2$ the tabulated phase is the one with the smallest factor of the mean, for $A = 4$ the twelfth smallest of sixteen; over the four phases of the period-2 array the factor of the mean averages 1.46.

Table S19: The clean lattice and 28 deterministic placements of blocked sites, $N = 35$, $q = 0.8$, CC, discrete time, all values exact: number M of blocked sites, number n of accessible sites, mean, mode, their ratio, their factors relative to the clean lattice (T_0 , t_0^* : first row), coefficient of variation and median in units of the mean. *Key* (sites (x_1, x_2) , start $(0, 0)$, target $(34, 34)$). *wall*: every site of the line $x_1 = 17$ (mid), 5 (*nearstart*) or 29 (*neartarget*) is blocked except a gap of g consecutive sites, centred or at the edge $x_2 = 0$ (*startedge*) or $x_2 = 34$ (*targetedge*). *serpentine_kK*: K equally spaced walls of this kind with three-site gaps at alternating edges. *target_box_RR_gG*, *start_box*: an L-shaped wall at Chebyshev distance R from the corner, with a door on one arm: for $G = 1$ one open site, the second site from the lattice edge (the edge site itself stays blocked); for $G = 3$ the three sites nearest the edge. *both_box*: the two together. *blockK*: a solid $K \times K$ block occupying, in both coordinates, the sites 26–31 (*near_target*), 3–8 (*near_start*), or 14–19 ($K = 6$) and 11–22 ($K = 12$) in the centre. *periodic_aA*: the sites whose two coordinates are both congruent to a fixed offset modulo A (offsets 1, 2, 1 for $A = 4$, 3, 2; for $A = 3$ and $A = 2$ this is the phase with the smallest factor of the mean).

geometry	M	n	T	t^*	t^*/T	T/T_0	t^*/t_0^*	CV	median/ T
clean	0	1225	14 100.8	2493	0.1768	1.000	1.000	0.918	0.719
wall_mid_gap_centre_g1	34	1191	24 822.5	7826	0.3153	1.760	3.139	0.855	0.752
wall_mid_gap_centre_g3	32	1193	19 366.8	5711	0.2949	1.373	2.291	0.866	0.745
wall_mid_gap_centre_g9	26	1199	16 363.7	3996	0.2442	1.160	1.603	0.891	0.732
wall_mid_gap_startedge_g3	32	1193	24 599.2	6166	0.2507	1.745	2.473	0.889	0.736
wall_mid_gap_targetedge_g3	32	1193	21 117.2	6921	0.3277	1.498	2.776	0.839	0.762
wall_nearstart_gap_centre_g3	32	1193	15 306.9	4665	0.3048	1.086	1.871	0.850	0.749
wall_neartarget_gap_centre_g3	32	1193	32 212.4	4341	0.1348	2.284	1.741	0.949	0.710
serpentine_k1_g3	32	1193	21 343.1	6750	0.3163	1.514	2.708	0.847	0.758
serpentine_k2_g3	64	1161	43 869.9	14 302	0.3260	3.111	5.737	0.826	0.755
serpentine_k3_g3	96	1129	49 880.9	16 886	0.3385	3.537	6.773	0.820	0.757
serpentine_k4_g3	128	1097	93 318.3	30 424	0.3260	6.618	12.204	0.823	0.755
serpentine_k6_g3	192	1033	165 747.9	54 165	0.3268	11.754	21.727	0.822	0.756
target_box_R3_g1	6	1219	30 307.9	3029	0.0999	2.149	1.215	0.960	0.706
start_box_R3_g1	6	1219	14 125.0	2610	0.1848	1.002	1.047	0.911	0.722
target_box_R5_g1	10	1215	32 882.3	3040	0.0925	2.332	1.219	0.963	0.705
start_box_R5_g1	10	1215	14 354.9	3112	0.2168	1.018	1.248	0.895	0.728
target_box_R8_g1	16	1209	35 239.3	3188	0.0905	2.499	1.279	0.962	0.705
start_box_R8_g1	16	1209	15 101.7	4444	0.2943	1.071	1.783	0.855	0.745
target_box_R5_g3	8	1217	22 256.5	2782	0.1250	1.578	1.116	0.946	0.710
start_box_R5_g3	8	1217	14 166.5	2740	0.1934	1.005	1.099	0.907	0.723
both_box_R5_g1	20	1205	33 014.3	3799	0.1151	2.341	1.524	0.951	0.709
block6_near_target	36	1189	17 302.6	2536	0.1466	1.227	1.017	0.934	0.714
block6_centre	36	1189	13 951.0	2674	0.1917	0.989	1.073	0.910	0.722
block6_near_start	36	1189	13 689.9	2498	0.1825	0.971	1.002	0.913	0.721
block12_centre	144	1081	13 458.9	3074	0.2284	0.954	1.233	0.890	0.730
periodic_a4	81	1144	16 422.8	2863	0.1743	1.165	1.148	0.920	0.719
periodic_a3	121	1104	15 272.7	2945	0.1928	1.083	1.181	0.909	0.723
periodic_a2	289	936	17 588.7	3494	0.1987	1.247	1.402	0.905	0.724

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